User Manual

Overview

What Is Tier0?

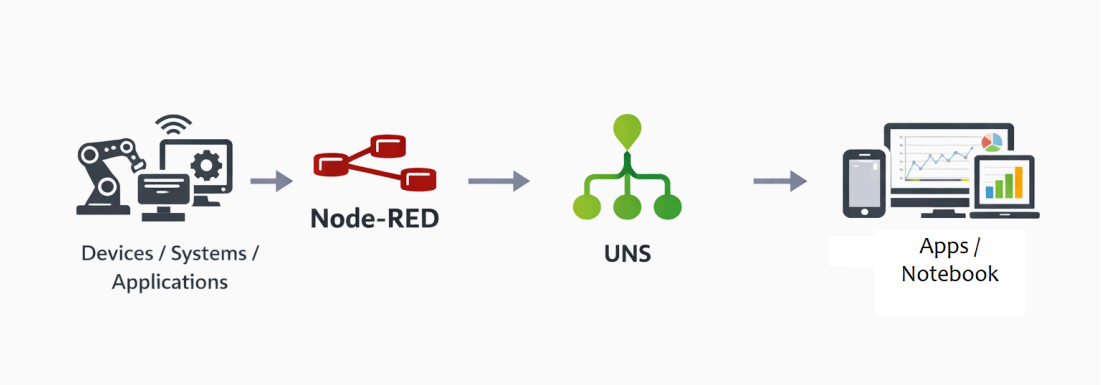
Tier0 is a Unified Namespace (UNS)-based industrial data platform.

It connects data through Node-RED, organizes it into a unified structure, and makes it available for applications and analytics.

How It Works?

Tier0 moves and structures data in a single workflow:

- Node-RED connects and processes data from devices and systems
- Data is published into the Unified Namespace (UNS), structured and stored in Tier0 database
- Data is consumed by dashboards, applications and analytics

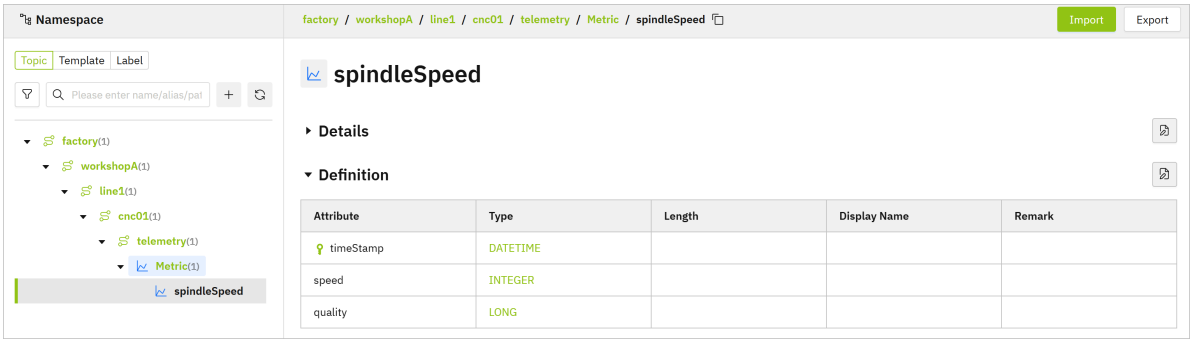


Key Components

Unified Namespace (UNS)

Defines a hierarchical structure for all industrial data.

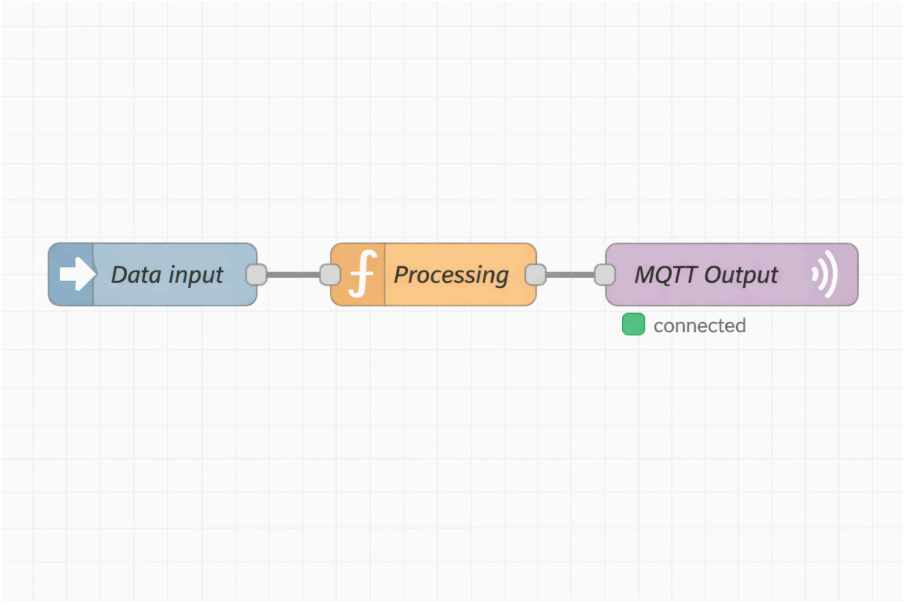
`factory/workshopA/line1/cnc01/telemetry/spindleSpeed`



Node-RED Integration

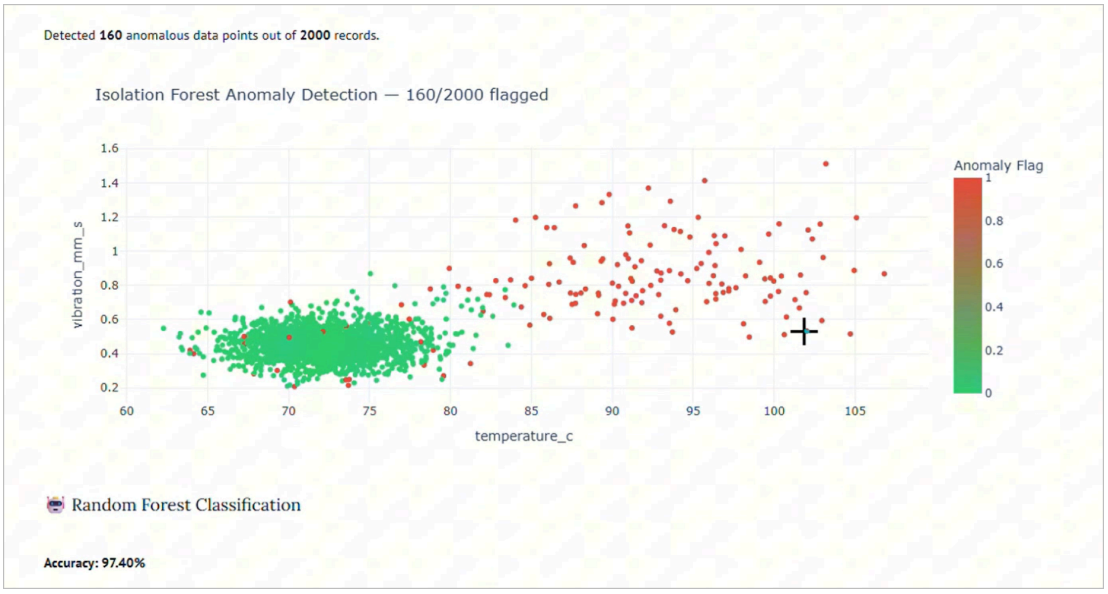
The core engine for data connection and processing.

- Connect devices, applications, and systems
- Format and publish data into the namespace



Python Notebook

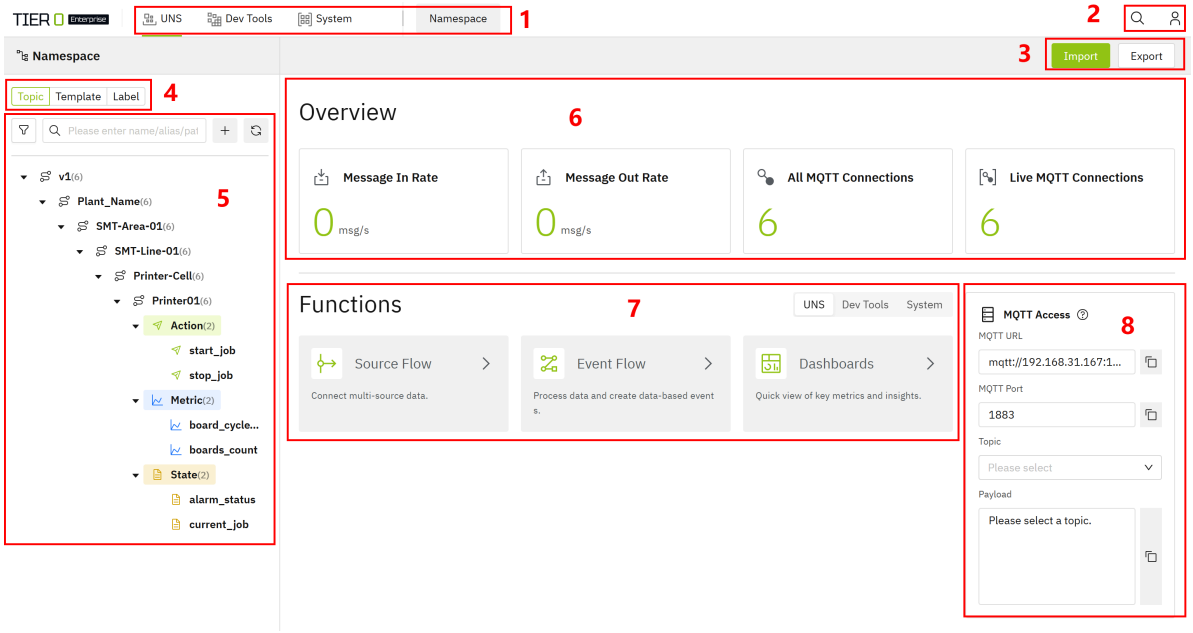
Analyze data with full flexibility using Python with an interactive environment.



Agentic App Builder

Create industrial applications using natural language without consuming any of your local resources. The whole process is done on cloud, and provide a package for local installation.

Homepage Introduction


TIER
Enterprise
UNS
Dev Tools
System
Namespace
1
2
Namespace
3
Import
Export
Topic
Template
Label
4
Please enter name/alias/pat
v1(6)
Plant_Name(6)
SMT-Area-01(6)
SMT-Line-01(6)
Printer-Cell(6)
Printer01(6)
Action(2)
start_job
stop_job
Metric(2)
board_cycle...
boards_count
State(2)
alarm_status
current_job
5
Overview
6
Message In Rate
0 msg/s
Message Out Rate
0 msg/s
All MQTT Connections
6
Live MQTT Connections
6
Functions
7
UNS
Dev Tools
System
Source Flow
Connect multi-source data.
Event Flow
Process data and create data-based event s.
Dashboards
Quick view of key metrics and insights.
MQTT Access
8
MQTT URL
mqtt://192.168.31.167:1...
MQTT Port
1883
Topic
Please select
Payload
Please select a topic.

Homepage

No.	Name	Description
1	Menu Bar	Lists the function menus of the platform, enabling quick access to various function modules.
2	Tool Bar	Common operations on the platform, including search and user information.
3	Import/Export	Integrate the data import/export function from **Namespace/Source Flow/Event Flow/Dashboard** together.
		Go back to the homepage.
4	Topic	Tree view of data models already built in UNS
	Template	Tree view of data model templates already built in UNS
	Label	Displays labels, and models under each label
5		Displays data models by category: **Text/Template/Label**.
	/	Search data models / Refresh data model list.
		Create folder-file structured data models based on data hierarchy.
6	Overview	Number of connections between data sources and the internal MQTT broker and messages in/out.
7	Functions	Lists all function modules of the platform.
8	MQTT Access	Detailed information of the internal MQTT broker. Select existing topics to see detailed payload.

Function Module Description

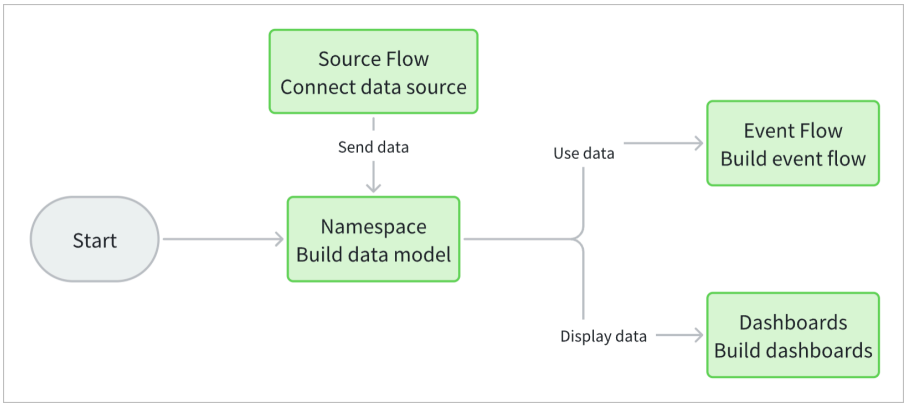
Module	Name	Description
UNS	Namespace (homepage)	Models industrial data into a hierarchical structure (folder-file) for clarity.
	Source Flow	Uses Node-RED to define the data source of the created model.
	Event Flow	Process Namespace data and configure data events via Node-RED.
	Dashboards	Displays data from **Namespace** on dashboards according to specific design.
Dev Tools	Containers	Integrates Portainer to manage docker containers.
	Notebook	Adopts Marimo for easy python script usage. For example, train models, run SQL queries and others.
	Advanced Use	Provides account and password of integrated application for deeper exploration.
System	Routing Management	Customizes Tier0 function module routing.
	User Management	Manage access control via users and roles.
	Open Data	Provide different types of data subscription, including API, MQTT and MCP.
	Menu Config	Supports menu changes on all existing function modules.

> ⓘ **info**
>
> - Private deployment. For detailed specifications and steps, see Deploy Tier0.
> - Applications built through natural language in **App Builder** is generated on cloud, and a package is available for local installation.

User Journey

User Journey

Follow the order displayed in the image, you will complete your first data flow in Tier0.



- Tier0 builds data models in a folder-file structure, making it easy to track and understand;
- It collects various types of data from diverse sources in **Source Flow**;
- Data collected to Tier0 can be processed in many ways in **Event Flow**;
- Processed data can be displayed on dashboards for better understanding through both **Event Flow** and **Dashboards**.
- Data collected to and processed in Tier0 can be obtained by third parties through Open API or MQTT subscription.

> ⓘ **note**
>
> - **Source Flow** and **Event Flow** are based on **Node-RED**. For detailed instructions, please see NodeRed.
> - **Dashboards** is based on **Grafana**. For detailed instructions, please see Grafana.

Quick Start Guide

Deploy Tier0

System Requirements

Configuration Type	CPU	Memory	HDD/IOPS
Basic Configuration	4 cores	8 GB	100 GB, 2000 IOPS (30% write)
Recommended Configuration	8 cores	16 GB	1 TB, 2000 IOPS (30% write)

Installing Tier0

Linux

1. Make sure you have the following system and components ready.

- Ubuntu Server 24.04
- Docker

Component	Version
Docker Engine - Community	27.4.0
Docker Buildx	v0.19.2
Docker Compose	v2.31.0
Containerd	1.7.24

> ⓘ **note**
>
> Above system and components have been tested. You can try on other versions if you are interested.

2. Clone the project from Github.

```
git clone https://github.com/FREEZONEX/Tier0-Edge.git
```

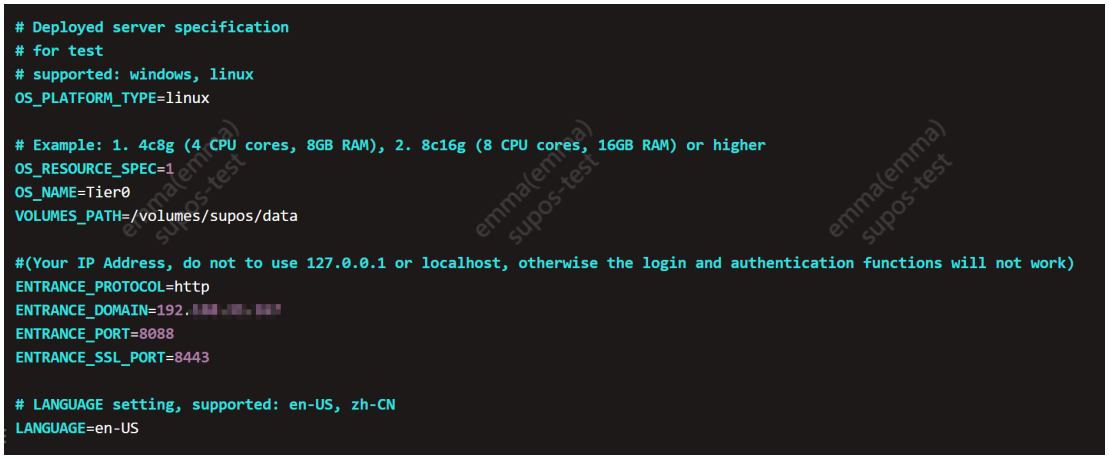
3. Go to the folder **Tier0-Edge/deploy** inside the project.

```
cd Tier0-Edge/deploy
```

4. Enter the **.env.default** file and press the **i** key to edit configuration items as needed.

```
vi .env.default
```

Item	Description
VOLUMES_PATH	The storage path for project data.
ENTRANCE_DOMAIN	Tier0 access address. Normally the IP or domain of the server where you install Tier0. ⚠ **warning** Do not use 127.0.0.1 or localhost, otherwise login and authentication functions **will NOT** work.
LANGUAGE	Tier0 display language. Chinese (zh-CN) and English (en-US) are supported.



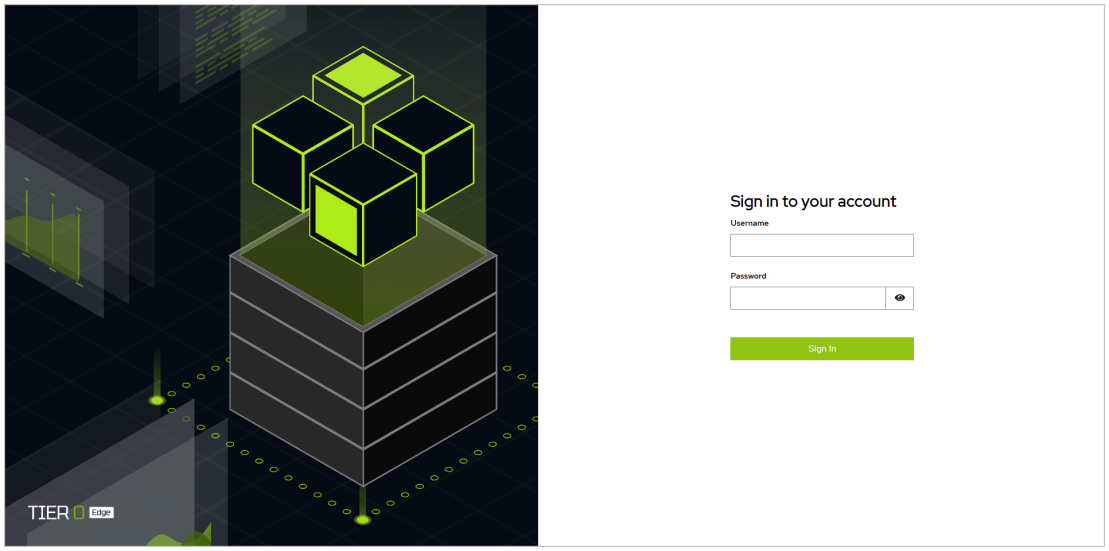
5. Press **ESC** to stop editting, save the file and install Tier0.

```
:wq!
bash bin/install.sh
```

6. Select whether to keep the entered entrance address and whether to use default setup, and then wait for the installation to complete.

```
Available options for ENTRANCE_DOMAIN:
0). Keep current: office.unibutton.com (default)
1). 192.168.31.167
2). 172.17.0.1
3). 172.18.0.1
Select option (0-3), or press Enter to keep current: 0
Keeping current ENTRANCE_DOMAIN: office.unibutton.com
Docker and Docker Compose meet the required versions.
Do you need to customize which services to install? [1] No, use defaults(default) [2] Yes 1
Info: success to generate /root/supOS-CE/bin/init/../../mount/postgresql/init-scripts/create_keycloak_database.sql
Info: success to generate kong_config.yml (auth enabled = true)
Info: loading npm cache complete.
golioth-websocket-datasource/
golioth-websocket-datasource/CHANGELOG.md
golioth-websocket-datasource/gpx_websocket_darwin_amd64
golioth-websocket-datasource/gpx_websocket_darwin_arm64
golioth-websocket-datasource/gpx_websocket_linux_amd64
golioth-websocket-datasource/gpx_websocket_linux_arm
golioth-websocket-datasource/gpx_websocket_linux_arm64
golioth-websocket-datasource/gpx_websocket_windows_amd64.exe
golioth-websocket-datasource/img/
golioth-websocket-datasource/img/assets/
golioth-websocket-datasource/img/assets/golioth-grafana-websockets-plugin-datasource.png
golioth-websocket-datasource/img/assets/grafana-websockets-graphing.png
golioth-websocket-datasource/img/assets/grafana-websockets-plugin-streaming.png
golioth-websocket-datasource/img/logo.svg
golioth-websocket-datasource/LICENSE
golioth-websocket-datasource/MANIFEST.txt
golioth-websocket-datasource/module.js
golioth-websocket-datasource/module.js.LICENSE.txt
golioth-websocket-datasource/module.js.map
```

7. Access Tier0 on a browser and log in.



Windows

> ⓘ **info**
>
> - Make sure you have installed the lastest **Docker Desktop** and **Git** on Windows 10 or 11.
> - Docker must be started.

1. Clone the project from Github in **Git Bash**.

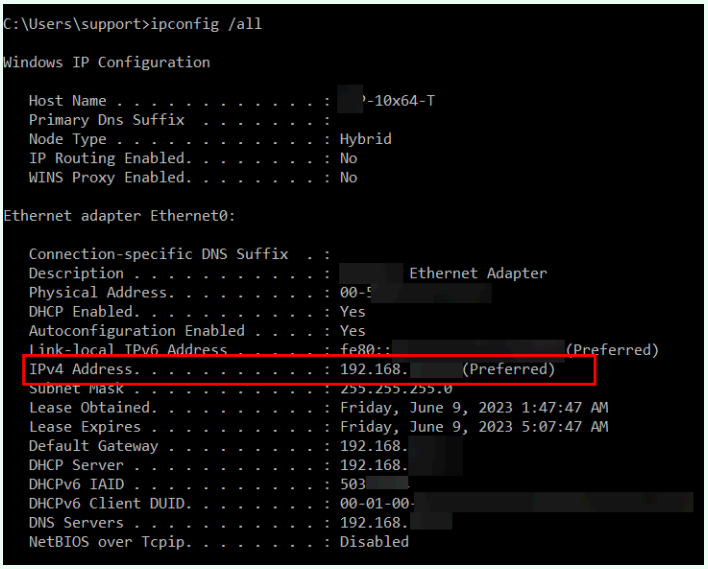
```
git clone https://github.com/FREEZONEX/Tier0-Edge.git
```

2. Go to the folder **Tier0-Edge/deploy** inside the project.

```
cd Tier0-Edge/deploy
```

3. Edit the following attributes in file **.env.default**.

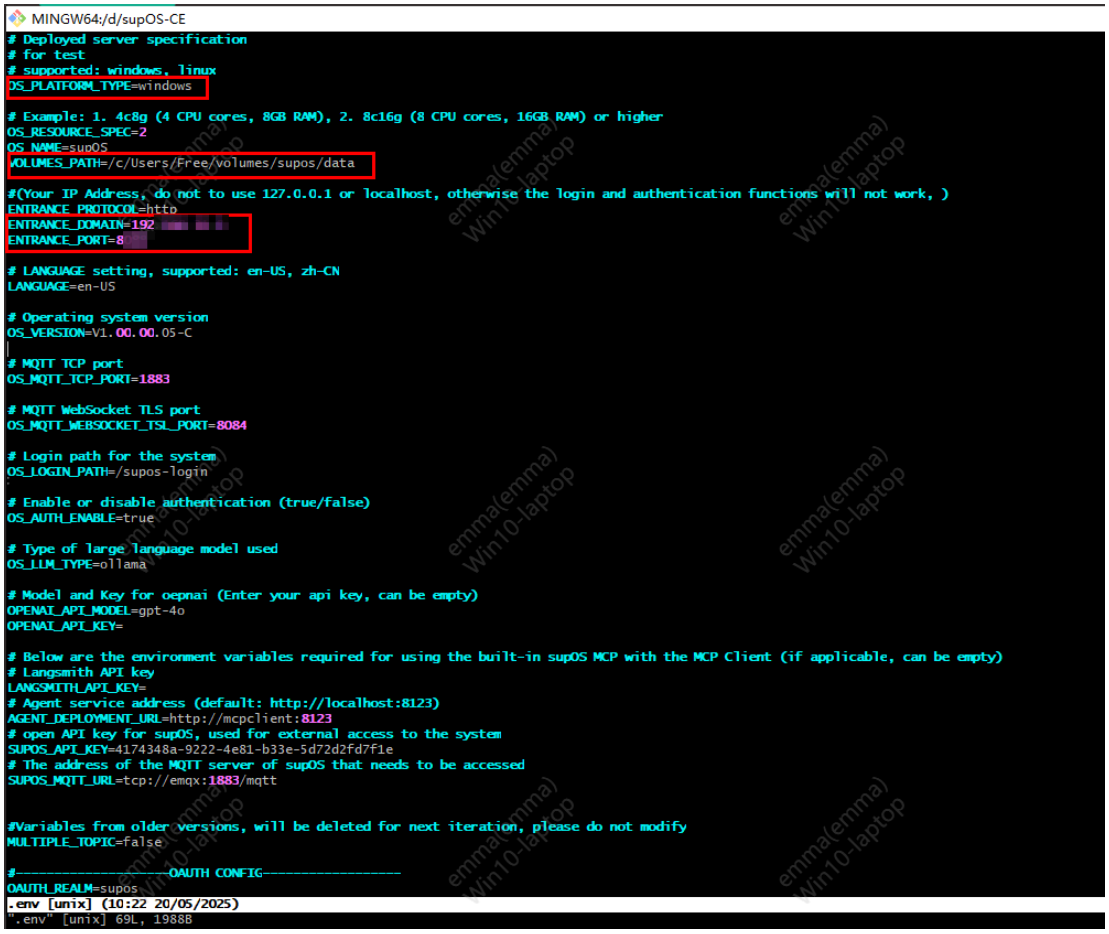
> ✓ **tip**
>
> Before editing .env.default file, execute `ipconfig` in Git Bash to get the IP of the server.



- **OS_PLATFORM_TYPE**: The operating system where Tier0 is installed. Change it to **windows**
- **VOLUMES_PATH**: The storage path for project data.
- **ENTRANCE_DOMAIN/ENTRANCE_PORT**: Tier0 access address. Normally the IP or domain of the server where you install Tier0, and make sure the port is not occupied.

```
vi .env.default
```

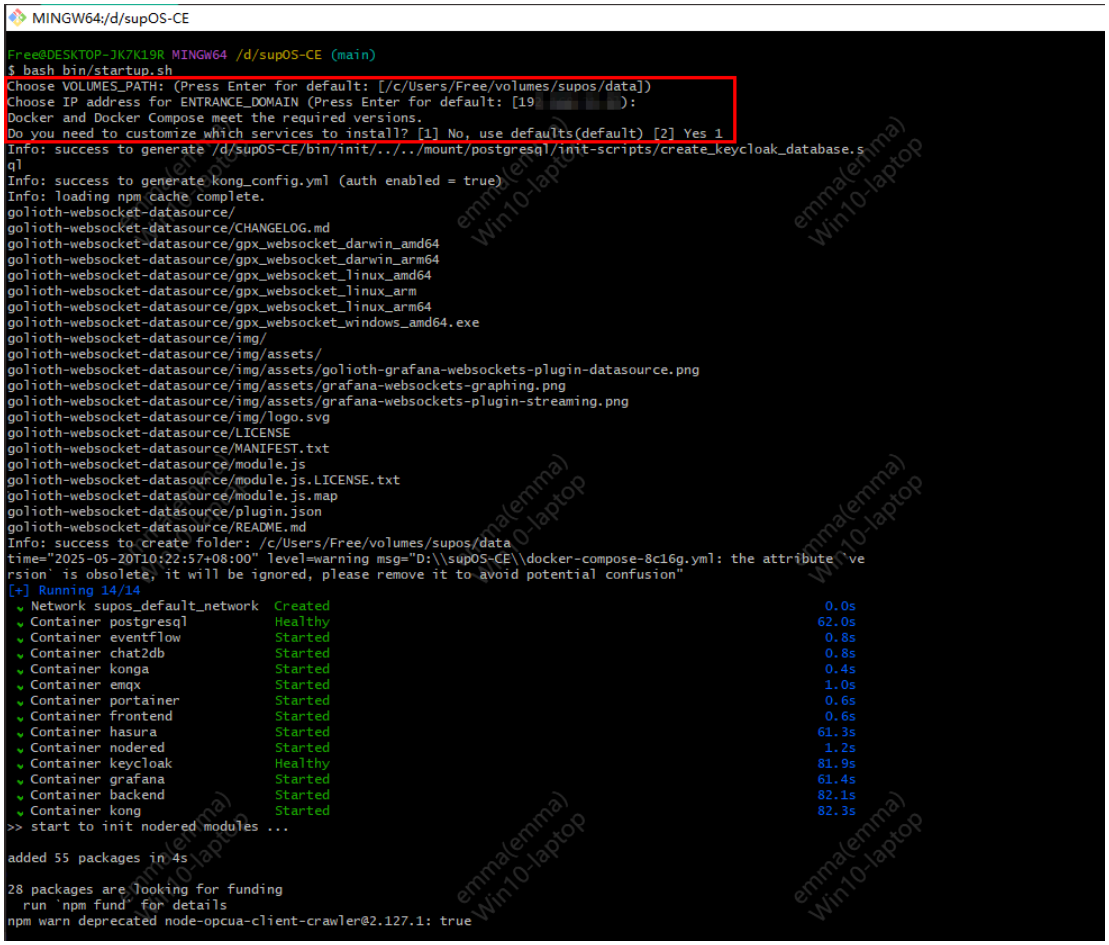
> ⓘ **note**
>
> **.env.default** file is hidden, directly enter the command to open it.



4. Save the file and start installing Tier0.

```
:wq!
bash bin/install.sh
```

5. Select the default configurations and wait for the installation to complete.



6. Access Tier0 on a browser and log in.

Uninstalling Tier0

1. Access the path **Tier0-Edge/deploy/bin**.
2. Execute the script **uninstall.sh**.

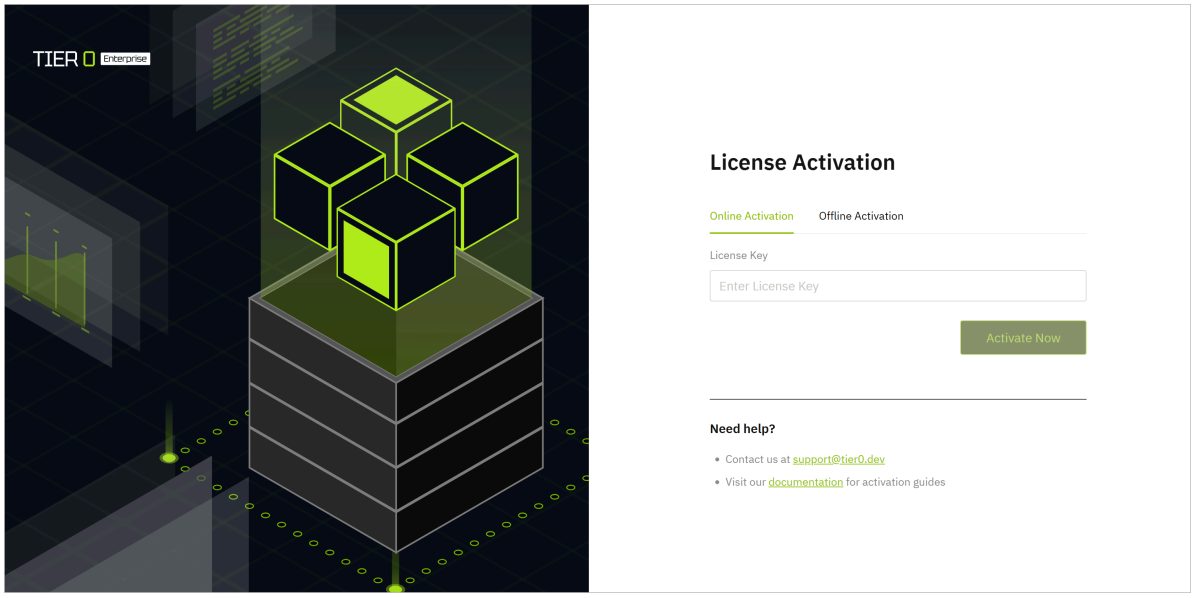
```
bash uninstall.sh
```

> ⓘ **info**
>
> - **clean-all.sh** will wipe out all project data and uninstall Tier0.
> - Images will not be deleted.
> - If you want to install Tier0 again, make sure you have pulled all images with the newest versions (based on image tags).

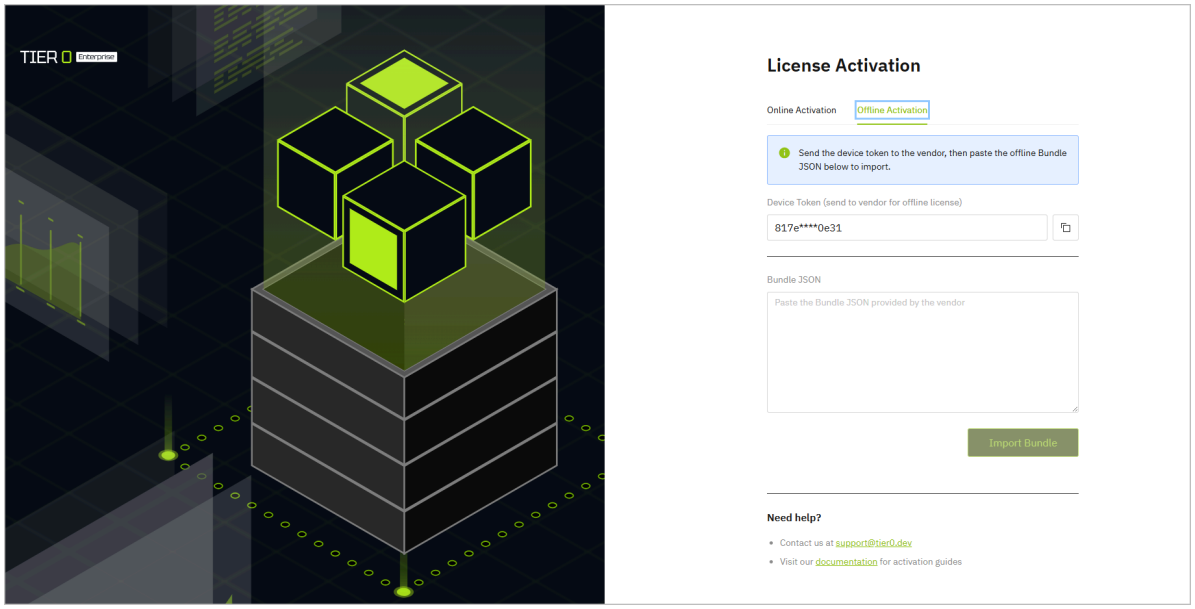
Login

1. Access Tier0 with the set domain and port on a browser.
2. On the login page, use your license to activate Tier0.

- **Online Activation**: Enter the license key, and click **Activate Now**.



- **Offline Activation**: Send the token to your vendor to get the JSON file, and import the file.



3. Enter the default username and password (tier0/tier0).
4. Click **Sign In**.

How-to Guide

Build Data Models

Tier0 uses UNS (Unified Namespace) to organize data into a clear tree structure. Each path and topic in the model automatically generates an MQTT topic, making it easy to get real-time data.

Additional Parameter Definition

- **Path-Additional Settings**

Item	Description
Extended Attribute	Add extended attributes for the path. Eg. unit.
Template	Select a template to inherit attributes from it. See Building Template.
Generate Template	Click ⊕ to add an attribute to the path, at the same time you can select to generate a same template for future use.

- **Topic-Topic Type**

Type	Description
State	Works as an API to receive system state data.
Action	Works as an API to execute CRUD commands.
Metric	Real-time data. For example, equipment temperature.

- **Topic-Attribute Generation Method**

Attribute Generation Method	Description
Custom	Customize attributes.
Template	Inherit attributes from the selected template.
Auto Parsing	Use JSON text to generate attributes with the same structure.

- **Topic-Linked Operation**

Item	Description
Enable History	Enable historical data storage for the topic.

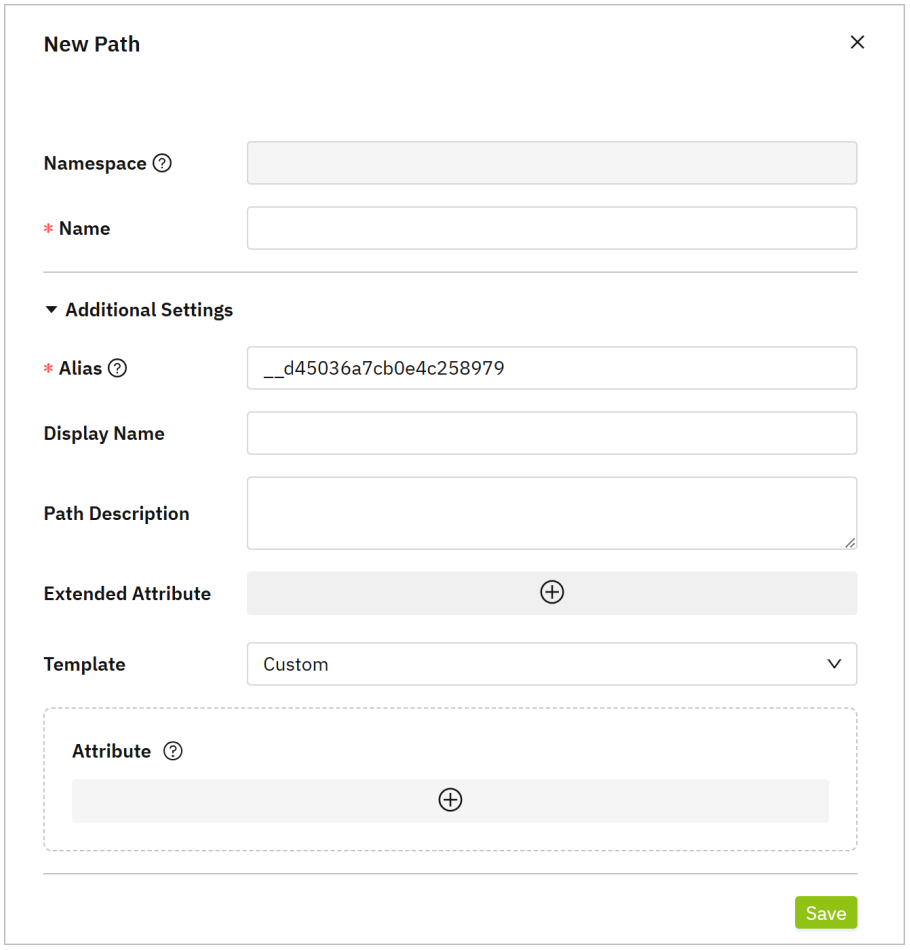
How to Build a Data Model

Based on simple folder-file structure, you can define the data hierarchy to a tree map.

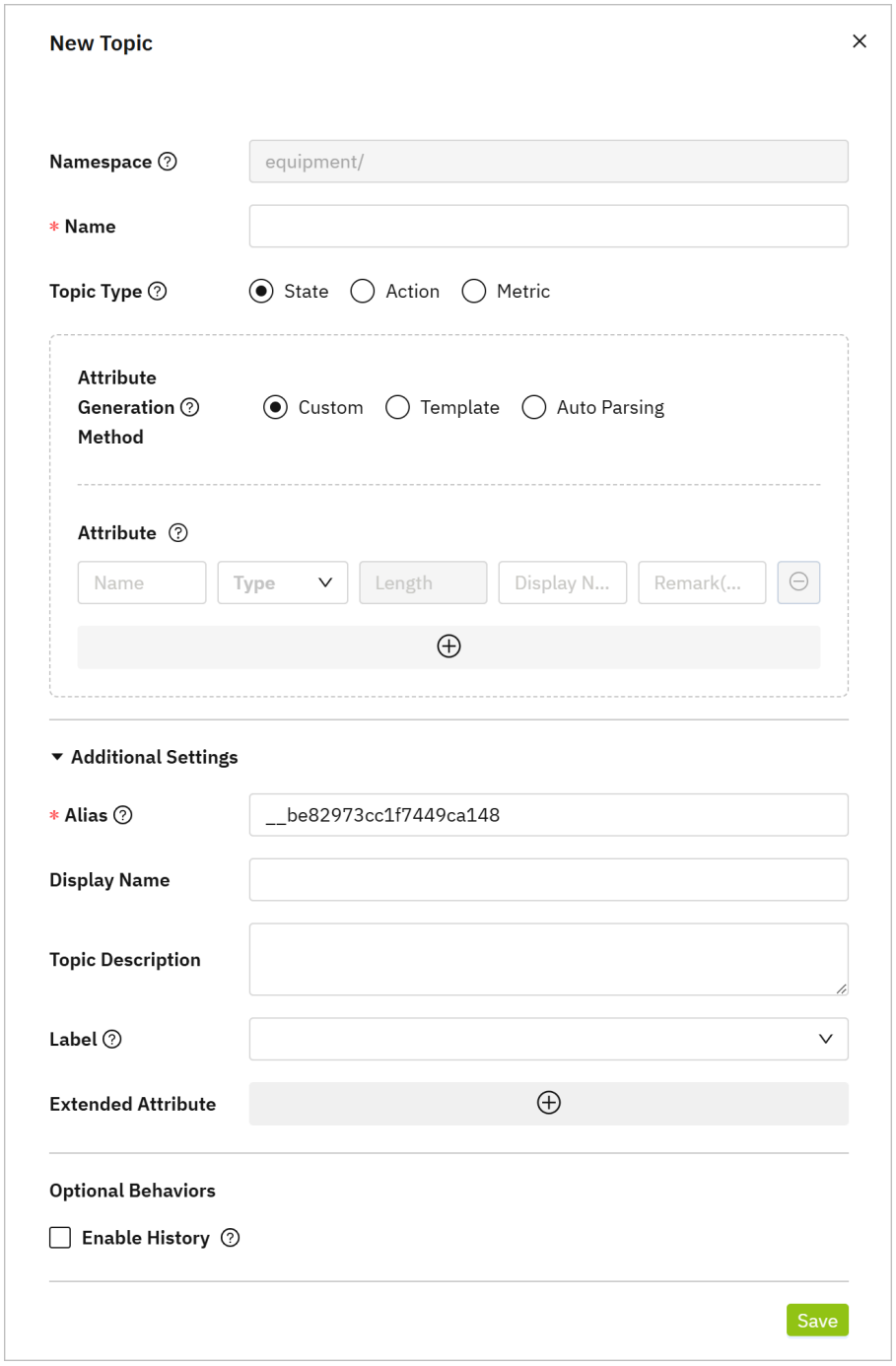
Building Models Manually

> ⓘ **note**
>
> **factory/equipment/CNC** will be used as an example, in which `factory` and `equipment` are paths and `CNC` is a topic.

1. Log in to Tier0, landed on the **Namespace** page.
2. Under **Topic**, click ＋ > **New Path** to add a path (e.g. `factory`).



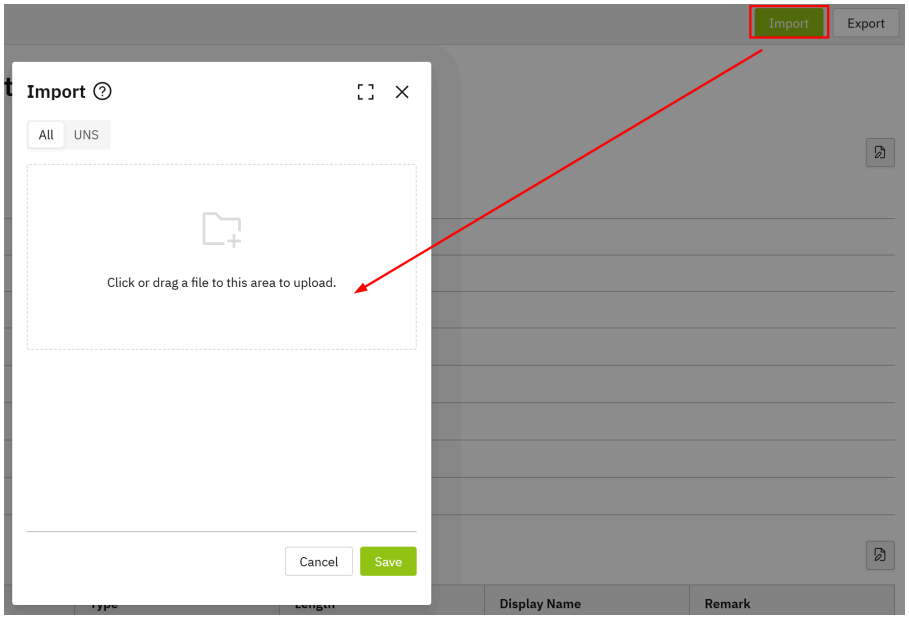
3. Enter the path information, set the alias as needed, and then click **Save**.
4. Select **equipment**, and then click ＋ > **New Topic** to add a topic (e.g. `CNC`) under it.



5. Enter the information of the topic, and then click **Save**.

Importing Models

1. Log in to Tier0, landed on the **Namespace** page.
2. Click **Import** at the upper-right corner.



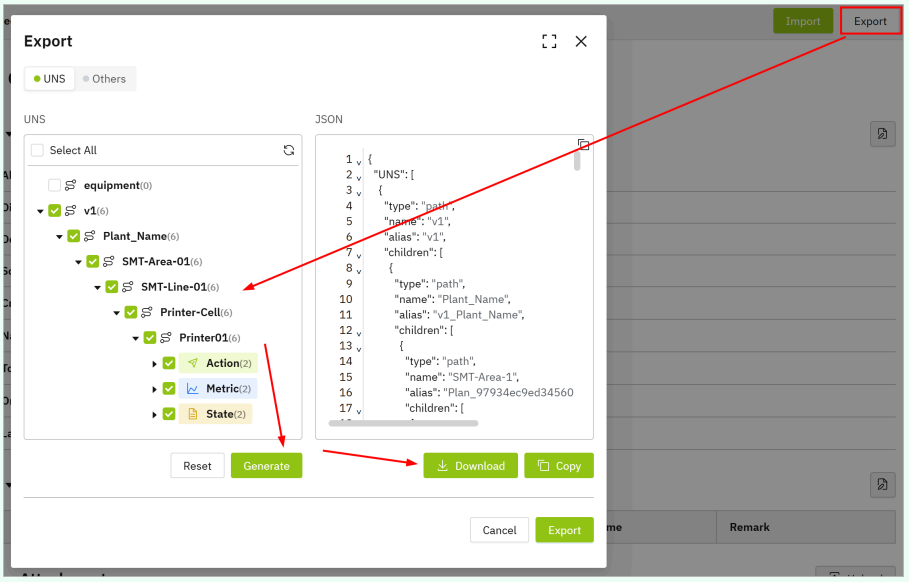
3. Import JSON to create models.

- Click **UNS**, and directly enter JSON.

- Click **Download Template** at the **UNS** tab, enter the model content and upload it at the **All** tab.


```
{
  "notes": "type:folder|file,topicType:STATE|ACTION|METRIC,dataType:TEMPLATE_TYPE|TIME_SEQUENCE_TYPE|RELATION_
  "Label": [{"name": "debug"},{"name": "dev"}],
  "Template": [],
  "UNS": [
    {
      "name": "v1",
      "type": "path",
      "children": [
        {
          "name": "Plant_Name",
          "type": "path",
          "children": [
            {
              "name": "SMT-Area-1",
              "type": "path",
              "children": [
                {
                  "name": "SMT-Line-1",
                  "type": "path",
                  "children": [
                    {
                      "name": "Printer-Cell",
                      "type": "path",
                      "children": [
                        {
                          "name": "Printer01",
                          "type": "path",
                          "children": [
                            {
                              "name": "State",
                              "type": "path",
```


> ✓ **tip**
>
> You can manually add a path and topic, export it and use it as an example for import.



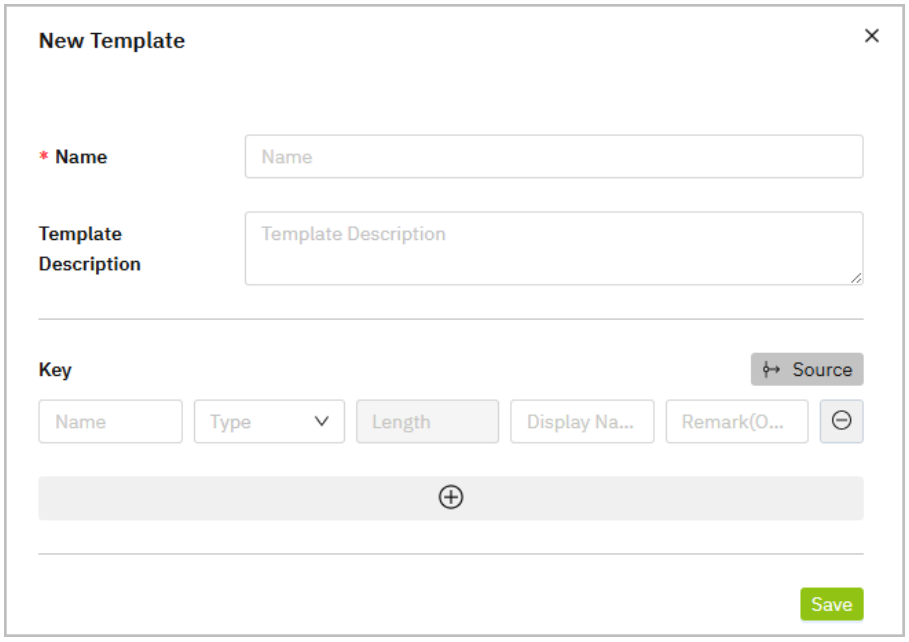
4. Click **Save**.

Optional Features

Building Template

1. On the **Namespace** page, click **Template**.
2. Click , and then enter the information of the template.

> ✓ **tip**
>
> Click **Source** to select added models and inherit their attributes.

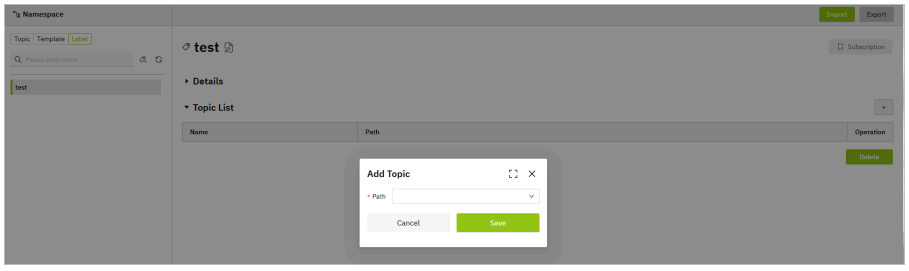


3. Click **Save**.

Creating Label

Labels are used for categorizing data models.

1. On the **Namespace** page, click **Label**.
2. Click , and then enter the label name.
3. Click **Save**, and then click + to add topics under the label.

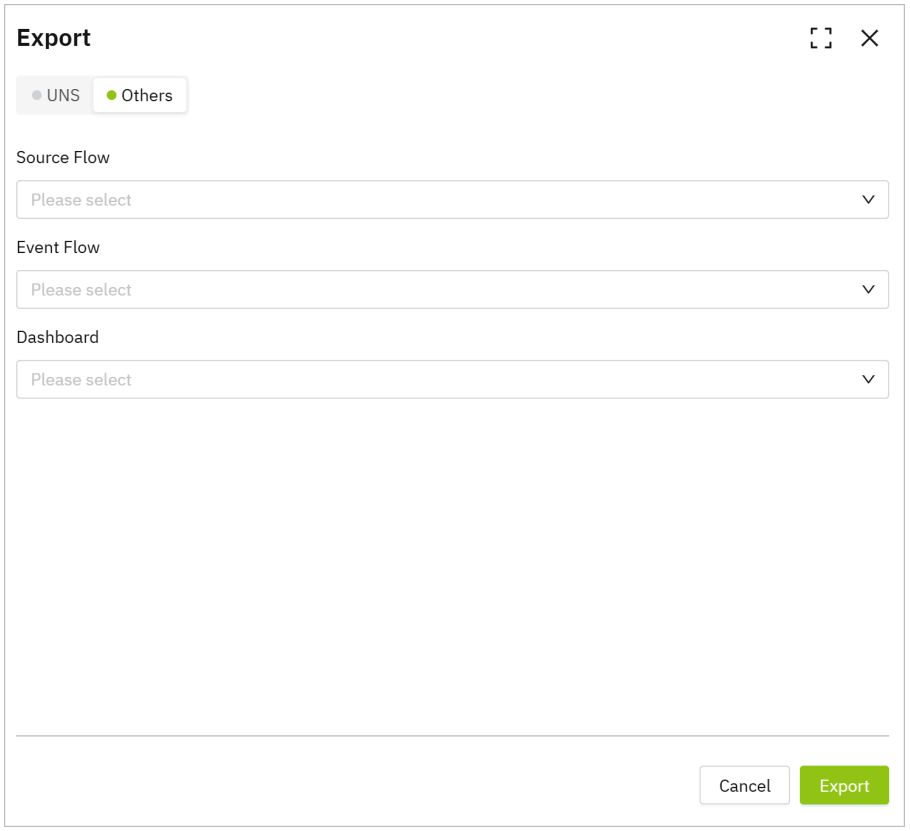


4. Click **Save**.

Exporting Data

Tier0 can export UNS models and other data, such as source flows, event flows, and data dashboards.

1. On the **Namespace** page, click **Export** at the upper-right corner.
2. Click the **Others** tab and select the corresponding data type.



3. Click **Export**.

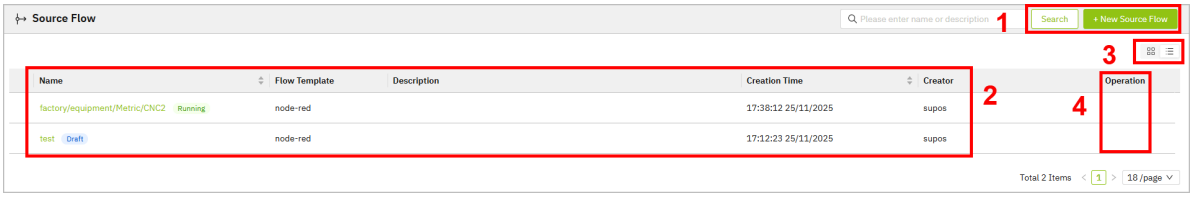
Connect Data Sources

How to Add Data Sources

Once the data model is built under **Namespace**, you need to connect real data to the model for subsequent processing and use.

Adding Data Flow Manually

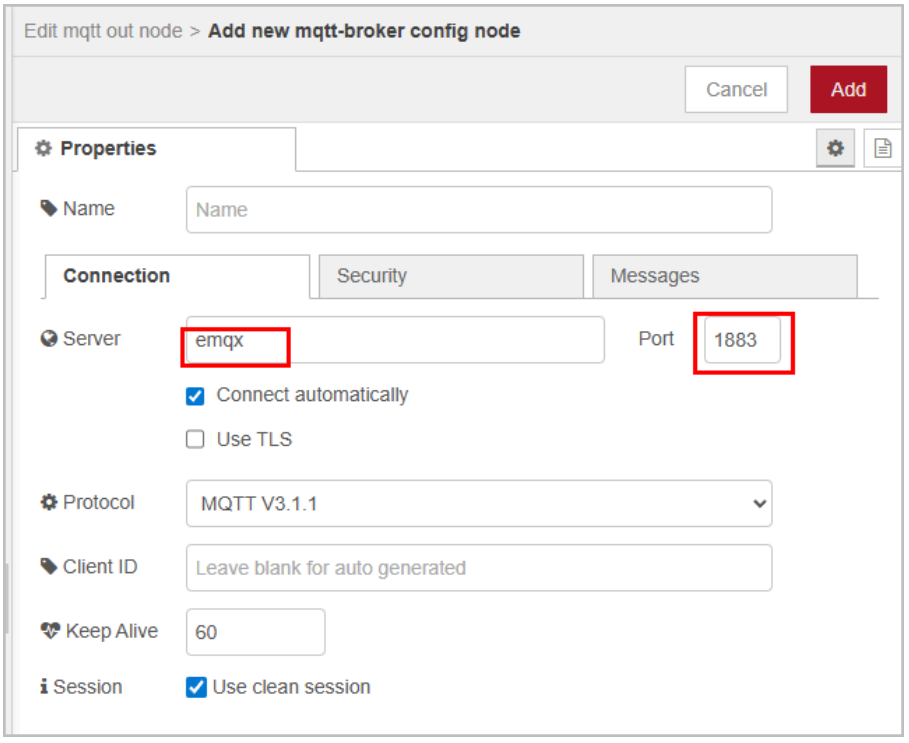
1. Log in to Tier0, go to **UNS** > **Source Flow**.



Index	Item	Description
1	Search	Search for existing data flows.
	New Source Flow	Add a data flow.
2	Flow List	Displays flow information. Click the flow name to start editing the data flow.
3	Display	Switch flow display between blocks and list.
4	Flow Operations	Copy / Edit / Delete the data flow.

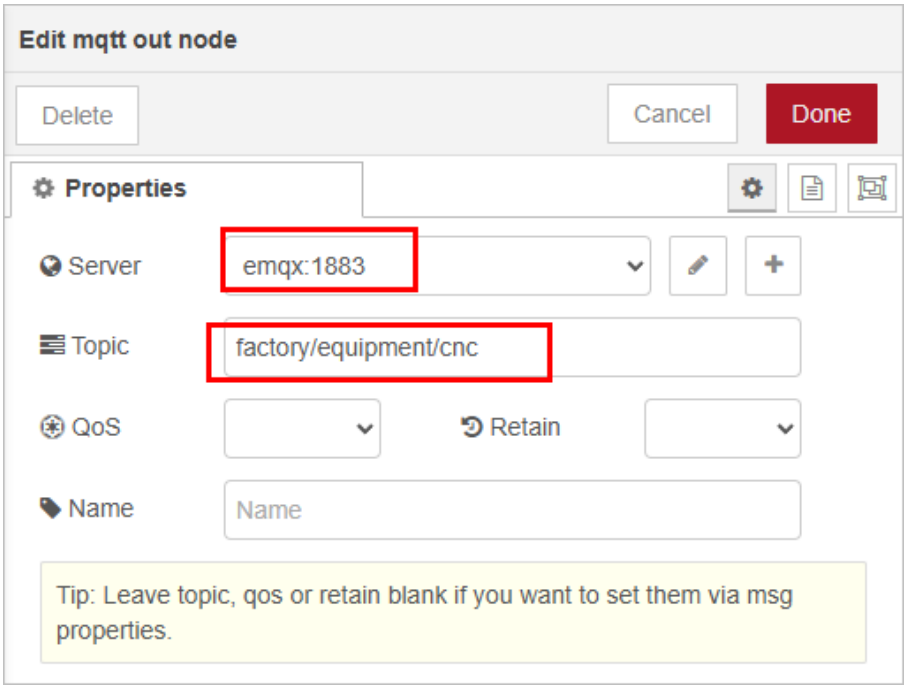
2. Click **New Source Flow**, enter a name and click **Save**.
3. Click the flow you just added, and drag in nodes according to your data source type (e.g. `modbus`).
4. Drag in an `mqtt out` node, and finish its configurations.

    1. Add the internal broker via `emqx:1883`.



    2. Set its subscription topic to the data model in **Namespace**.

For example, the data model structure is `factory/equipment/cnc`, the topic will be `factory/equipment/cnc`.

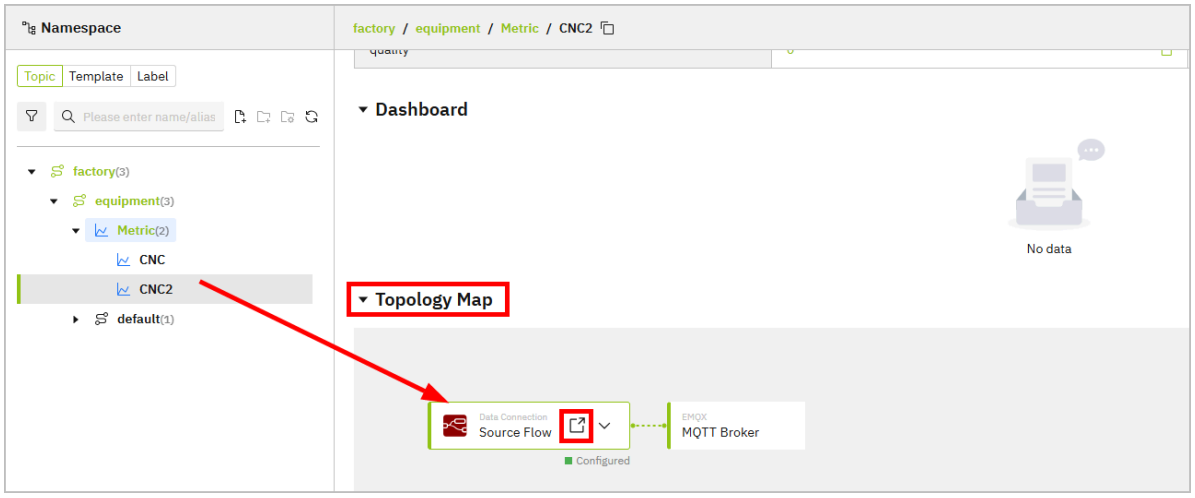


5. Click **Deploy** at the upper-right corner, trigger the flow and the data will be sent to corresponding model under **Namespace**.

Configuring Data Source through Generated Flow

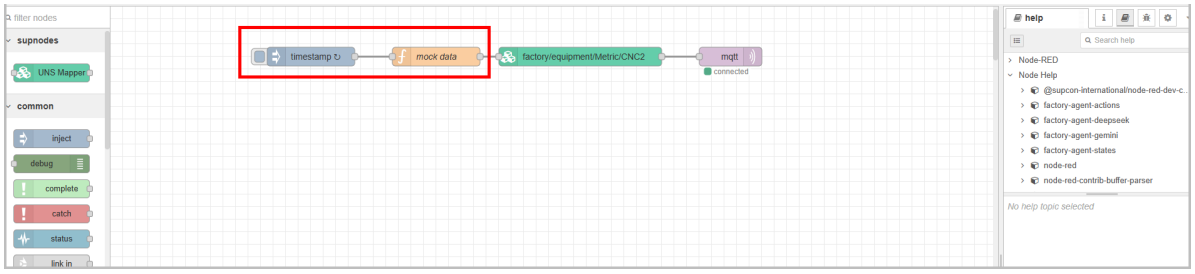
> ⓘ **note**
>
> Data flow will only be automatically generated when you select **Mock Data** while creating topics.

1. Log in to Tier0, landed on the **Namespace** page, and under the **Topic** tab, select a file.
2. Scroll down to **Topology Map**, click the icon on **Source Flow** to redirect to the generated data flow under **Source Flow**.

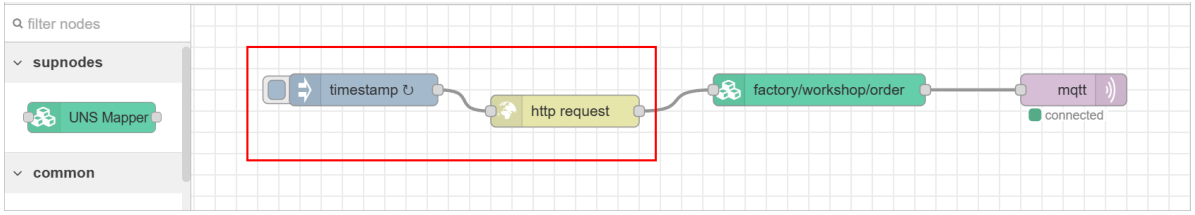


3. Change the data source of the generated flow.

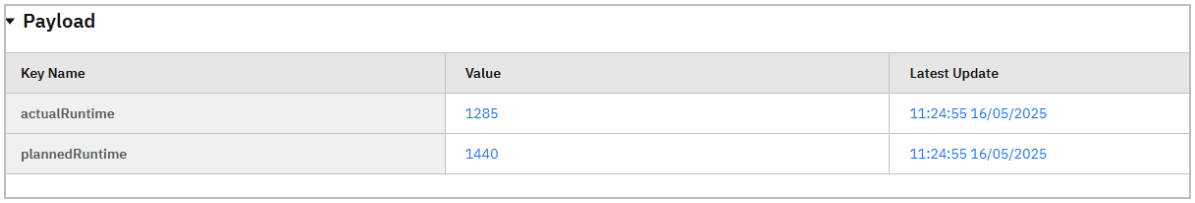
> ⓘ **info**
>
> Make sure the returned data of the source node is **JSON object**. Currently, Tier0 can only parse this data type.



4. Delete the `mock data` function node and connect the source node to the rest of the flow.



5. Click **Deploy** at the upper-right corner, and then trigger the flow.
6. Go back to **Namespace**, check whether the data is received.

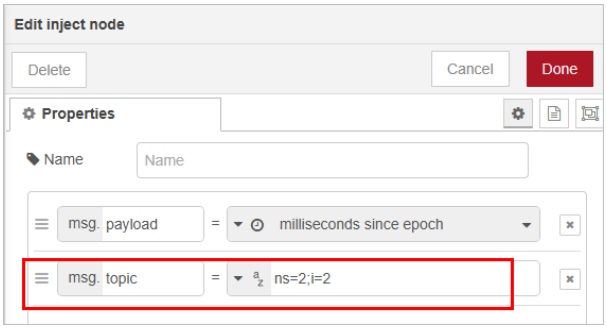


Common Data Sources

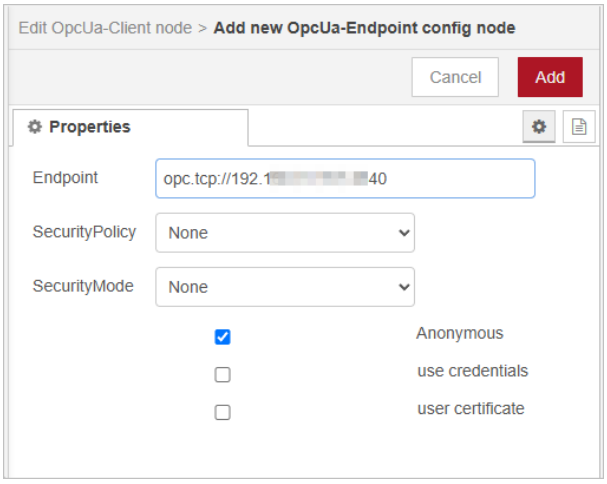
Connecting OPC UA

Read Data from OPC UA Server

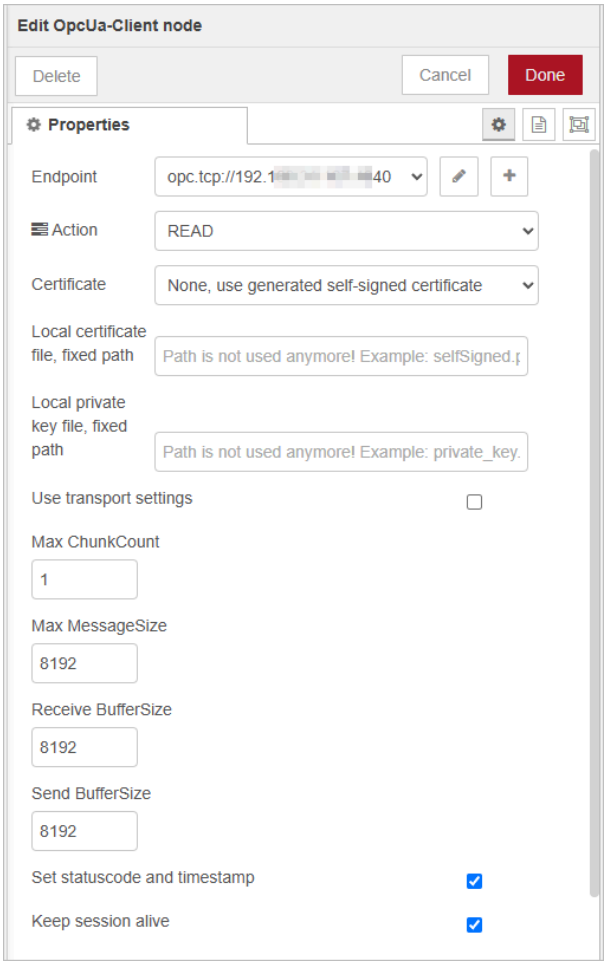
1. In **Source Flow**, drag an **inject** node.
2. Double-click the node, and then enter the **NodeId** from your OPC UA server.



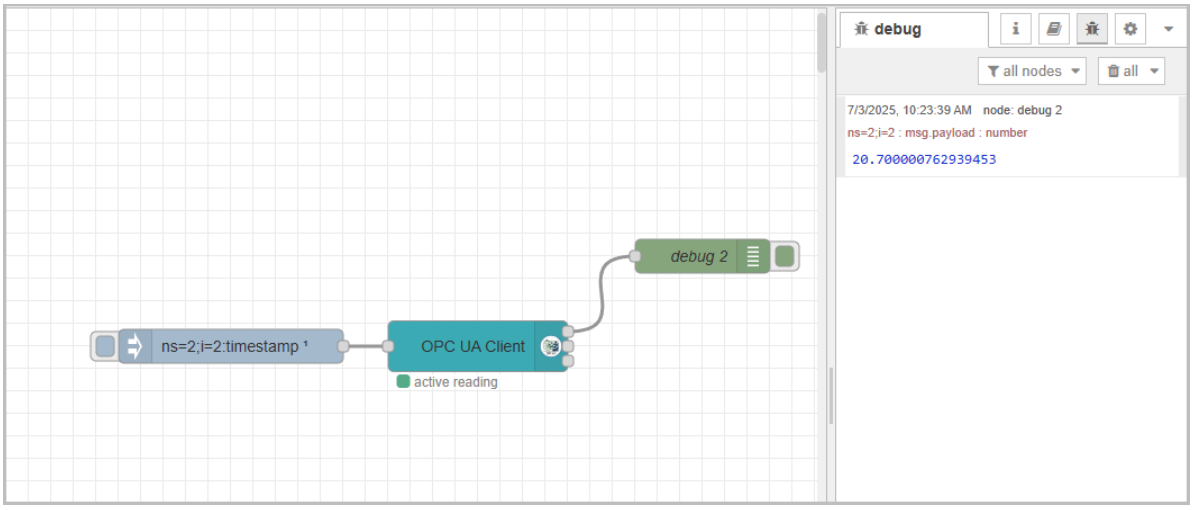
3. Drag an **OpcUa-Client** node, and then double-click it.
4. Click ➕ next to **Endpoint** to enter the information of your OPC UA server.



5. Click **Add**, and set **Action** to **READ**.
6. Leave everything else as default, and then click **Done**.



7. Connect the **inject** node to the **OpcUa-Client** node, and then connect a debug node at the end.
8. Click **Deploy** at the upper-right corner, and then click **inject** node to trigger the flow.



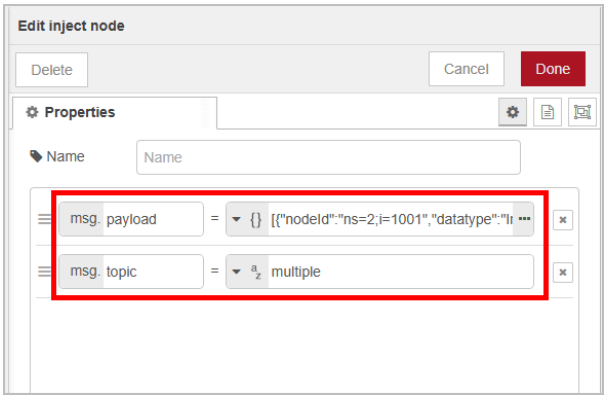
Subscribing Data from OPC UA Server

1. In **Source Flow**, drag an **inject** node.
2. Double-click the node, and then enter the **NodeId** from your OPC UA server.

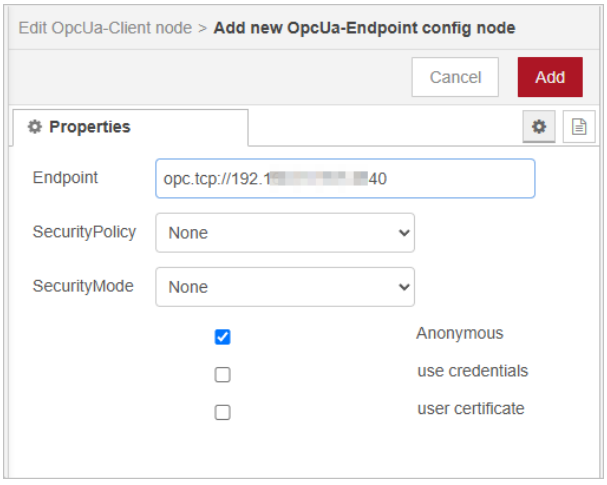
- msg.payload:

```
[
    {
        "nodeId": "ns=2;i=1001",
        "datatype": "Int32"
    },
    {
        "nodeId": "ns=2;i=1002",
        "datatype": "Int32"
    }
]
```

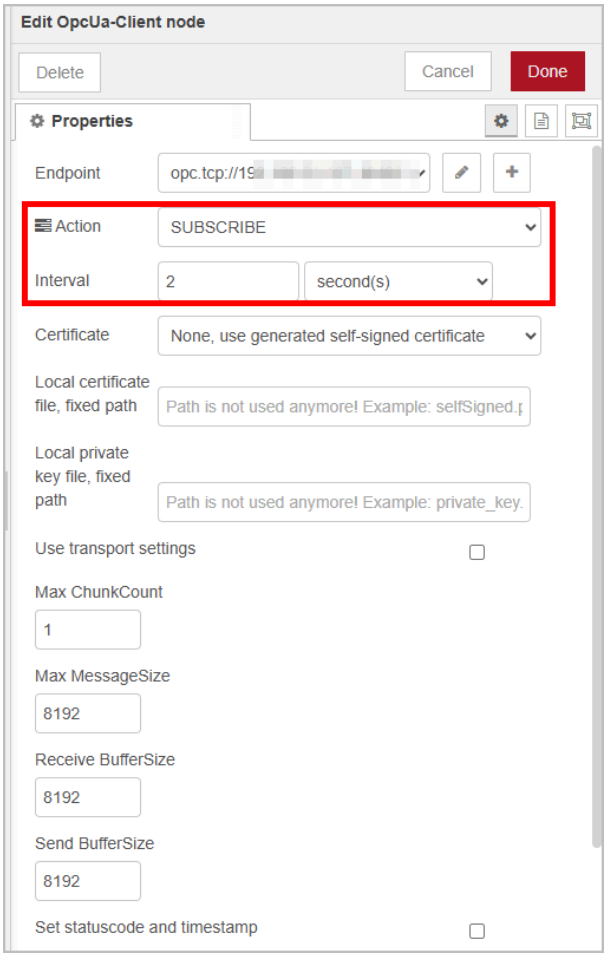
- msg.topic: multiple



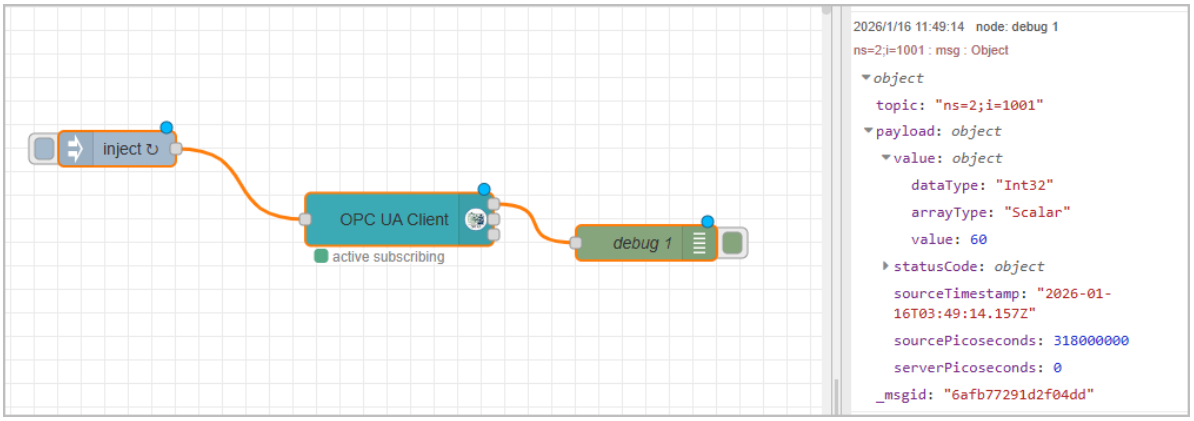
3. Drag an **OpcUa-Client** node, and then double-click it.
4. Click ➕ next to **Endpoint** to enter the information of your OPC UA server.



5. Click **Add**, and set **Action** to **SUBSCRIBE**.
6. Set the subscription interval, and leave everything else as default, and then click **Done**.

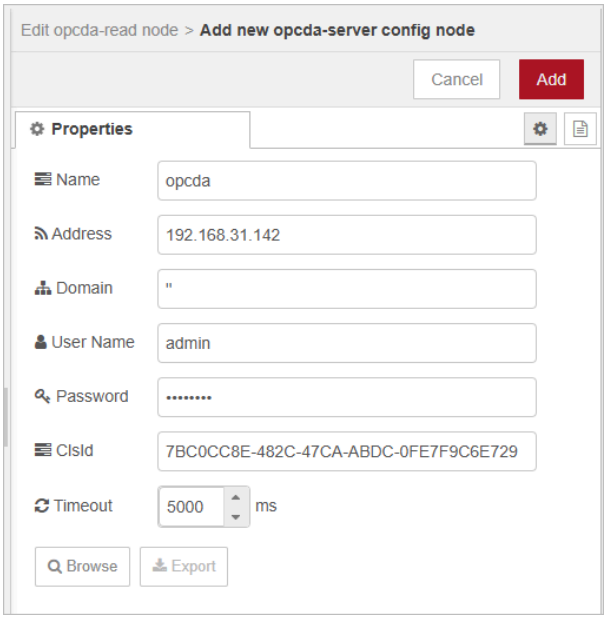


7. Connect the **inject** node to the **OpcUa-Client** node, and then connect a debug node at the end.
8. Click **Deploy** at the upper-right corner, and then click **inject** node to trigger the flow.



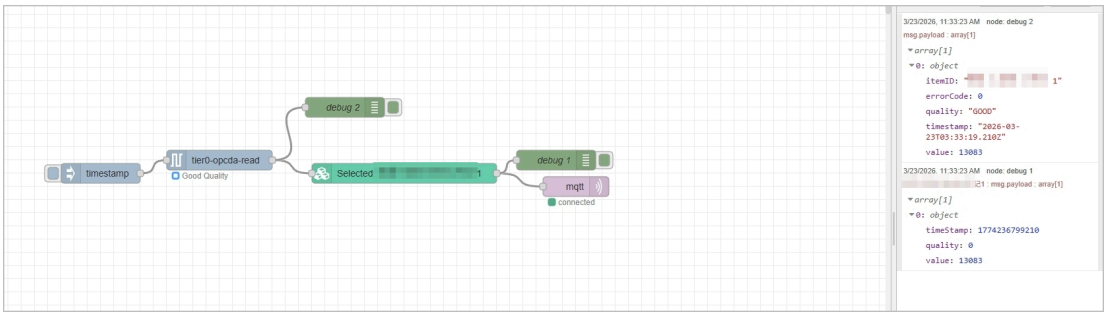
Connecting OPC DA

1. In **Source Flow**, drag an **inject** node.
2. Drag an **opcda-read** node, and then double-click it.
3. Click ✚ next to **Server** to enter the information of your OPC DA server.



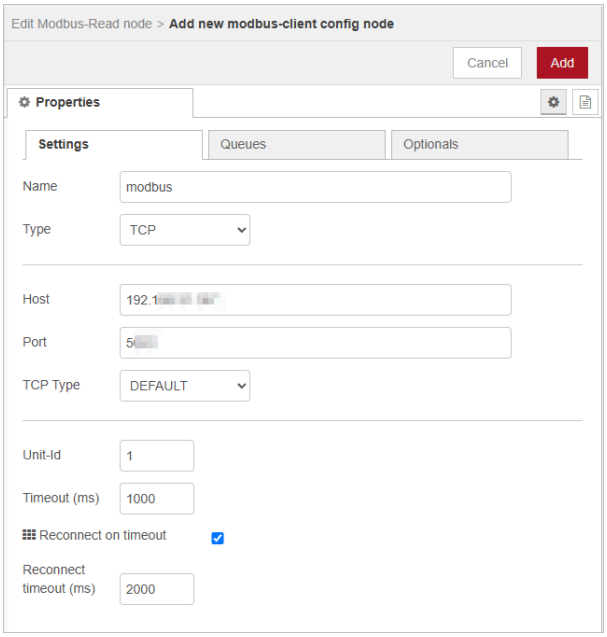
Parameter	Description
Address	OPC DA server IP address.
Domain	The domain name of your OPC DA server.
User Name/Password	The username and password you use to log in to the DA server.
ClsId	The application ID of the DA connetion service you use. For example, KepServer.

5. Click **Browse** to see the connected channels.
6. Click **Add**, and then under **Items**, add the items needed.
7. Connect the **inject** node to the **opcda-read** node, and then connect a debug node at the end.
8. Click **Deploy** at the upper-right corner, and then click **inject** node to trigger the flow.

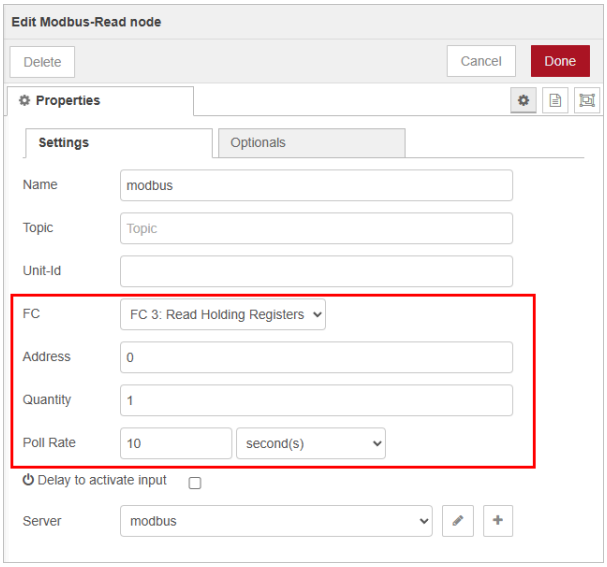


Connecting Modbus

1. In **Source Flow**, drag a **Modbus Read** node.
2. Double-click the node, and then click ✚ next to **Server** to enter the information of your modbus server.

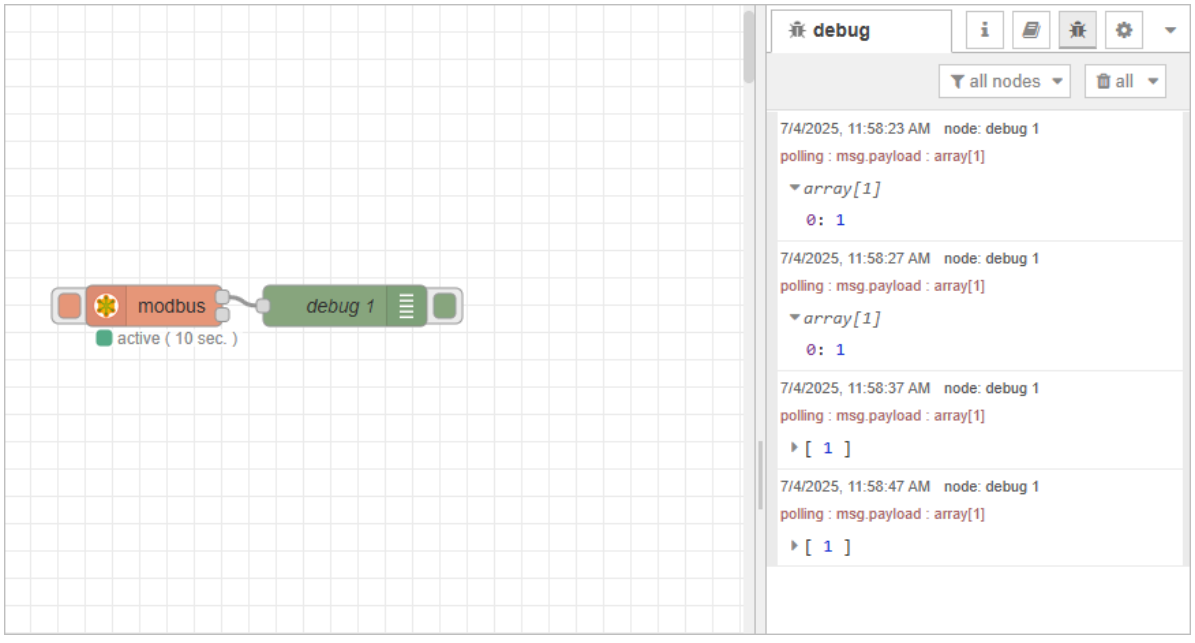


3. CLick **Add**, and then enter the information of the data to be connected.



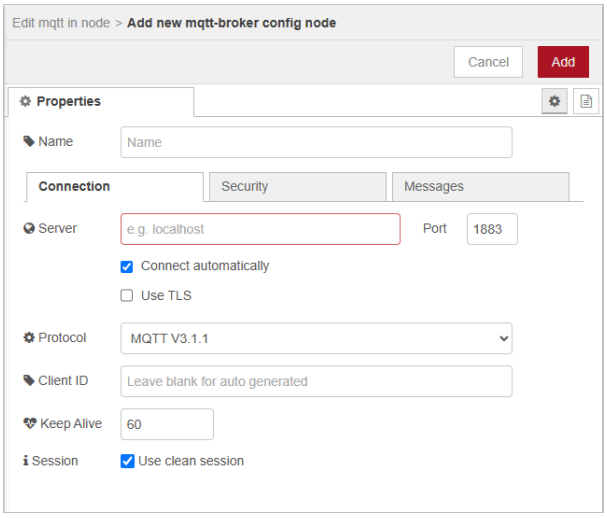
Parameter	Description
Name	The display name of the node (for visual identification only).
Topic	Optional topic string passed to output messages.
Unit-Id	The Modbus slave ID of the target device (typically 1–247).
FC	Function code that specifies the Modbus action, e.g., FC 3 = Read Holding Registers.
Address	The starting register address to read from (usually zero-based).
Quantity	The number of consecutive registers to read.
Poll Rate	How often the node polls the Modbus device (e.g., every 10 seconds).
Delay to activate input	Optional delay before activating the input signal for polling.
Server	Reference to a configured Modbus server (IP, port, protocol, etc.).

4. Click **Done**.
5. Connect a debug node at the end.
6. Click **Deploy** at the upper-right corner, and then click **Modbus Read** node to trigger the flow.



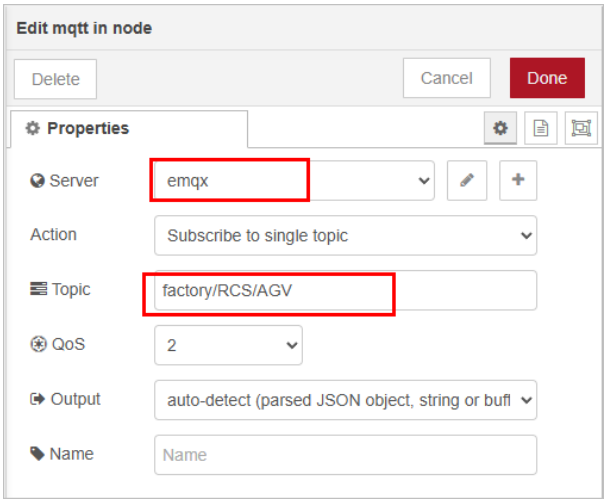
Connecting MQTT

1. In **Source Flow**, drag an **mqtt in** node.
2. Double-click the node, and then click + next to **Server**.

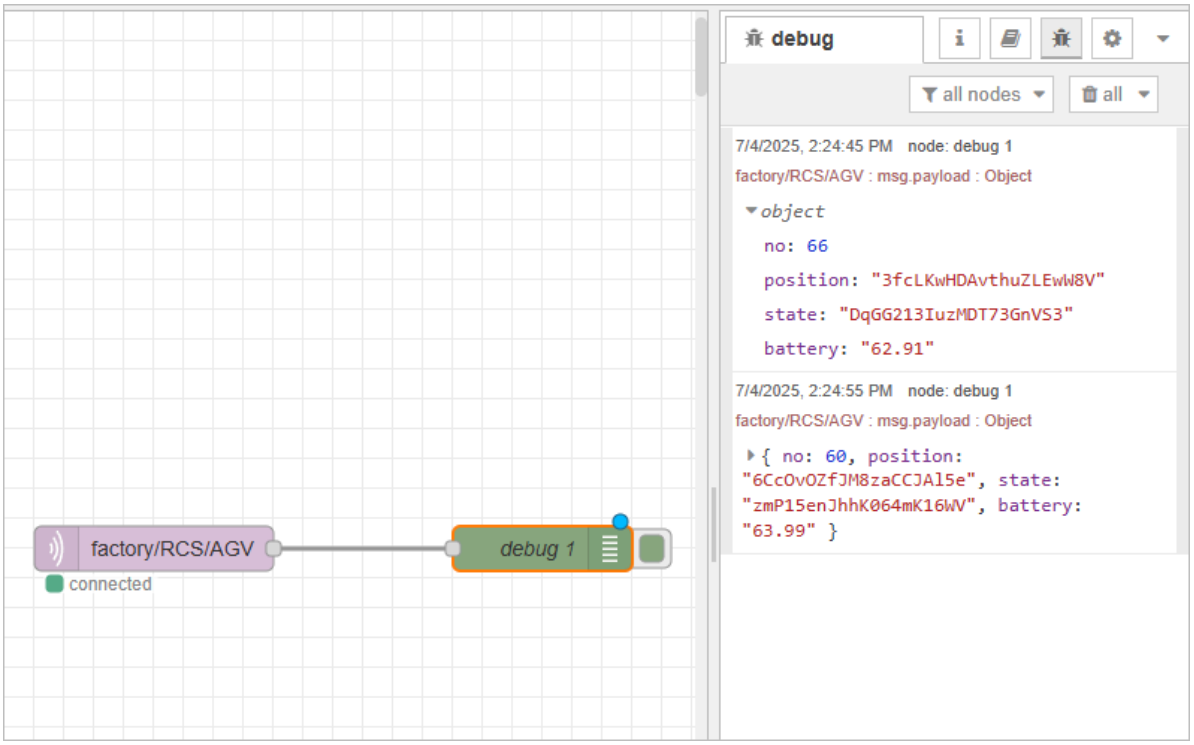


Parameter	Description
Name	Optional name for the MQTT broker configuration (for internal reference).
Server	The hostname or IP address of the MQTT broker (e.g., `localhost` or `broker.hivemq.com`).
Port	The port used to connect to the MQTT broker (default is `1883` for non-TLS, `8883` for TLS).
Connect automatically	If enabled, Node-RED will try to connect to the broker automatically when the flow starts.
Use TLS	Enable this option if the broker requires a secure TLS/SSL connection.
Protocol	The MQTT protocol version to use for connection (e.g., MQTT v3.1.1 or v5.0).
Client ID	Optional client ID. If blank, a unique one will be auto-generated. Must be unique per client.
Keep Alive	The maximum period (in seconds) between communications with the broker. Helps detect dropped connections.
Use clean session	If enabled, no persistent session is maintained with the broker; otherwise, session state (like subscriptions) is retained.
Username (Security tab)	Username for authenticating with the MQTT broker, if required.
Password (Security tab)	Password corresponding to the username for broker authentication. Stored securely.

3. Click **Add**, and then enter the topic to be subscribed.
4. Click **Done**.



5. Connect a debug node at the end.
6. Click **Deploy** at the upper-right corner, and then once the topic receives data, the flow will be triggered.



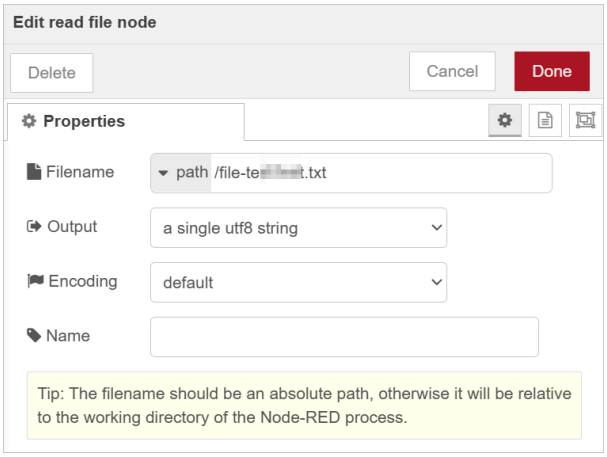
Connecting File

> ⓘ **info**
>
> Before processing files in **Source Flow**, you need to put the file in to the corresponding docker.
>
> ```
> scp C:\Users\you\Desktop\myfile.txt username@server IP:/home/username/
>  //copy the file from local to server
> docker cp /root/config.json <docker name or ID>:/app/config.json
>  //copy the file from server to docker
> ```

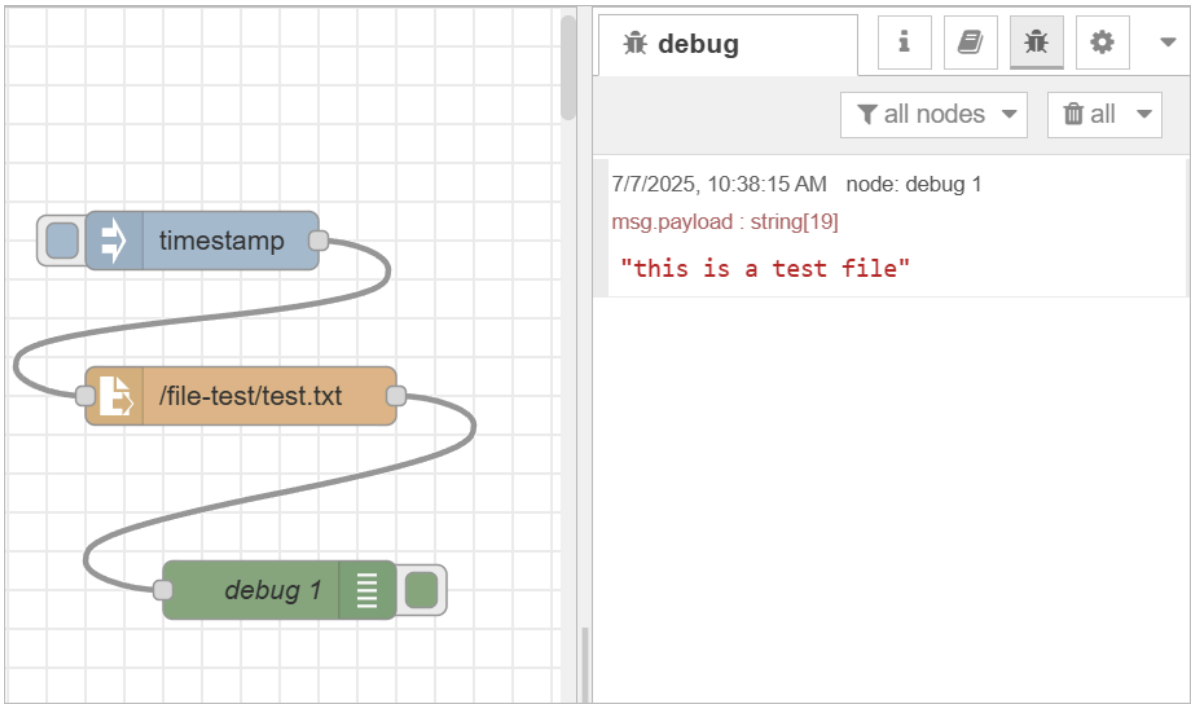
> ✓ **general docker commands**
>
> | Command | Description |
> | --- | --- |
> | `docker ps` | Lists all running Docker containers. |
> | `docker ps \| grep [container]` | Filters running containers based on the specified docker information. |
> | `docker exec -it [container] /bin/sh` | Starts a shell session inside a running container using `/bin/sh`. |
> | `mkdir [directory]` | Creates a new directory with the specified name inside a container. |

Connecting a Text File

1. In **Source Flow**, drag an **inject** node.
2. Drag a **read file** node, and then double-click it.
3. Enter the file path and output configuration.

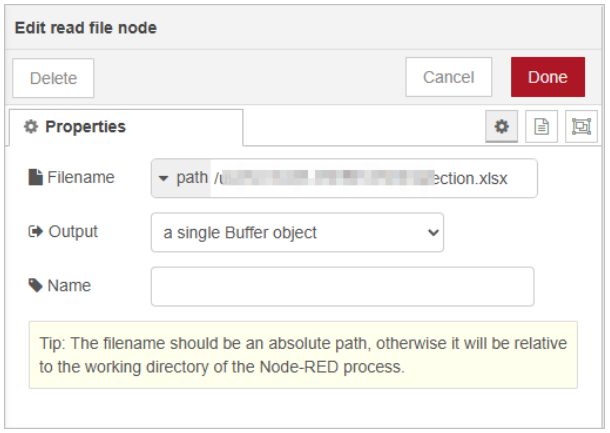


4. Click **Done**, and then connect a debug node at the end.
5. Click **Deploy** at the upper-right corner, and then click **inject** node to trigger the flow.

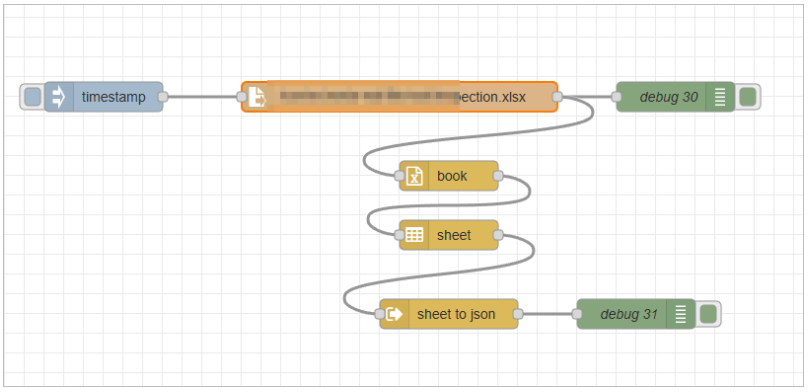


Connecting an Excel File

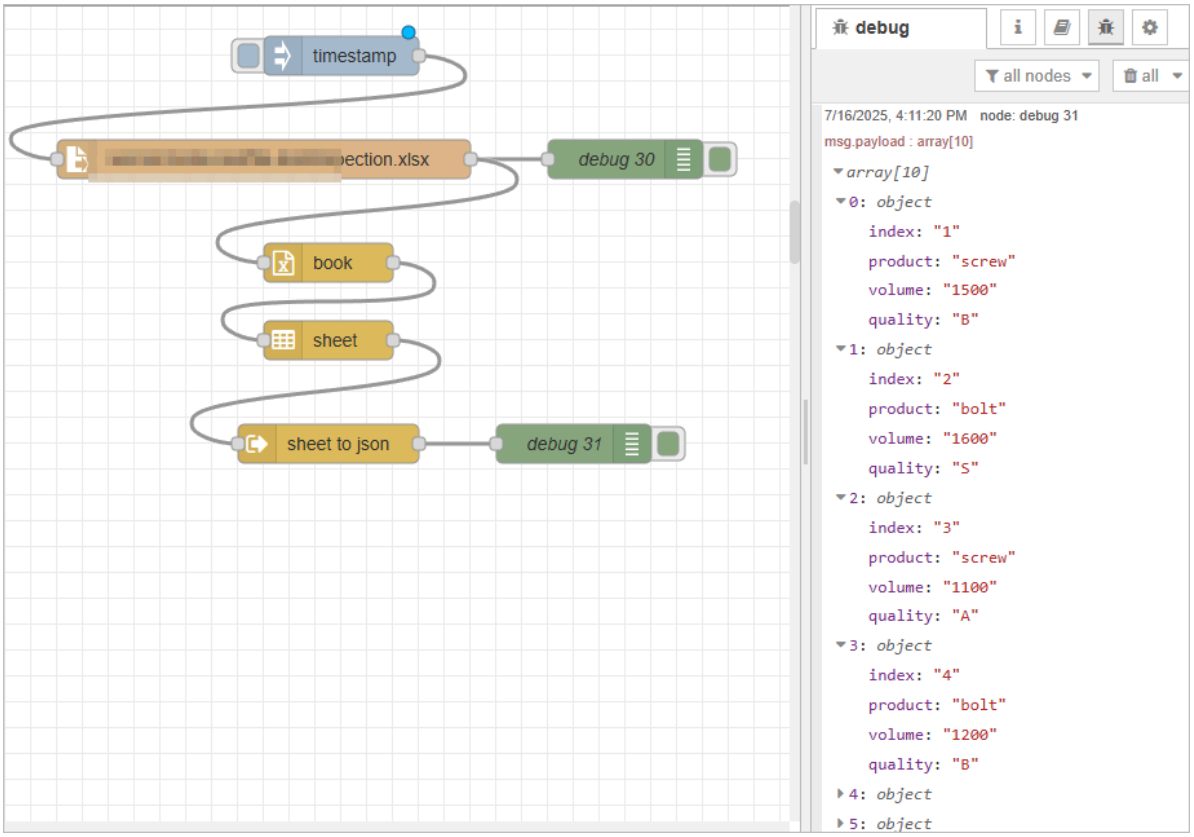
1. In **Source Flow**, drag an **inject** node.
2. Drag a **read file** node, and then double-click it.
3. Enter the file path and output configuration.



4. Click **Done**, and then connect a **book**, a **sheet** and a **sheet to json** node by order to convert table data to json.



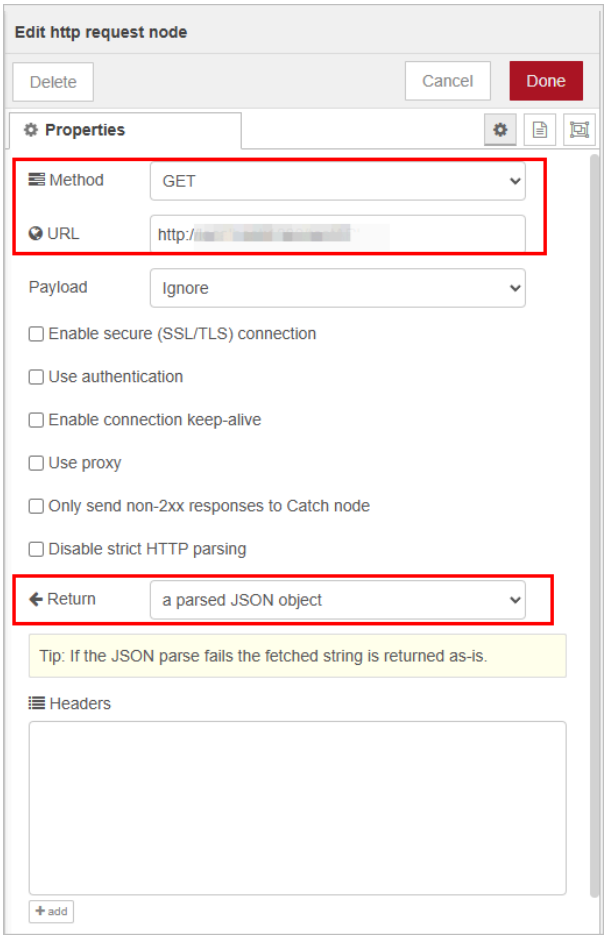
5. Click **Deploy** at the upper-right corner, and then click **inject** node to trigger the flow.



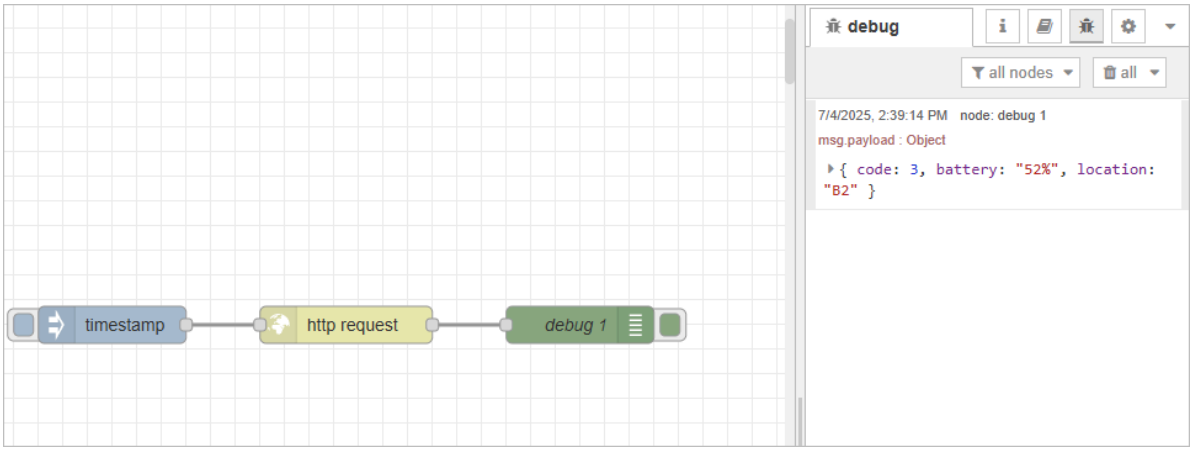
Connecting RestAPI

1. In **Source Flow**, drag an **inject** node.
2. Drag an **http request** node, and then double-click it.
3. Enter the API information.

> ✓ **tip**
>
> Set the **Return** to **a parsed JSON obbject** for Tier0 namespace to parse.

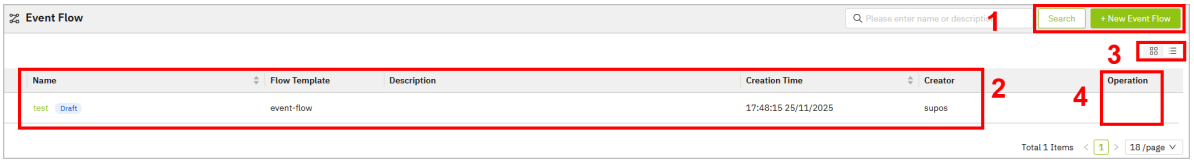


5. Click **Done**, and then connect a debug node at the end.
6. Click **Deploy** at the upper-right corner, and then click **inject** node to trigger the flow.



Process Data

1. Log in to Tier0, and then select **UNS** > **Event Flow**.



Index	Item	Description
1	Search	Search for existing data flows.
	New Event Flow	Add a data flow.
2	Flow List	Displays flows information. Click the flow name to start editing the data flow.
3	Display	Switch flow display between blocks and list.
4	Flow Operations	Copy / Edit / Delete the data flow.

2. Click **New Event Flow**, enter a name and click **Save**.
3. Click the flow you just added, and build data flows to analyze your data.

Common Data Processing Methods

Combining Multiple Sources

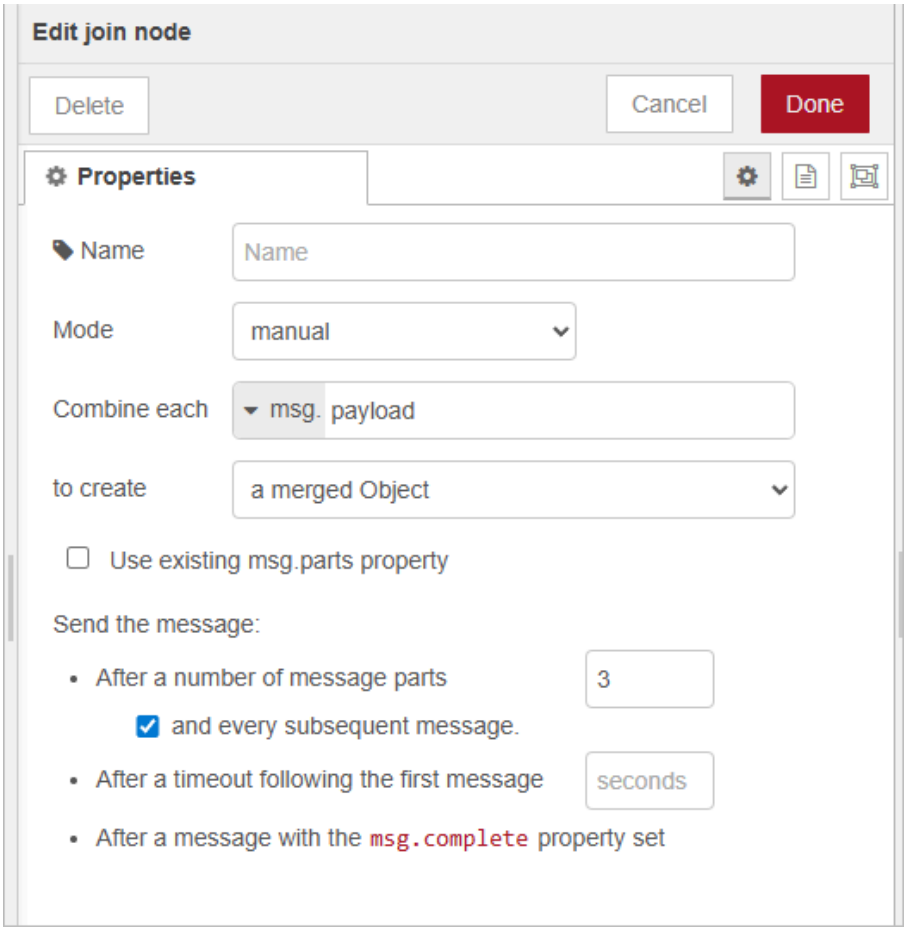
Example

There are 3 data sources connected to Tier0, and we need to combine them together for the next step.

```
{
  "workshop": {
    "furnace1": {
      "temp": 85
    },
    "furnace2": {
      "humidity": 64
    },
    "furnace3": {
      "pressure": 80
    }
  }
}
```


What Node to be Used?

In this example, we use **join** node to combine previous messages.

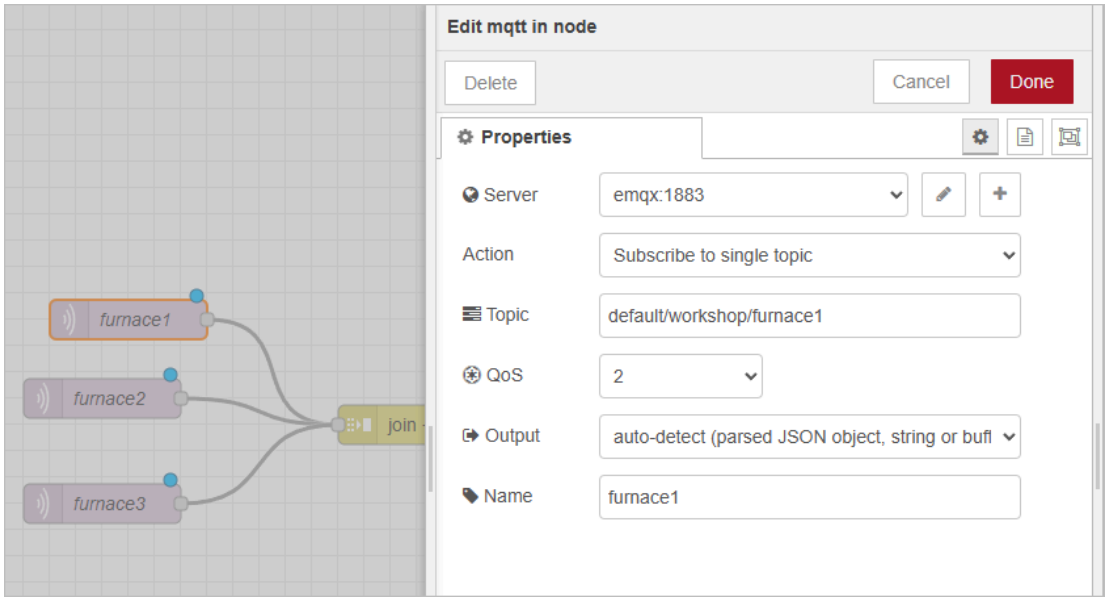


Parameter	Description
Mode	Join mode. • **manual**: You configure how to merge messages yourself. • **auto**: Automatically reassemble sequences split by a **split** node, based on `msg.parts`. • **reduce sequence** – Runs a JavaScript reduce function across the sequence to produce a single result.
Combine each	Message part you want to combine. Commonly `msg.payload`, but can be others (e.g. `msg.data.value`).
to create	Result type: • **a merged Object**: Merge all objects into a single object (later keys overwrite same-name keys). • **a key/value Object**: Use a **key property** (default `msg.topic`) as the key, and the **Combine each** property as the value. • **an Array**: Append each value to an array in order. • **a String / a Buffer**: Concatenate text or binary data (String can use a separator; Buffer is joined as binary).
Use existing `msg.parts` property	When selected, the node uses the `msg.parts` from previous nodes (usually created by a **split** node) to determine grouping and when to send the joined output. Fields like `msg.parts.count` and `msg.parts.index` are used.
Send the message → After a number of message parts	Outputs the joined result after receiving the set number of fields.
Send the message → and every subsequent message	When selected, after the initial output, the node outputs an updated result upon new messages arrival.
Send the message → After a timeout following the first message	Optional. Starts a timer when the first message arrives; once the timeout (seconds) is reached, the node outputs whatever has been collected.

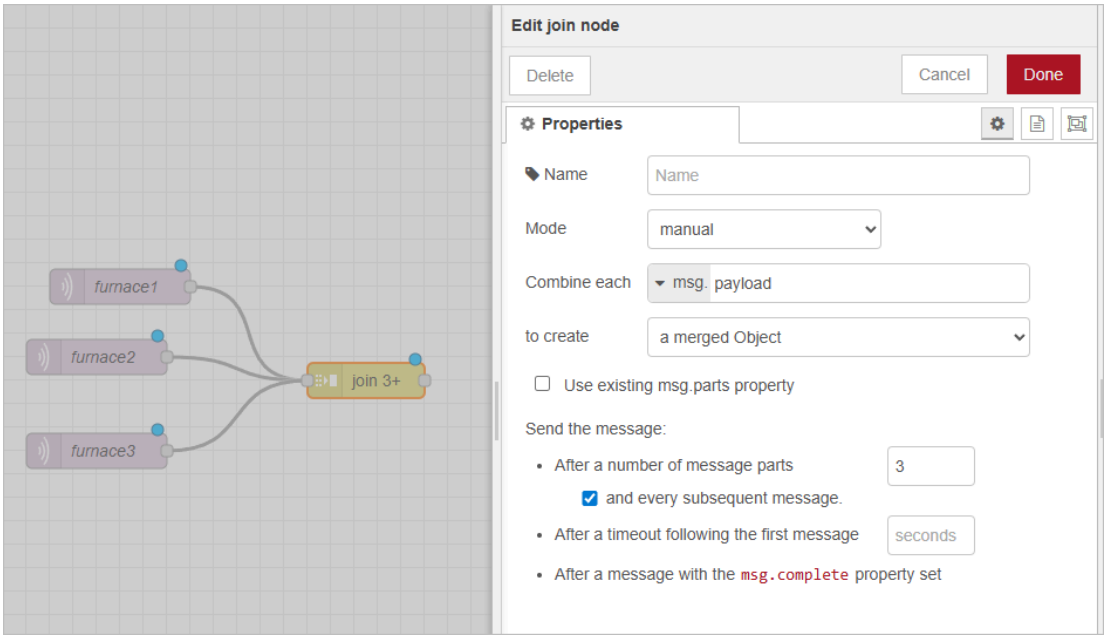
Parameter	Description
Send the message → After a message with the `msg.complete` property set	Outputs the joined result immediately when a message arrives with `msg.complete = true`.
(Advanced) String/Buffer separator	For **String** output, an optional separator can be set. For **Buffer** output, data is concatenated directly with no separator.

How to Combine Sources?

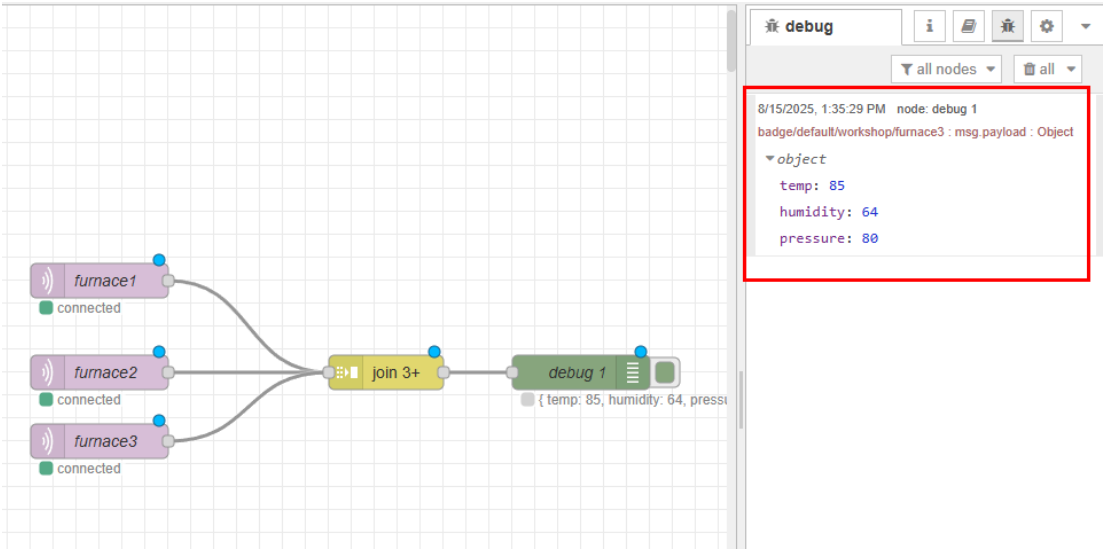
1. In **Event Flow**, drag in 3 **mqtt in** nodes, and subscribe to corresponding topics relatively.



2. Connect a **join** node to all 3 nodes, and merge them into an object.



3. Connect a **debug** node, and trigger the flow to check the results.



Filtering Data

Example

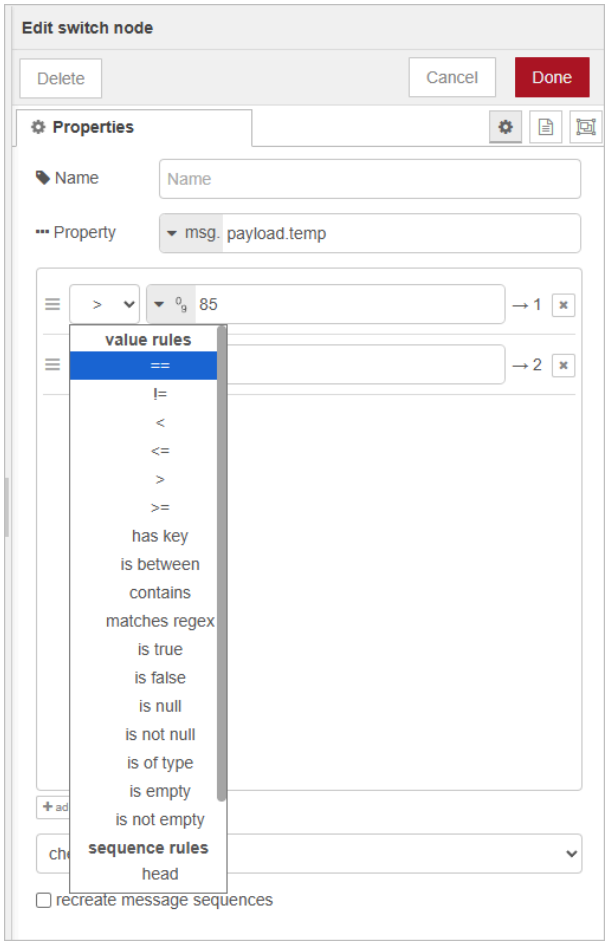
The **furnace** data source contains 3 fields, and we want to route messages based on the temperature threshold:

- If **temp > 85** → go to **Output 1**
- If **temp ≤ 85** → go to **Output 2**

```
{
  "furnace": {
    "temp": 86,
    "humidity": 64,
    "pressure": 80
  }
}
```

What Node to be Used?

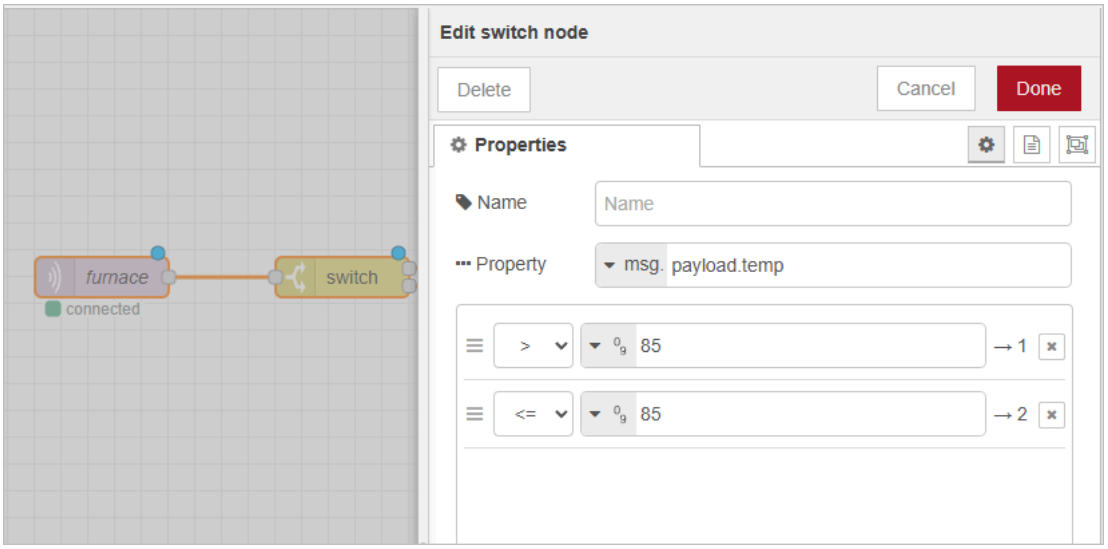
In this example, we use the **switch** node to route messages according to conditions.



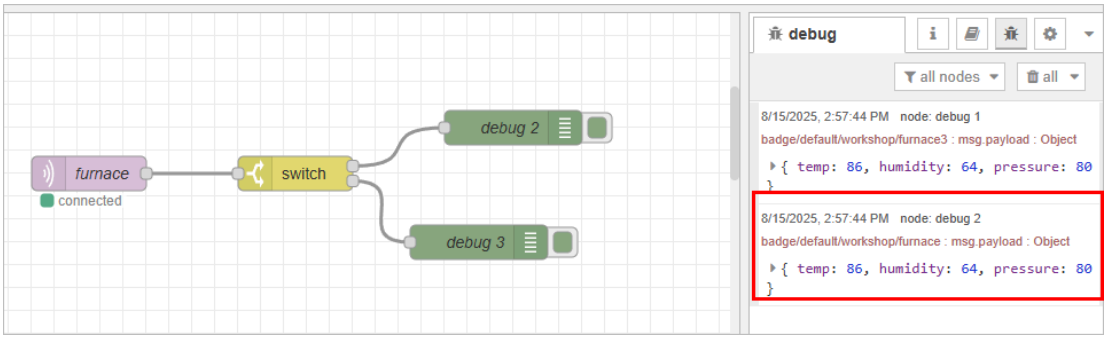
Parameter	Description
Property	The property path to evaluate. Commonly `msg.payload` or a deeper property, such as `msg.payload.temp` in this example.
Rules	A set of evaluation rules, each corresponding to one output port.
Type	The data type for comparison: number, string, boolean, JSONata, etc. ⓘ **info** Ensures numeric comparisons are not treated as string comparisons.
Value	The value to compare against can come from: a constant value (e.g. 85), `msg.` fields (e.g. `msg.threshold`), `flow.` / `global.` context, or a `JSONata` expression.
checking all rules	Evaluates all rules in order. If a message matches multiple rules, it will be sent to multiple outputs.
Stop after first match	Stops checking once the first rule matches, so the message is sent to only one output.
recreate message sequences	When enabled, preserves and rebuilds message sequence metadata (msg.parts)—usually created by a split node—so that the sequence is maintained even after branching.

How to Filter Data?

1. In **Event Flow**, add an **mqtt in** node to get the data from Tier0.
2. Drag in a **switch** node and configure its properties.



3. Connect a **debug** node to each output of the **switch** node.
4. Trigger the flow and check the results.



Changing Data

Example

We receive a message and need to rename fields, calculate new values, and remove unused fields for cleaner downstream processing.

Input message

```
{
  "furnace": {
    "temp_c": 30.5,
    "humidity": 64,
    "pressure": 80
  },
  "site": "A1"
}
```

Target transformation

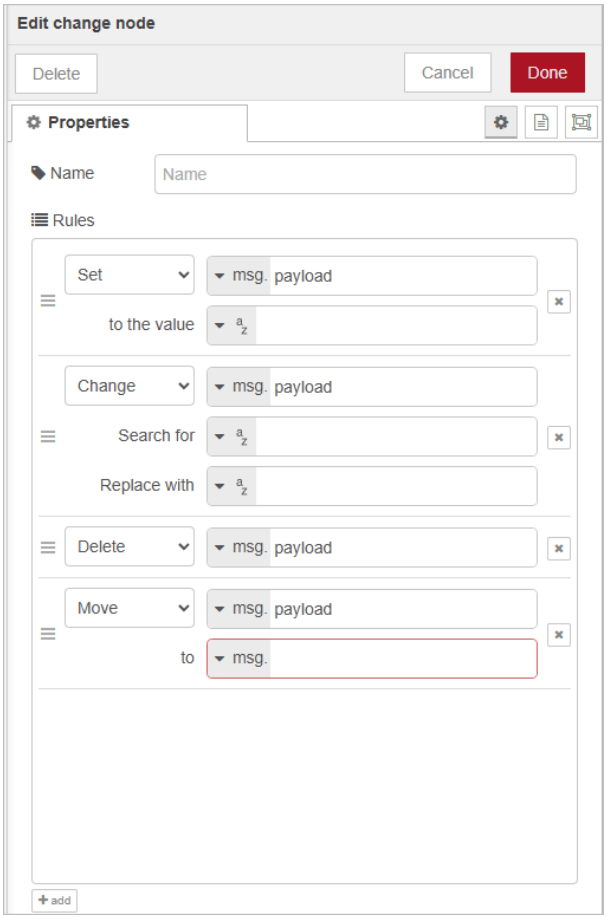
- Move furnace.temp_c to furnace.temperature.celsius
- Add furnace.temperature.fahrenheit (converted from Celsius)
- Remove pressure
- Change the site from A1 to B1

Output message (example)

```
{
    "furnace":{
        "humidity":64,
        "temperature":{
            "celcius":30.5,
            "fahrenheit":86.9
            }
        },
    "site":"B1"
}
```

What Node to be Used?

In this case, we use the **change** node to perform operations like **Set**, **Change**, **Delete** and **Move** on message properties.

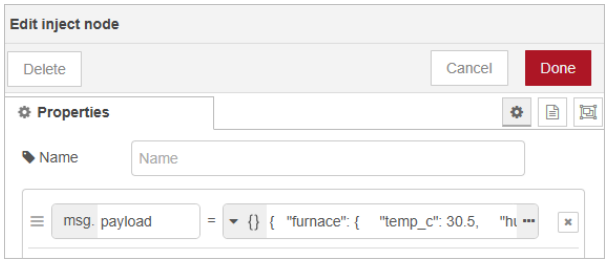


Parameter	Description
Rules	Each rule is executed in order, top to bottom. Multiple rules can be added to perform complex transformations.
Rule Type	• **Set**: Set or create a property value. • **Change**: Find and replace. • **Delete**: Remove a property. • **Move**: Move a property from one path to another.
Property/Source/Target	Defines which property to operate on, such as `msg.payload` or `msg.payload.furnace.temp_c`. Can also reference `flow.` or `global.` context variables.
Target Value	• **Set**: The value to assign. • **Change**: The value to find and replace.

How to Use the Change Node?

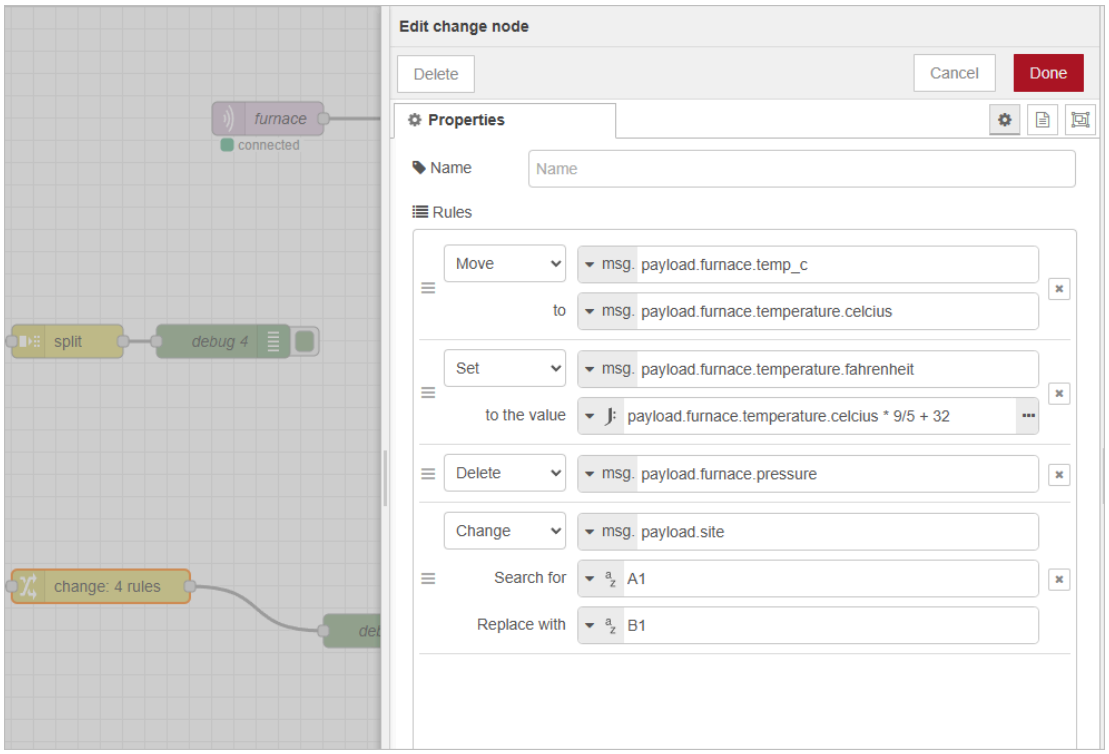
1. In **Event Flow**, drag in an **inject** node, and add the following json as the input.

```
{
  "furnace": {
    "temp_c": 30.5,
    "humidity": 64,
    "pressure": 80
  },
  "site": "A1"
}
```

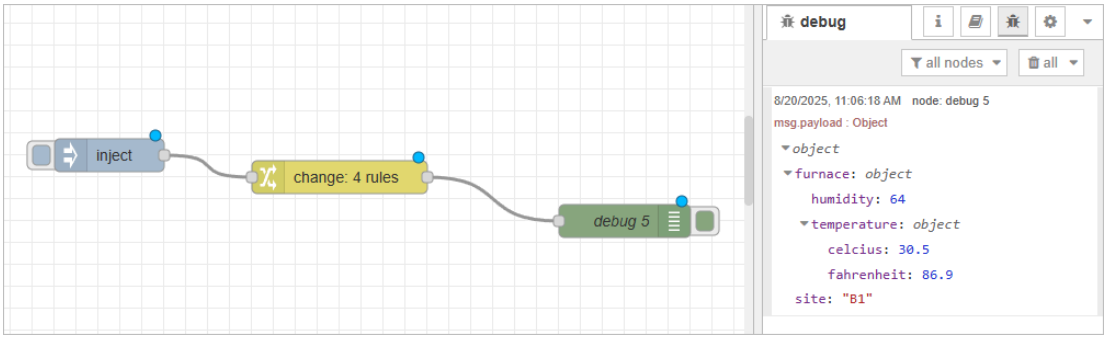


2. Connect it to a **change** node.
3. Open the **change** node configuration panel and add rules.

- Move msg.payload.furnace.temp_c → msg.payload.furnace.temperature.celcius
- Set msg.payload.furnace.temperature.fahrenheit → payload.furnace.temperature.celcius * 9/5 + 32
- Delete msg.payload.pressure
- Change value of msg.payload.site from A1 → B1



4. Connect the **change** node to a **debug** node to check the transformed output.



Spliting Data

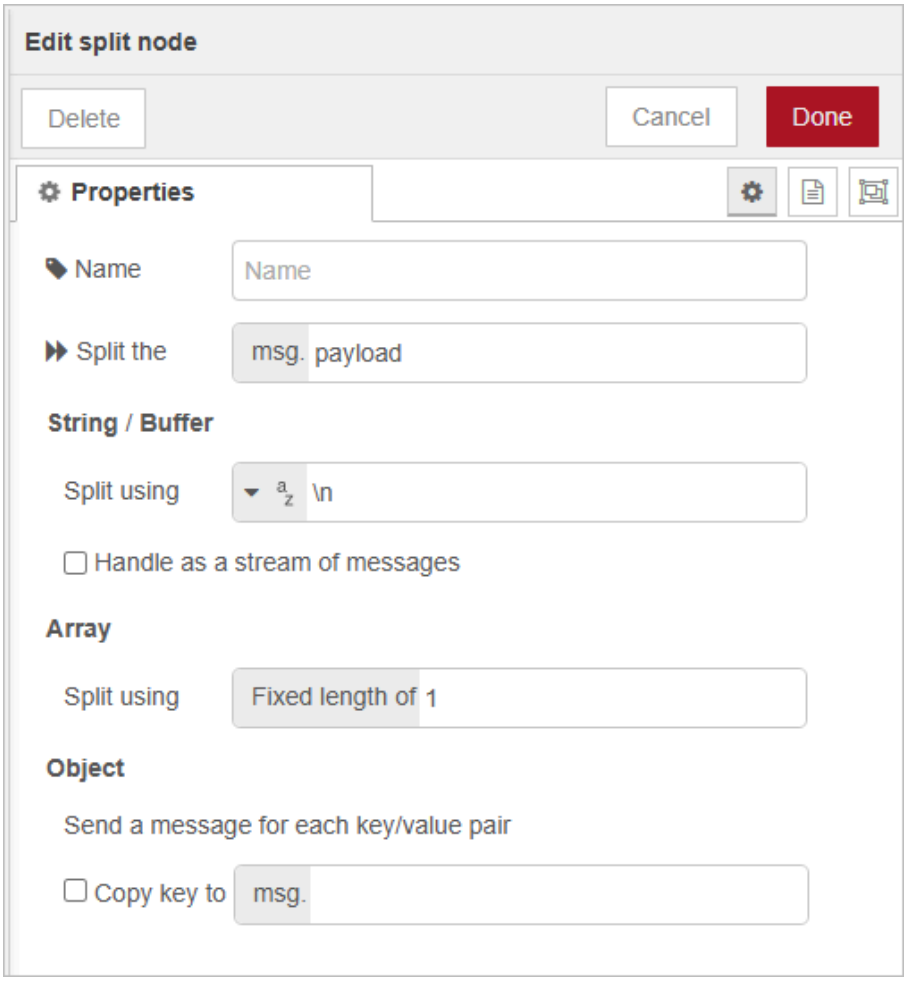
Example

We have a single message containing multiple furnace readings, and we need to split them into individual messages for further processing.

```
{
  "furnace": {
    "temp": 86,
    "humidity": 64,
    "pressure": 80
  }
}
```

What Node to be Used?

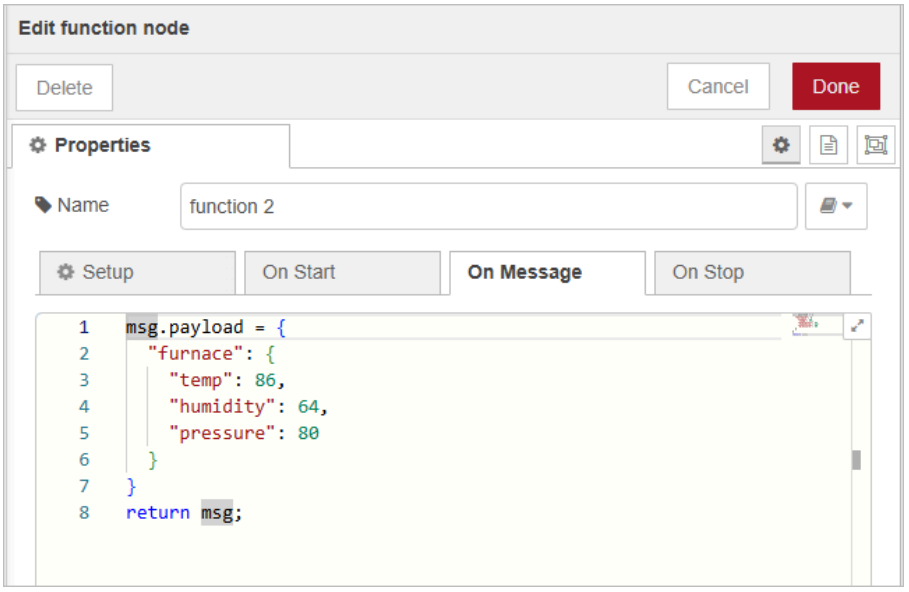
In this example, we use the **split** node to break one message into multiple smaller messages.


Edit split node
Delete
Cancel
Done
Properties
Name
Name
Split the
msg.
payload
String / Buffer
Split using
\n
Handle as a stream of messages
Array
Split using
Fixed length of
1
Object
Send a message for each key/value pair
Copy key to
msg.

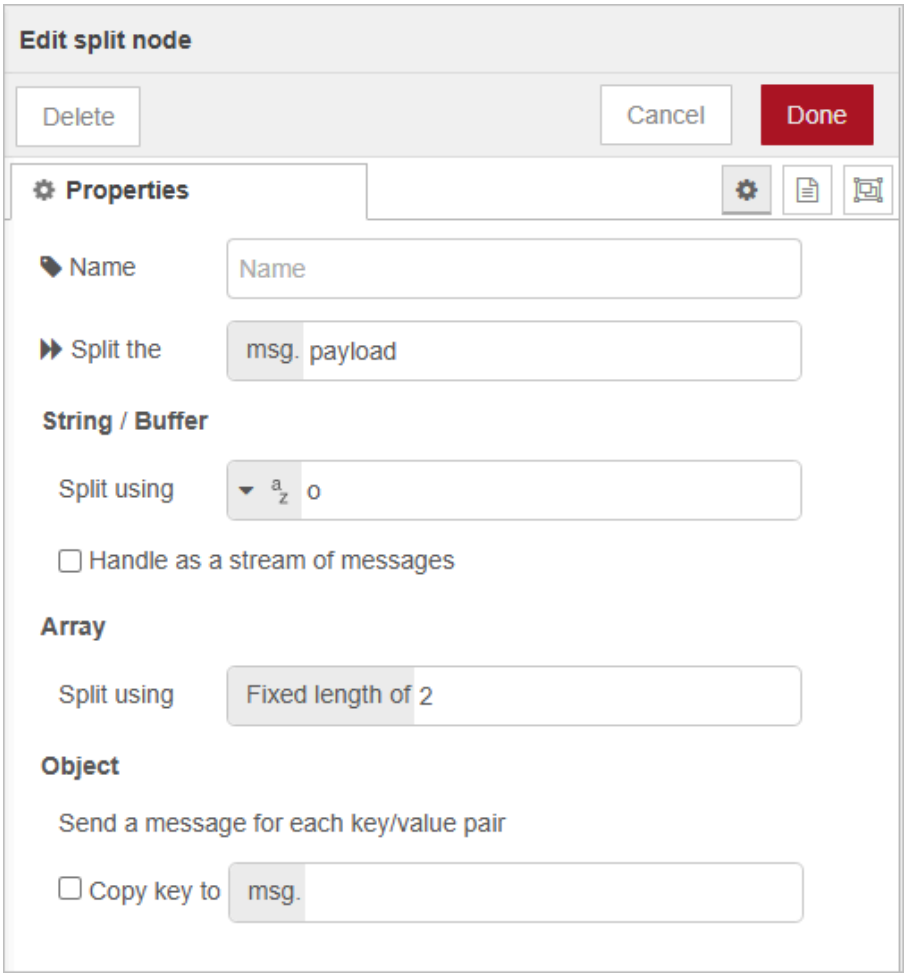
Parameter	Description
Name	Optional label for the node to help identify it in the flow.
Split the	The property of the message to split, usually `msg.payload`, but can be any property such as `msg.payload.data`.
String / Buffer - Split using	For string or buffer payloads, specifies how to split: • **string**: Use a specific string or regular expression pattern to split the message. • **buffer**: Defines chunk size by which the message is split. • **Fixed length of**: Split the string according to the set length. • Leave empty to return the tring unsplit.
Handle as a stream of messages	When checked, the node will process incoming data as a continuous stream, emitting split messages as data arrives, rather than waiting for the full payload.
Array - Split using	**Fixed length of N**: Splits the array into multiple messages with N elements each.
Object - Send a message for each key/value pair	The value of each key/value pair will be returned as payload.
Copy key to `msg.payload`	If checked, the original key will be returned as `msg.payload` of the split messages.

How to Split Messages?

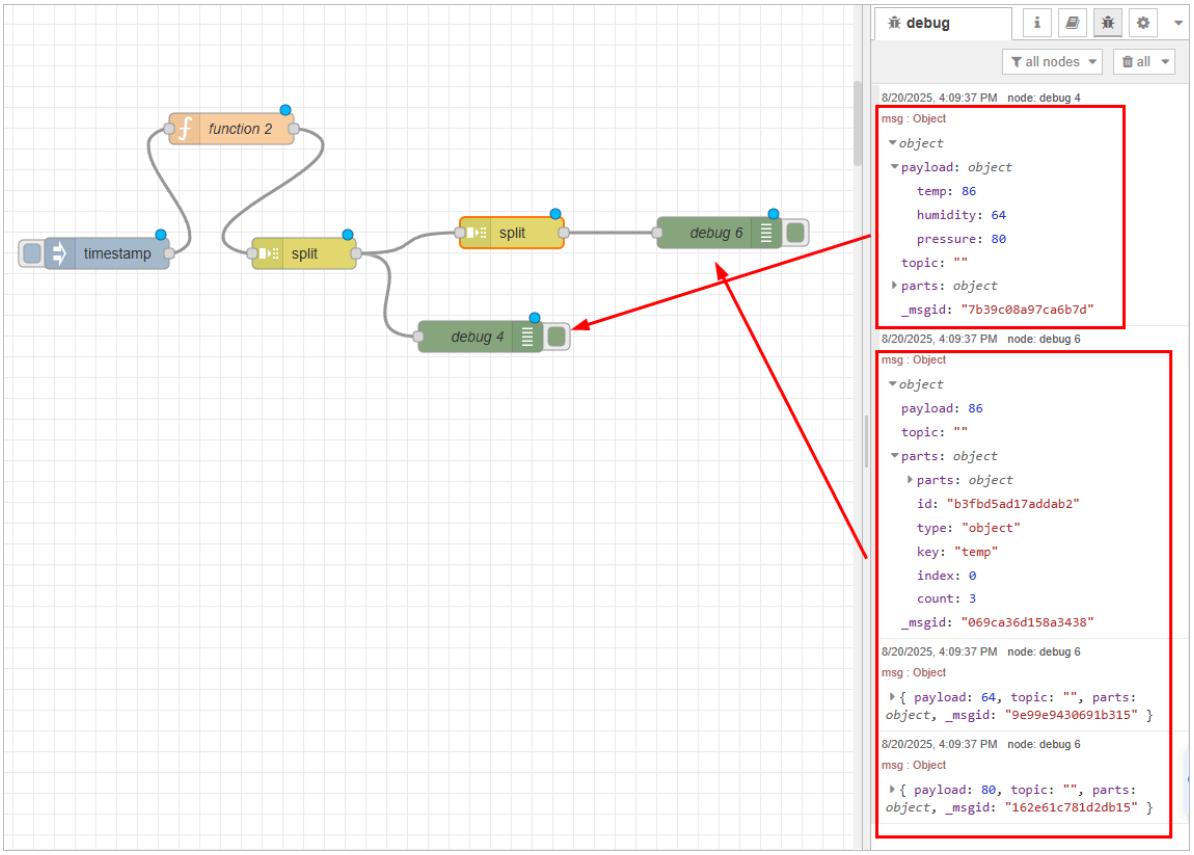
1. In **Event Flow**, drag in an **inject** node, and use a **function** node to transmit the object.



2. Connect it to 2 **split** nodes in order, and configure both properties as the same on the **Object** part.



3. Connect the **split** node to a **debug** node to inspect each individual output message.

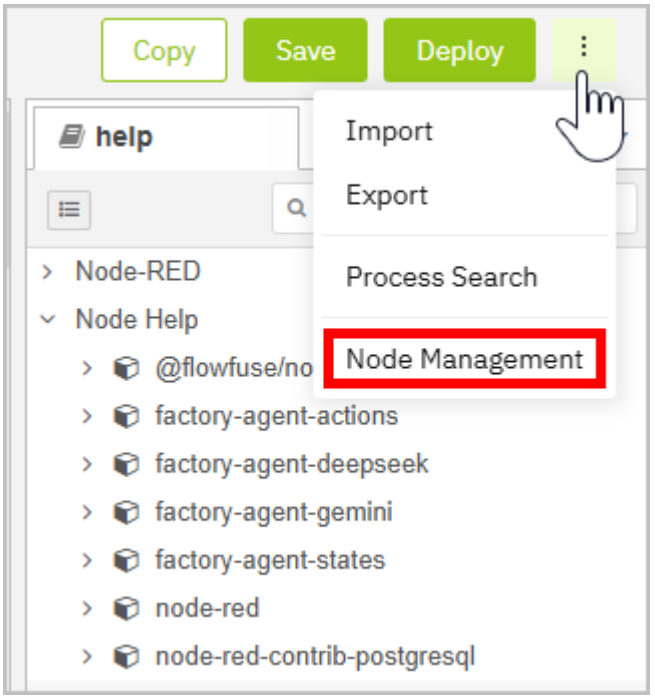


Visualize Data

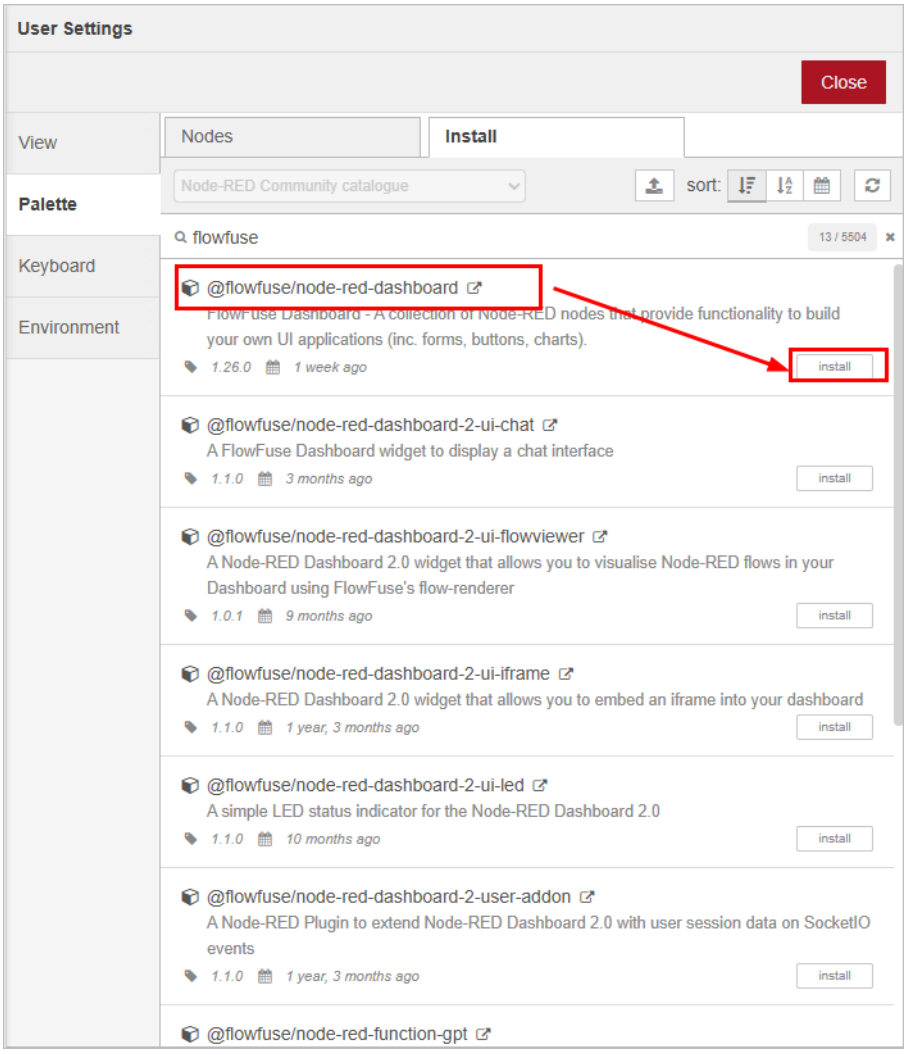
How to Build a Dashboard on Event Flow

Data collected to, processed on Tier0, can be visualized in charts, tables, texts in custom styles and other formats on Tier0 as well.

1. Log in to Tier0, and select **UNS** > **Event Flow**.
2. Click **New Event Flow** to add a flow.
3. Click the flow name to start editing.
4. Click ⋮ at the upper-right corner, and then click **Node Management**.



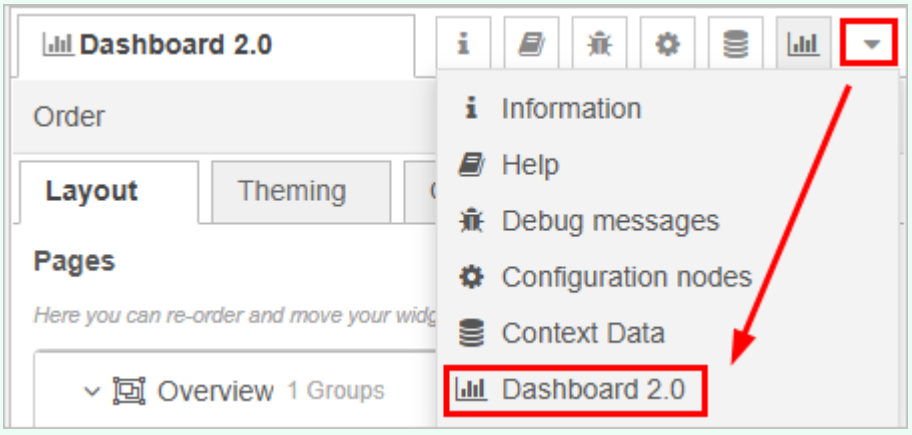
5. Under **Install**, search for and install `@flowfuse/node-red-dashboard`.

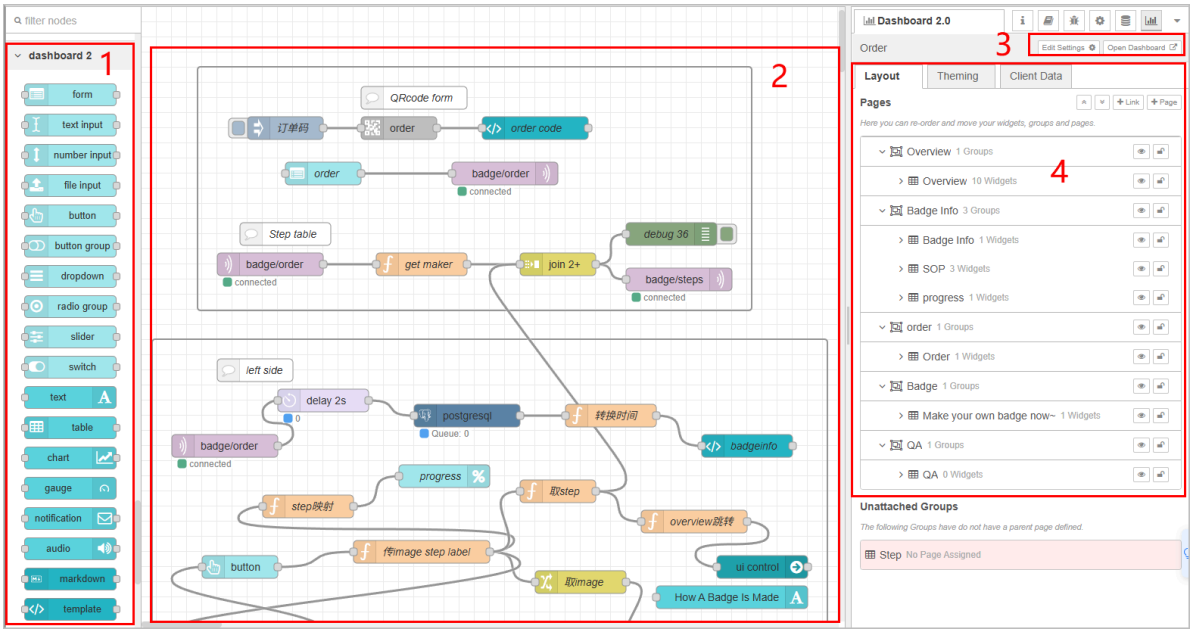


6. Use nodes under flowfuse to build a dashboard.

✓ **tip**

Click ▾ at the upper-right corner, and then click **Dashboard2.0** to adjust the layout.





Index	Item	Description
1	Node	Node of Dashboard 2.0, including gauge, form, progress bar and more.
2	Canvas	Where you drag in nodes to build a dashboard.
3	Operations	Edit the `ui-base` node and view the dashboard.
4	Layout	Displays all nodes in the corresponding hierarchical layout.

> ⓘ **info**
>
> For details on how to use each node, see Flowfuse Node Docs.

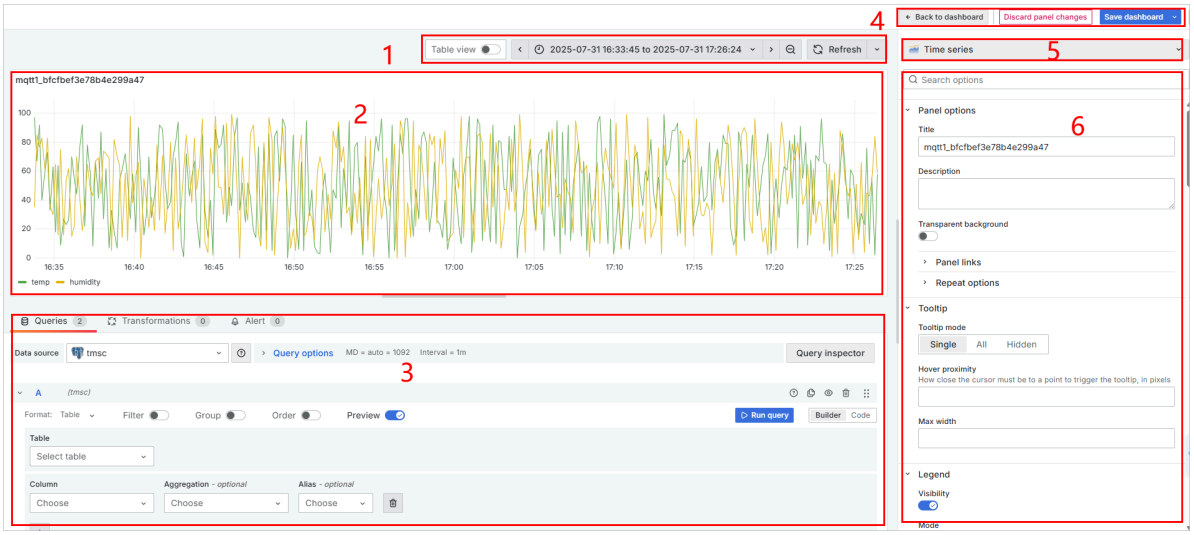
Display Data on Dashboards

Data collected to, processed on Tier0, can be visualized in charts, tables, heat maps and other formats on a dashboard to provide a comprehensive insight.

1. Log in to Tier0, and select **UNS** > **Dashboards**.

Index	Item	Description
1	Search	Search for existing dashboards.
	New Dashboard	Add a dashboard.
2	Dashboard List	Displays dashboard information. Click the name to start editing.
3	Display	Switch dashboard display between blocks and list.
4	Operations	Copy / Edit / Delete the dashboard.

2. Click **New Dashboard** to add a dashboard.
3. Click the dashboard name to start editing.
4. Click **Add Visualization**, and then select a data source for the dashboard in the pop-up window.
    - The default **tmsc** is the internal database where the data collected on **Namespace** is stored.
    - Click **Configure a new data source** at the lower-right corner to configure other data sources. e.g. MQTT.



Index	Item	Description
1	Tools	Switch between chart and table views.
	Time Range	Select the time range where the data is displayed.
	Refresh	Refresh the element.
2	Display Area	Display the current data source via the selected element.
3	Data Source Configuration	Change data source, and you can either select columns from the database or directly use SQL query to handle complex data.
4	General Operations	Go back to dashboard list/discard the changes on the current panel/save the dashboard.
5	Element	Select different elements for displaying the data.
6	Element Configuration	Configurations of the current element.

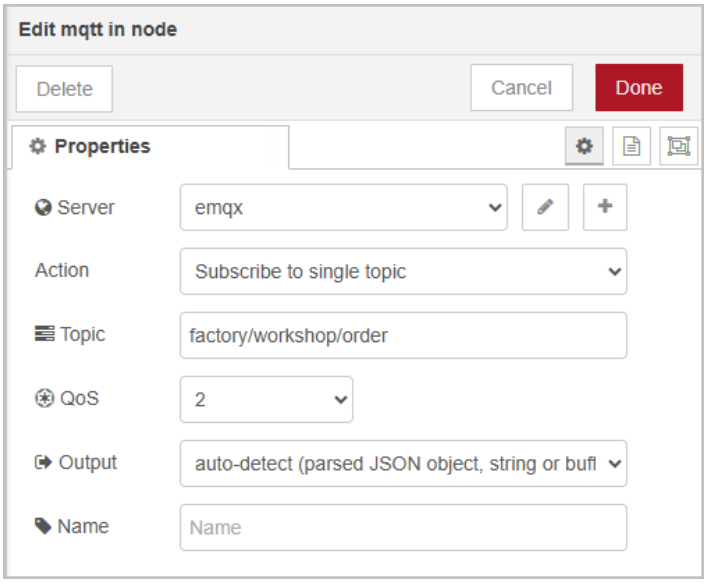
Obtain Data from Tier0

Data can be collected to Tier0 through various protocols, it can also be obtained through MQTT subscription, or even directly from Tier0 database.

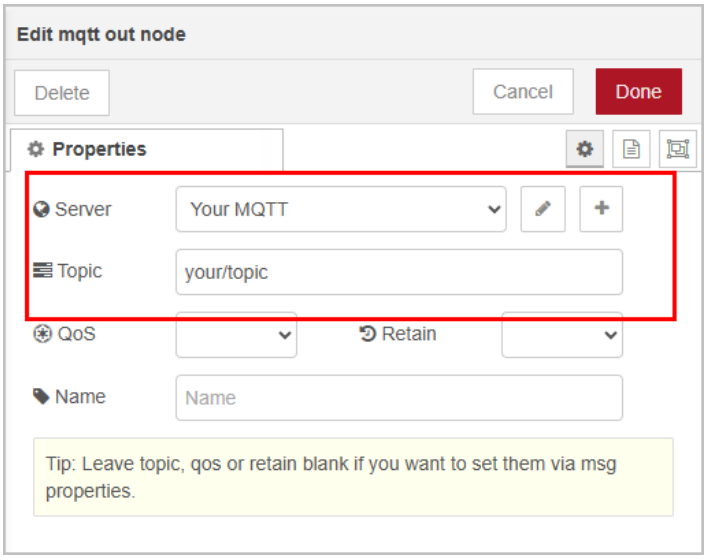
Obtaining Data through MQTT Subscription

> ⓘ **info**
>
> This method is only suitable for third parties who have experience in MQTT protocol.

1. Log in to Tier0, and then select **UNS** > **Event Flow**.
2. Click **New Event Flow**, enter a name and click **Save**.
3. Click the flow you just added, drag an `mqtt in` node, and enter the configurations.
   - Tier0 MQTT broker address: `emqx:1883`
   - Topic: The topic to obtain data from, for example, `factory/workshop/order`.



4. Process the data obtained, or directly use an `mqtt out` node to transmit the data to the third party MQTT.

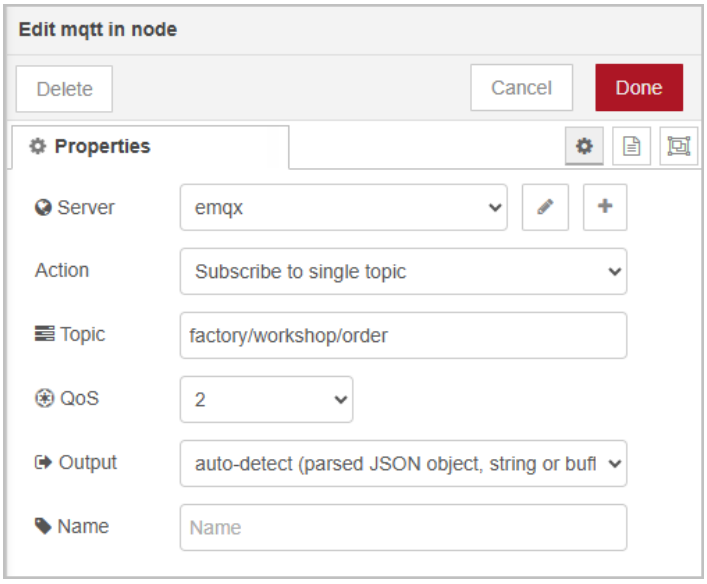


5. Deploy the flow, and data will be pushed to the third party when the topic data changes.

Obtaining Data through API

> ⓘ **info**
>
> This method is suitable for third parties who do not support MQTT and need event-driven data.

1. Log in to Tier0, and then select **UNS** > **Event Flow**.
2. Click **New Event Flow**, enter a name and click **Save**.
3. Click the flow you just added, drag an `mqtt in` node, and enter the configurations.
   - Tier0 MQTT broker address: `emqx:1883`
   - Topic: The topic to obtain data from, for example, `factory/workshop/order`.



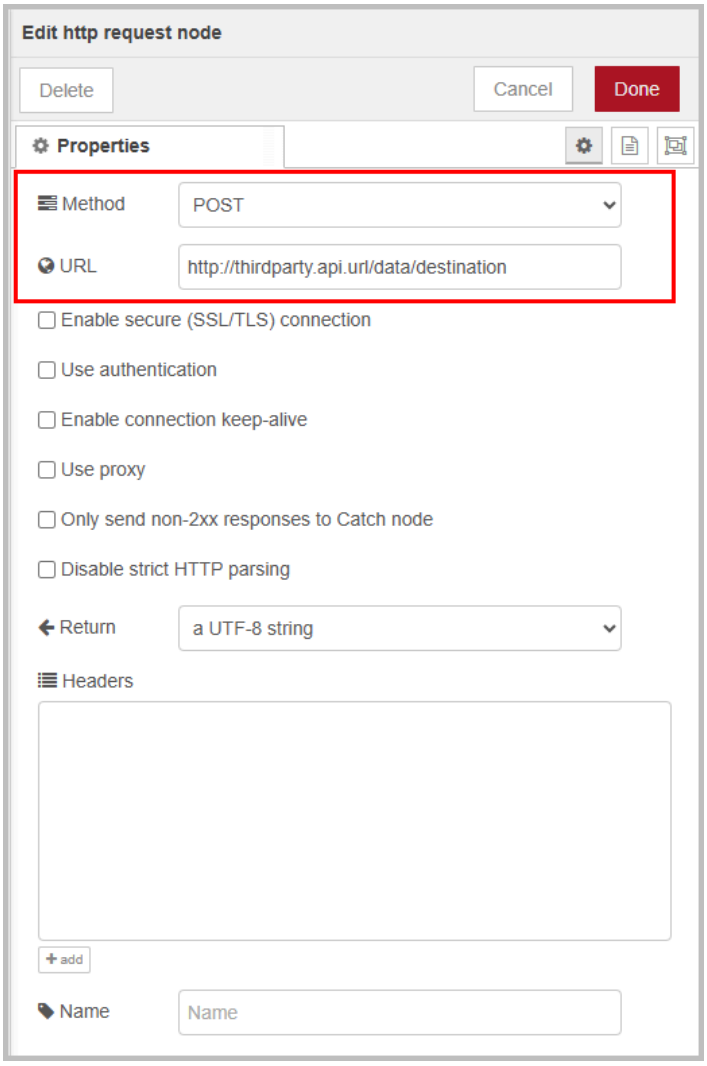
4. Drag a `function` node to the canvas, and write a script to format the data obtained according to third-party receiver API.

```
// parse payload if it's a string
if (typeof msg.payload === "string") {
    msg.payload = JSON.parse(msg.payload);
}

// set headers
msg.headers = {
    "Content-Type": "application/json"
};

return msg;
```

5. Use an `http request` node to send the data to the third party through API.



6. Deploy the flow, and data will be pushed to the third party when the topic data changes.

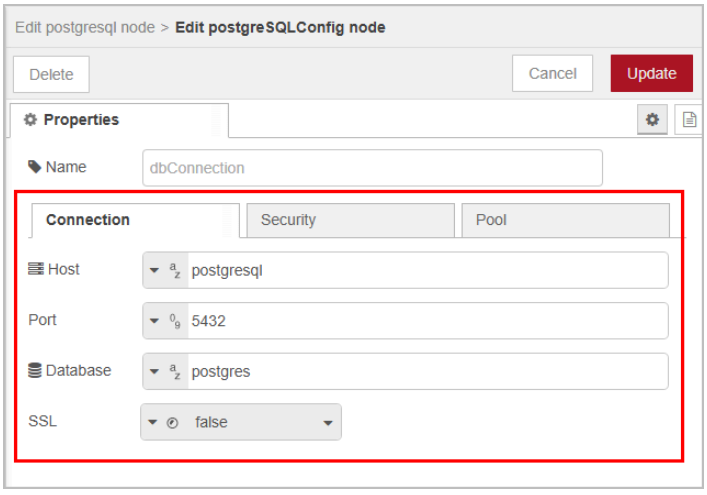
Obtaining Data Directly from Tier0 Database

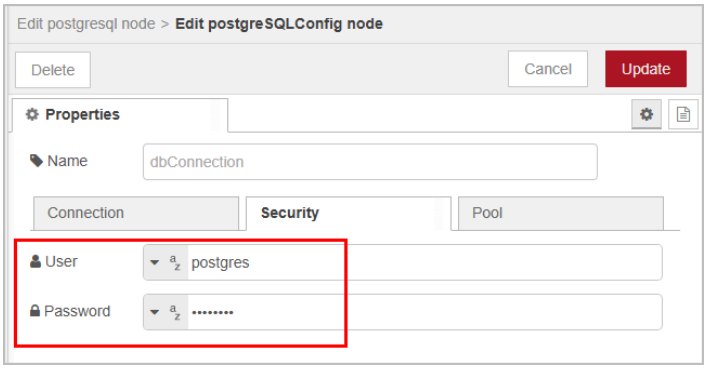
> ⓘ **info**
>
> This method is suitable for requests on history data.

1. Log in to Tier0, and then select **UNS** > **Event Flow**.
2. Click **New Event Flow**, enter a name and click **Save**.
3. Click the flow you just added, drag an `inject` node.
4. Drag a `postgres` node to the canvas, and connect to Tier0 PostgreSQL database.
    - **Connection**
        - Host: `postgresql`
        - Port: `5432`

- Database: `postgres`
- SSL: `false`

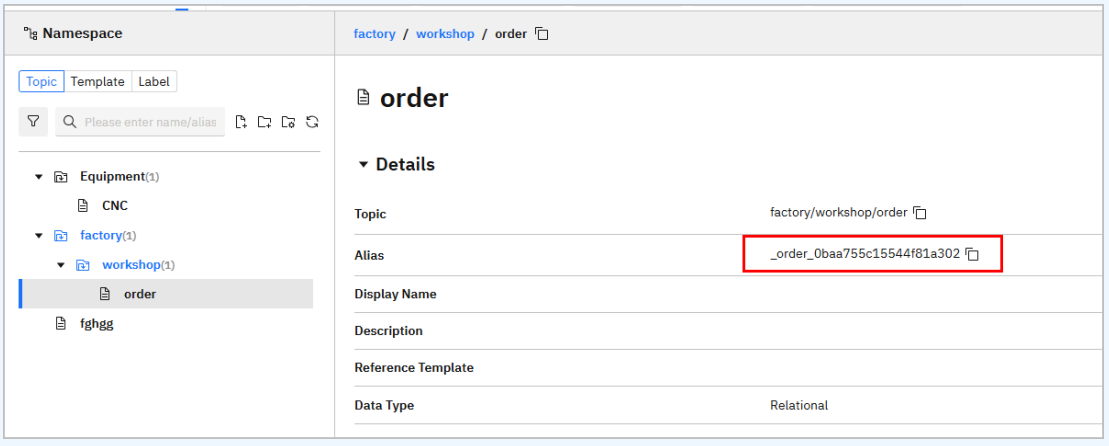
- **Security**
  - User: `postgres`
  - Password: `postgres`



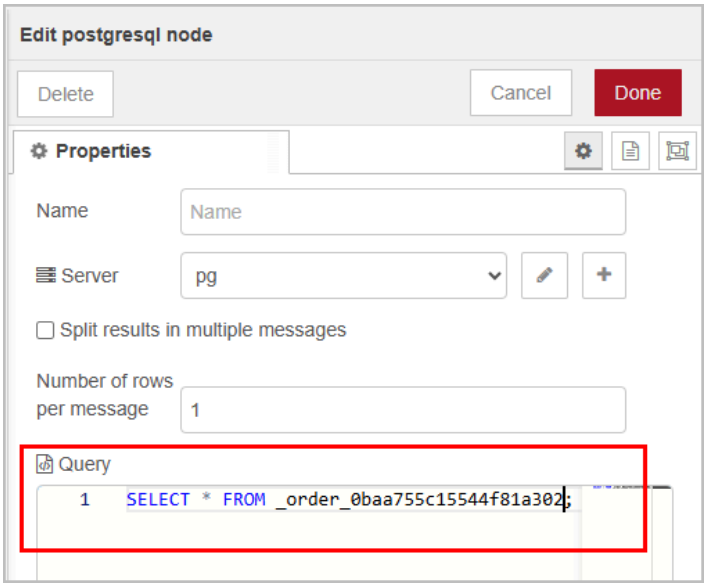


5. Write SQL statements to query data from Tier0 database.

> ⓘ **info**
>
> Get the table name from topic **Details** under **Namespace**.



```
SELECT * FROM _order_0baa755c15544f81a302;
```



6. Add an `http request` or `mqtt out` node to transmit the queried data to the third party.

Advanced Guide

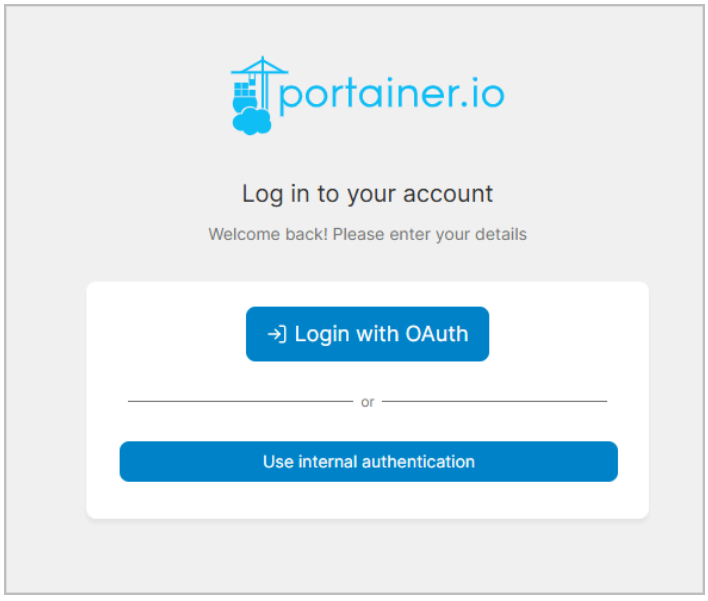
Container Management

Tier0 uses Portainer to manage containers, making it easy to deploy, manage, and scale applications.

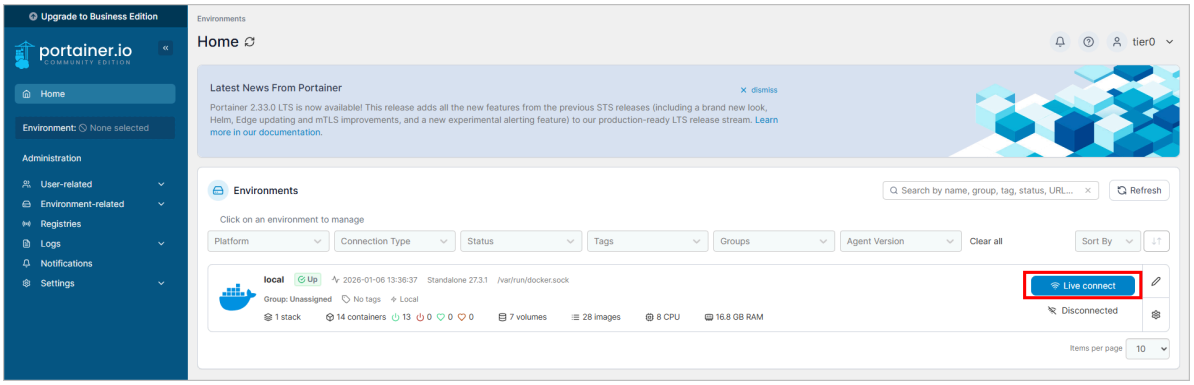
> ⓘ **info**
>
> In this guide, simple operations that relate to general container management are introduced. For detailed Portainer usage, see Portainer Documentation.

Accessing Container Management

1. Log in to Tier0, and then select **Dev Tools** > **Containers**.

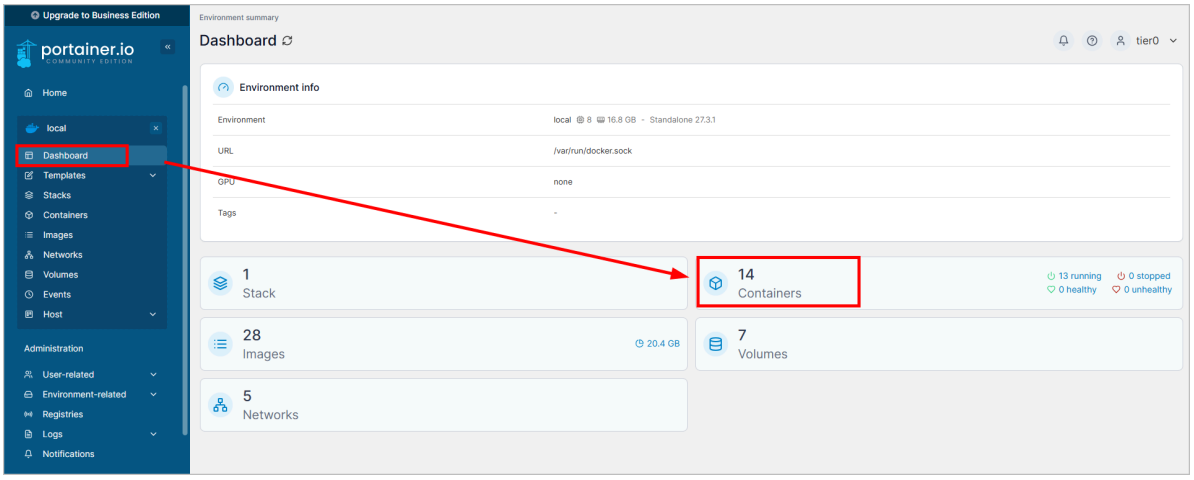


2. Click **Login with OAuth**, and then click **Live connect** to start managing containers.

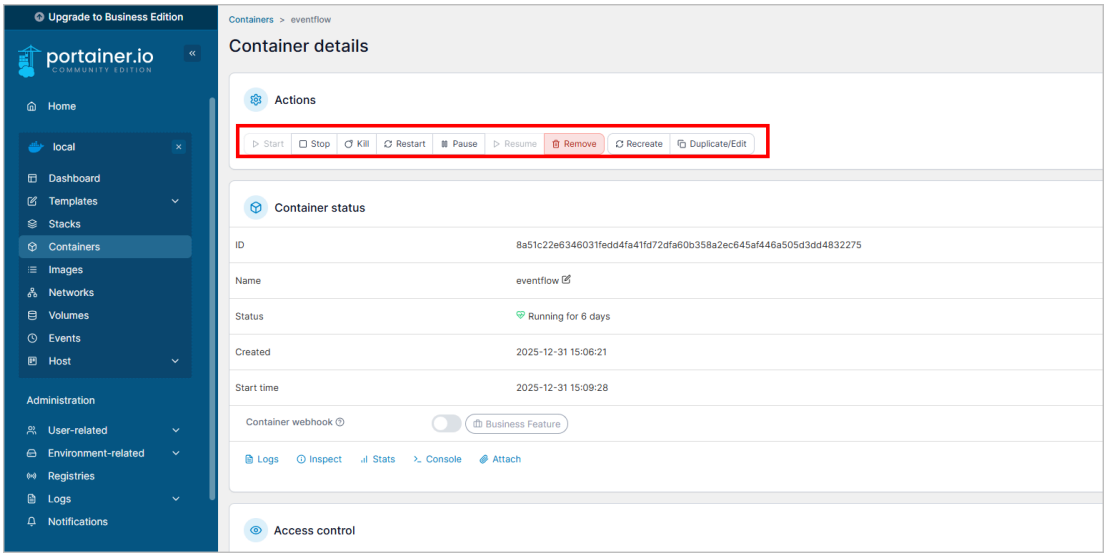


General Operations on Containers

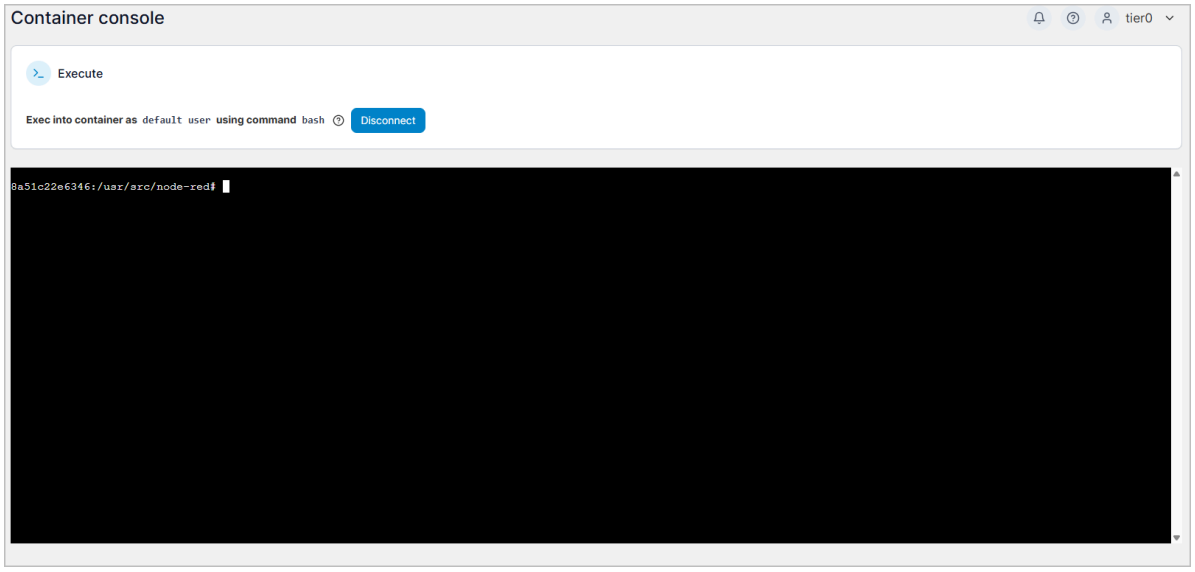
1. On the dashboard, click **Containers**.



2. Select a container, and then you can start, stop, restart, pause, or remove the container.

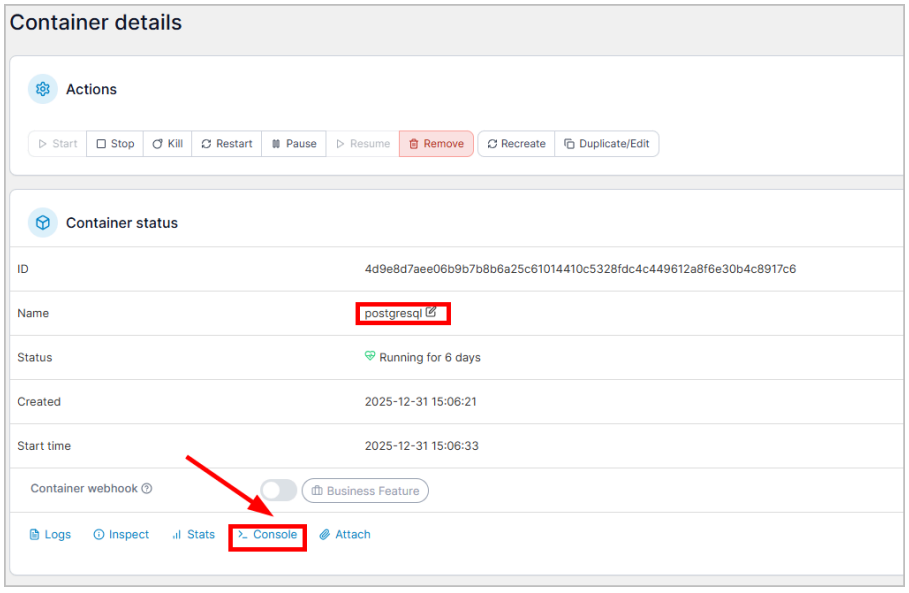


3. (Optional) Click **Console** to access the container terminal for more complex operations.



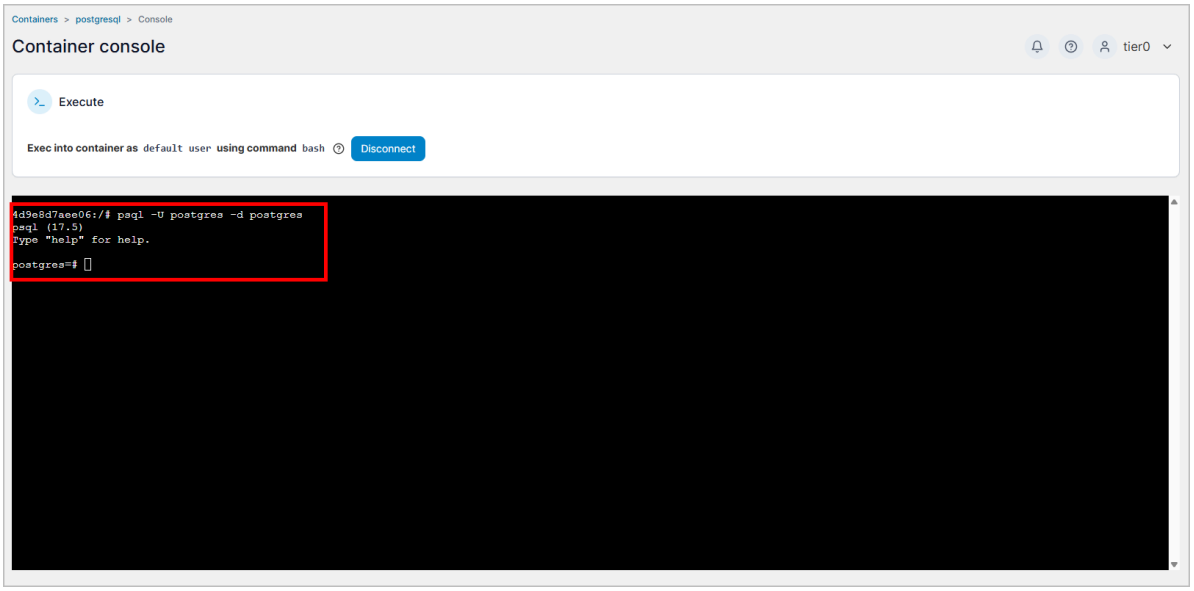
Accessing Database on Container

1. On the dashboard, click **Containers**.
2. On page 2, select **postgresql**, and then click **Console**.



3. Click **Connect**, and then use the following command to access the database.

```
psql -U postgres -d postgres
```



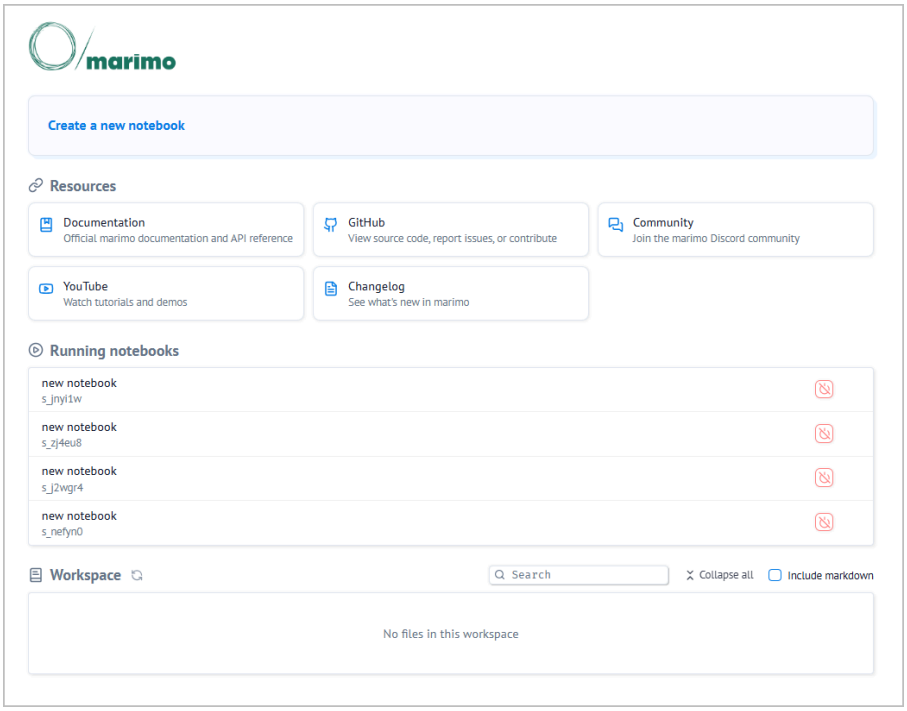
Notebook

Tier0 adopts Marimo for easy python script usage. For example, you can train models, run SQL queries and design dashboards with data connected in **Namespace**.

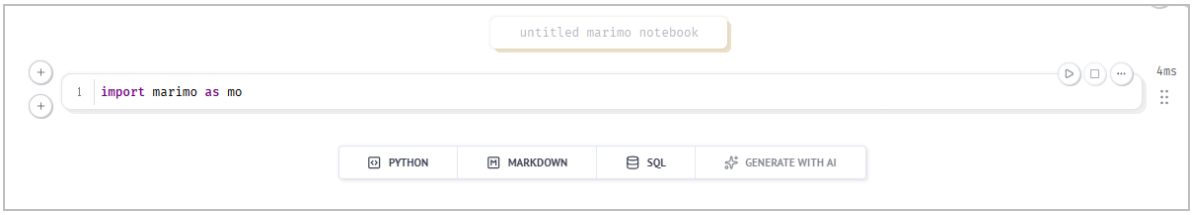
Using Notebook to Run SQL Queries

> ⓘ **note**
>
> Tier0 databases are initially connected.

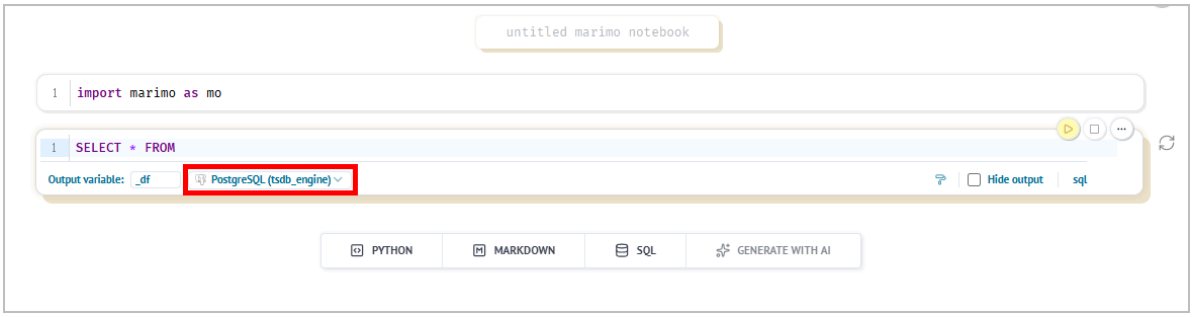
1. Log in to Tier0, and then select **Dev Tools** > **Notebook**.



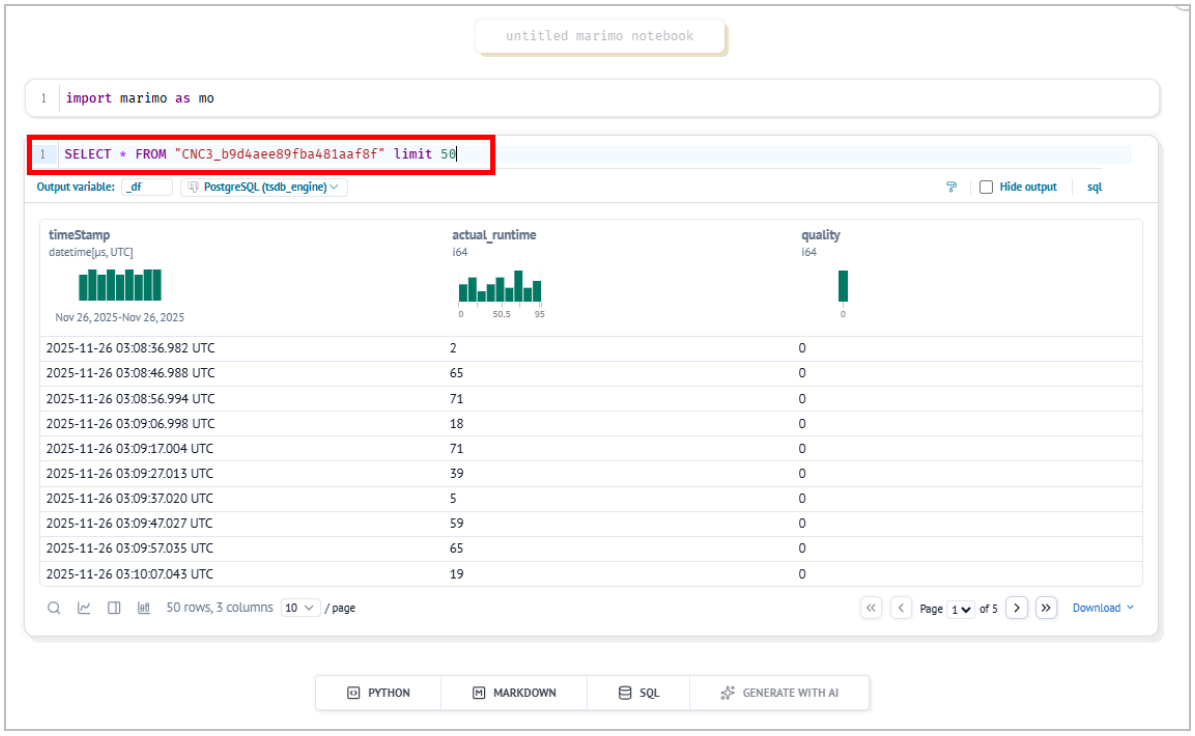
2. Click **Create a new notebook**, and then click ▷ to run the first cell.



3. Click **SQL** to add a new cell, and then select a connected database to run SQL statements.



4. Get the table name from topic **Details** under **Namespace**, and then write SQL statements.



> ✓ **tip**
>
> - Click on the left panel to view the list of connected databases.
> - Find the table you need, click next to the table name to generate SQL statements automatically.

Using Notebook to Draw Charts

1. Log in to Tier0, and then select **Dev Tools** > **Notebook**.
2. Click **Create a new notebook**, and then add the following 3 cells to draw a simple line chart.

- Cell 1: Import libraries

```
import marimo as mo
```

- Cell 2: Query data from the database

```
_df = mo.sql(
    f"""
    SELECT *
    FROM "table_name"
    LIMIT 100
    """,
    engine=tsdb_engine
)
data = _df
```

> ✓ **tip**
>
> Use a SQL cell to write queries and then click  on the cell end to switch the cell to Python. Make sure you assign the query result to a variable (e.g. `data`).

- Cell 3: Draw a line chart

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(data["actual_runtime"])
plt.title("Actual Runtime")
plt.xlabel("Index")
plt.ylabel("actual_runtime")
plt.grid(True)
plt.show()
```

3. Click at the lower-right corner to run all cells and view the chart.

untitled marimo notebook

```python
import marimo as mo
```

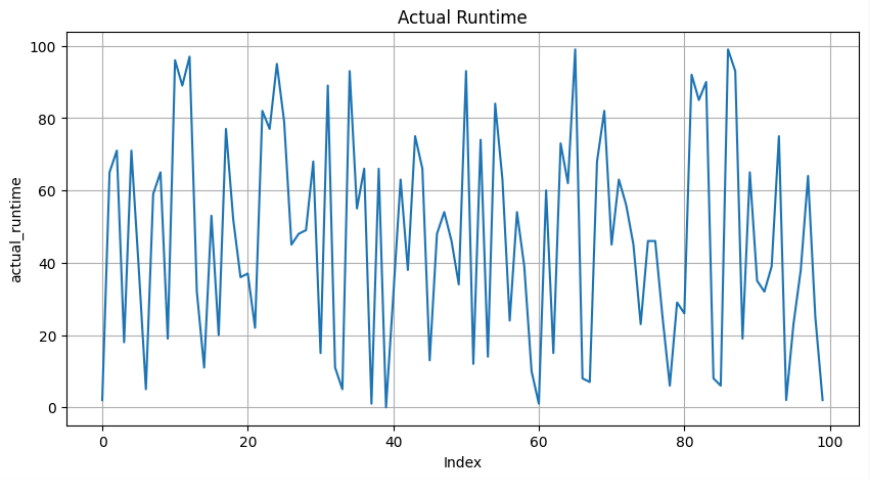
```python
_df = mo.sql(
    f"""
    SELECT *
    FROM "CNC3_b9d4aee89fba481aaf8f"
    LIMIT 100
    """,
    engine=tsdb_engine
)
data = _df
```

timeStamp datetime[µs, UTC] Nov 26, 2025-Nov 26, 2025	actual_runtime i64 0 46 93	quality i64 0
2025-11-26 03:08:36.982 UTC	2	0
2025-11-26 03:08:46.988 UTC	65	0
2025-11-26 03:08:56.994 UTC	71	0
2025-11-26 03:09:06.998 UTC	18	0
2025-11-26 03:09:17.004 UTC	71	0
2025-11-26 03:09:27.013 UTC	39	0
2025-11-26 03:09:37.020 UTC	5	0
2025-11-26 03:09:47.027 UTC	59	0
2025-11-26 03:09:57.035 UTC	65	0
2025-11-26 03:10:07.043 UTC	19	0

100 rows, 3 columns 10 / page Page 1 of 10 Download

```python
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(data["actual_runtime"])
plt.title("Actual Runtime")
plt.xlabel("Index")
plt.ylabel("actual_runtime")
plt.grid(True)
plt.show()
```

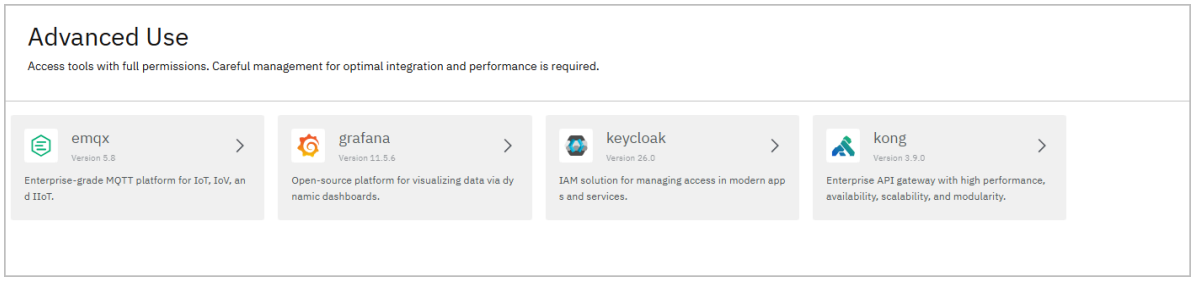


PYTHON MARKDOWN SQL GENERATE WITH AI

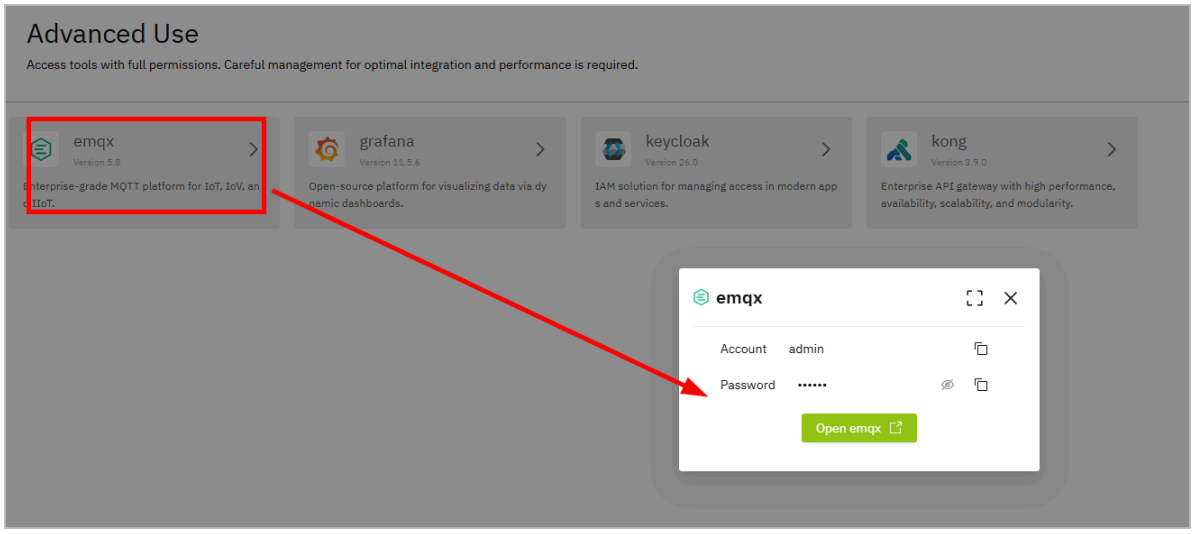
Advanced Use

Tier0 empowers you to use open source products that are closely integrated with it, with full permissions.

1. Log in to Tier0, and then select **Advanced Use** under **Dev Tools**.



2. Click an open source product to access its management page.



System

Routing Management

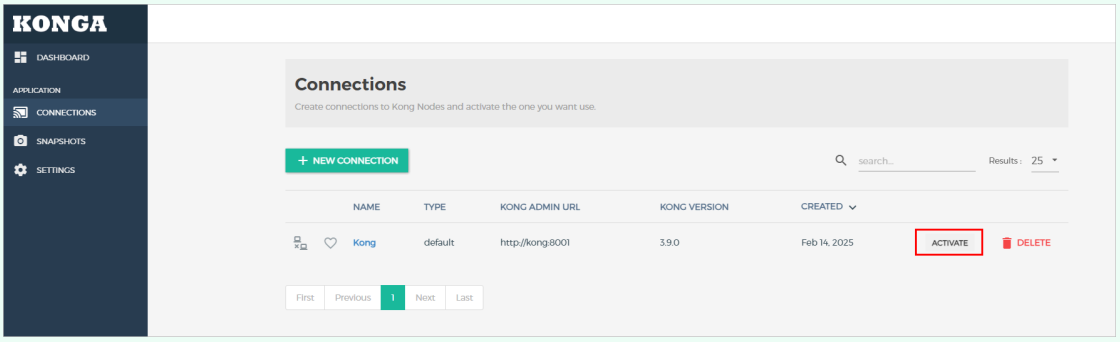
Tier0 supports service routing based on Konga, and you can add your application to Tier0 for integrated entry. Following certain rules, you can:

- Set the routing path to your backend service.
- Fashion access authorization mechanisms.
- Add external service to Tier0 frontend as one of the applications.

Creating Service

1. Select **System** > **Routing Management**, and then click **SERVICES**.

> ✓ **tip**
>
> If the **DASHBOARD** shows no active connection, go to **CONNECTIONS**, and then click **ACTIVATE** next to **Kong**.

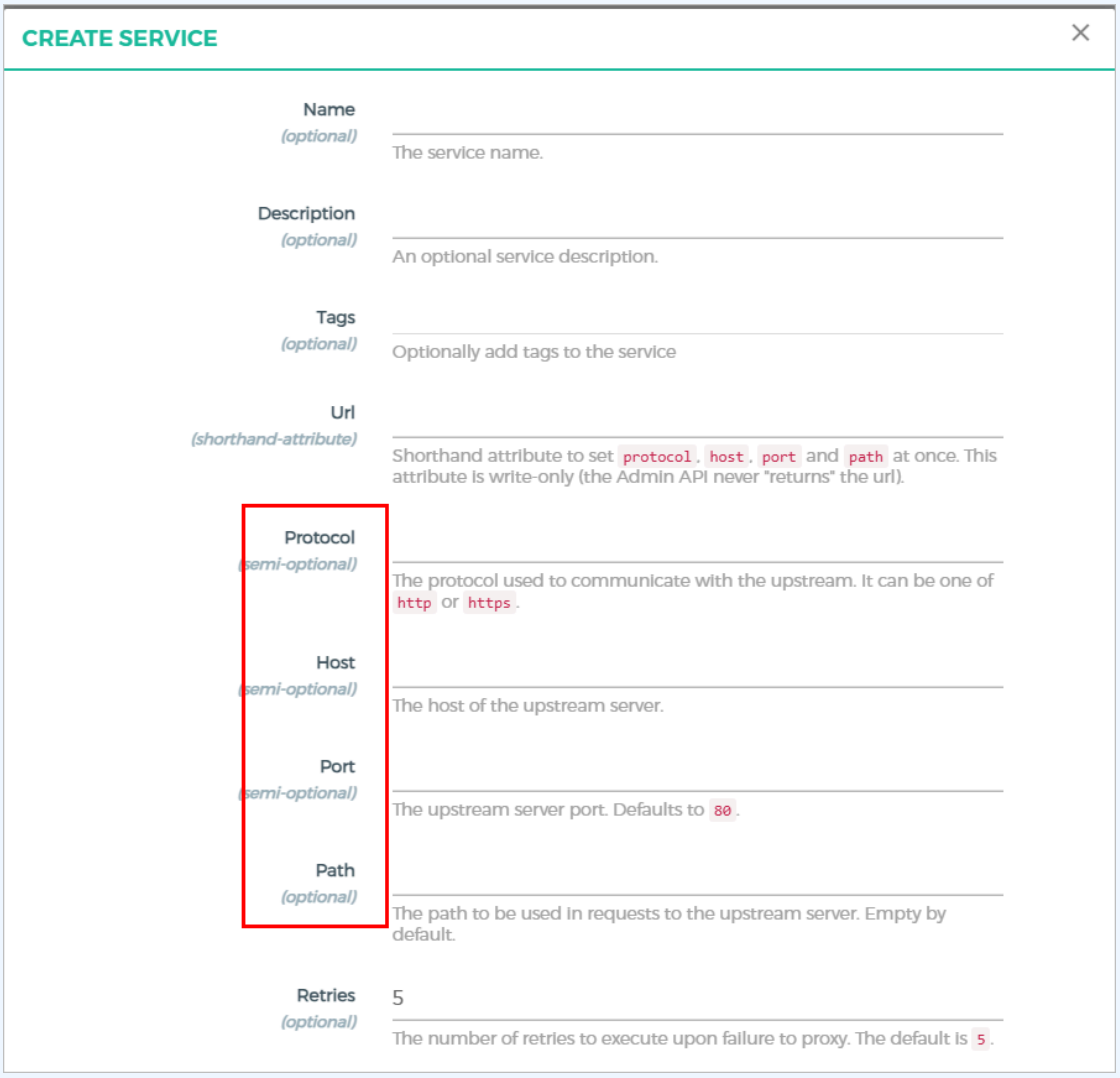


2. Click **ADD NEW SERVICE**, and then enter the information of your backend service.

ⓘ **info**

These parameters are necessary. For instance, you have a backend resource **https://example.com:8080**.

- **Protocol**: https
- **Host**: example.com
- **Port**: 8080
- **Path**: Request path prefix, which will added to the path of request from client before the request gets forwarded to your service.



3. Click **SUBMIT SERVICE**.

Building Routes

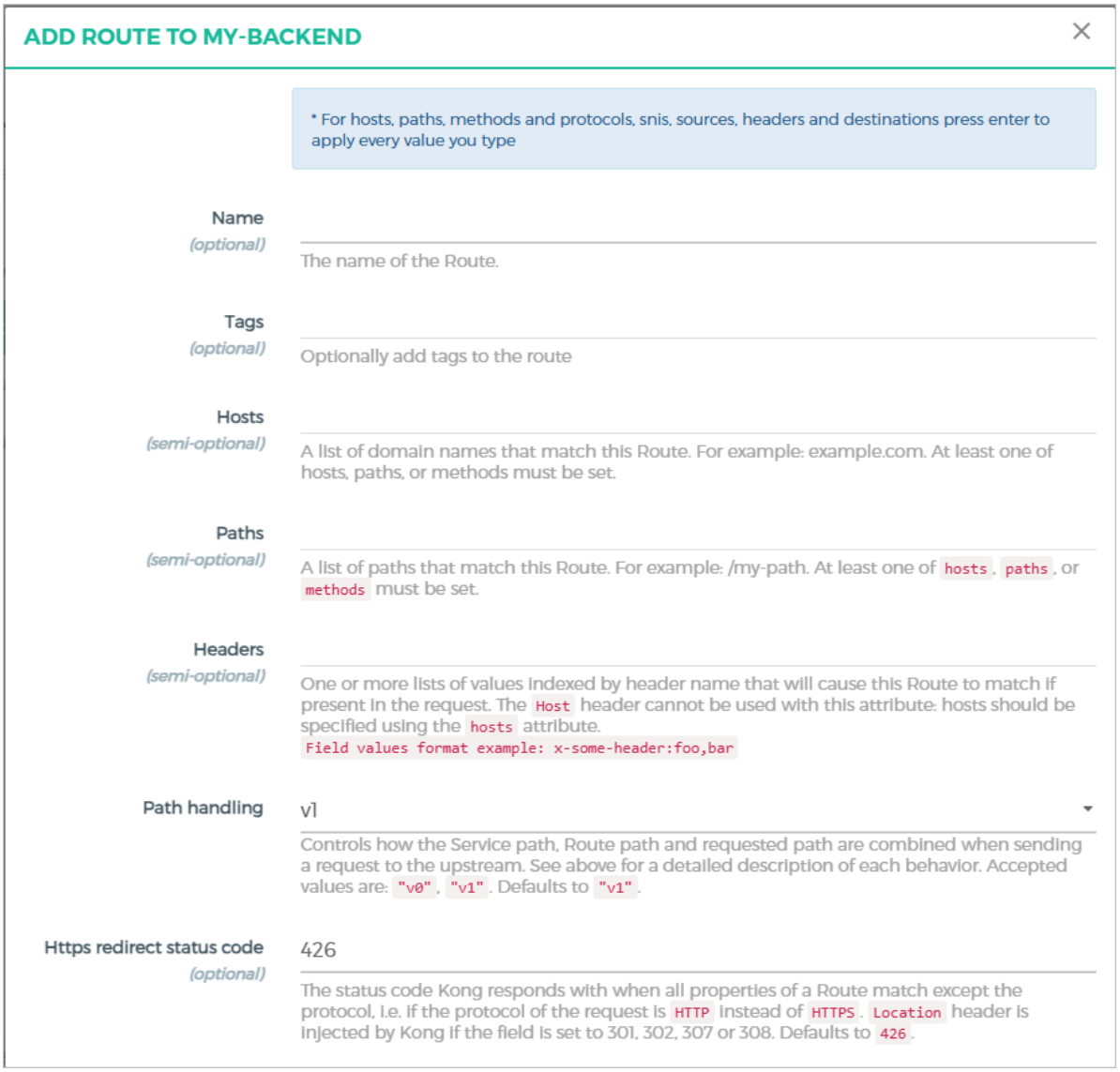
Build the routing path for accessing the service.

1. On the **Service Details** page, click **Routes**.
2. Click **ADD ROUTE**, and then enter the information of the route to access your service.

- **Tags**: We have mapped some tags with fixed meanings.

Tag	Description
menu	The service will appear as an application on Tier0 homepage.
parentName:menu.tag.xxx	xxx is the name of an existing first-level menu in Tier0 (all in lowercase). It indicates that the current route appears as a second-level menu of xxx.
homeParentName:menu.tag.xxx	xxx is the name of an existing first-level menu in Tier0 (all in lowercase). It indicates that the current route appears in the navigation bar and serves as a second-level menu of xxx.
description:menu.desc.xxx	xxx is the name of the route. It represents the description information corresponding to xxx.

- **Paths**: Request paths that will be recognized by Kong and Kong will match and forward the request to your service.



3. Click **SUBMIT ROUTE**.

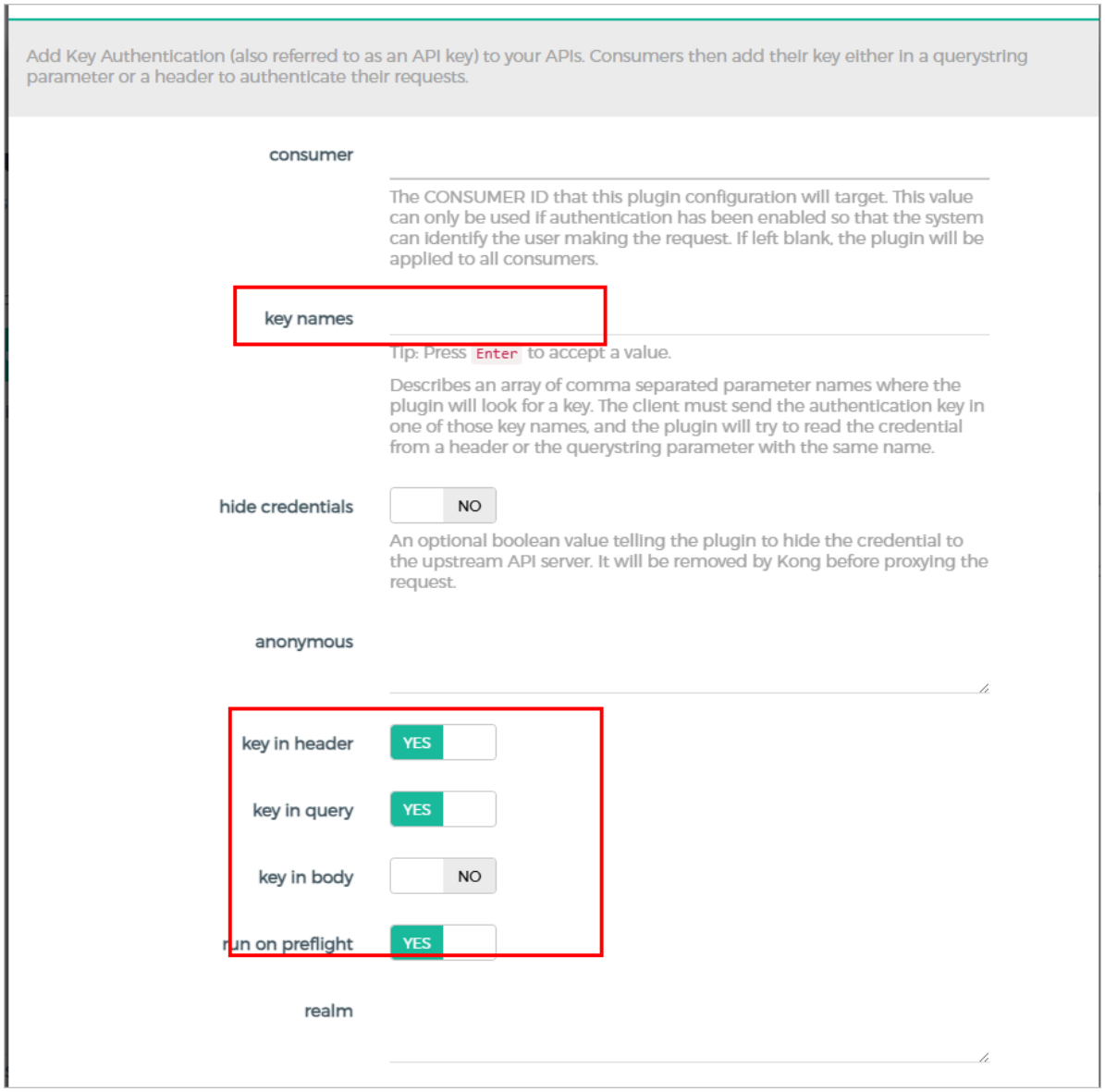
Setting Authentication

Set authentication mechanisms for your service.

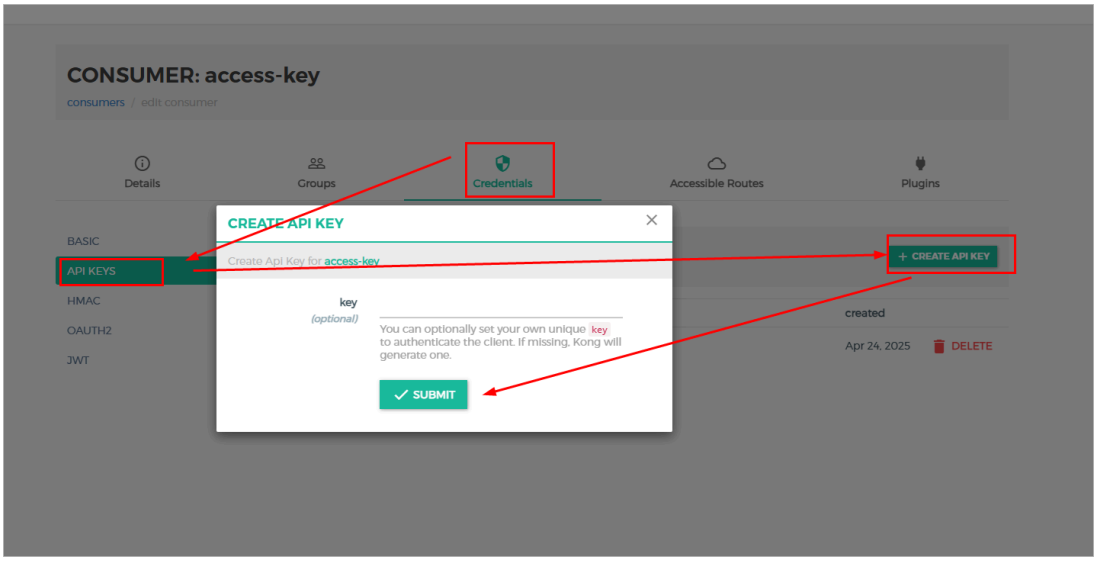
1. On the **Service Details** page, click **Plugins**.
2. Click **ADD PLUGIN**, and select **Key Auth**.

> ⓘ **note**
>
> You can add other authentication methods. For more details, see Konga.

3. Enter the **Key names** and decide how the anthentication information should be transmitted.



4. Click **ADD PLUGIN**.
5. Click **CONSUMERS** on the left side menu to add an authentication key.
    1. Click **CREATE CONSUMER**, and give it a username or ID.
    2. On the details page, select **Credentials**.
    3. Select **API KEYS**, and then create an API key.



4. Leave it empty and click **SUBMIT**, Konga will generate a custom key, which is necessary when accessing your service.

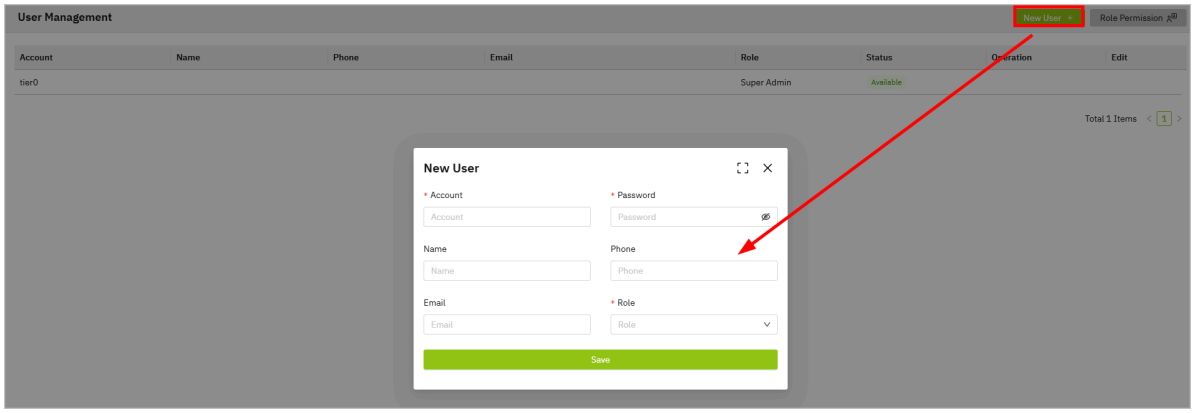
User Management

In Tier0, you can add users and assign different roles to them, easily managing module access. Meanwhile, you can also allow users to register account by themselves.

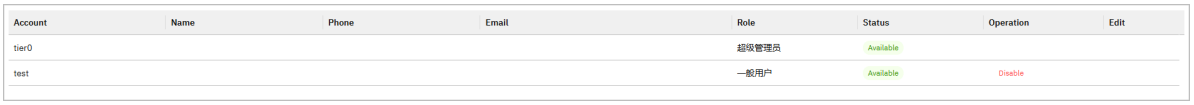
> ⓘ **info**
>
> We adopted **Keycloak** to achieve comprehensive user management. You can customize it under **System** > **Authentication**.

Adding User

1. Log in to Tier0, and then select **User Management** under **System**.
2. Click **New User** at the upper-right corner.



3. Enter the information and select a role for the user, and then click **Save**.

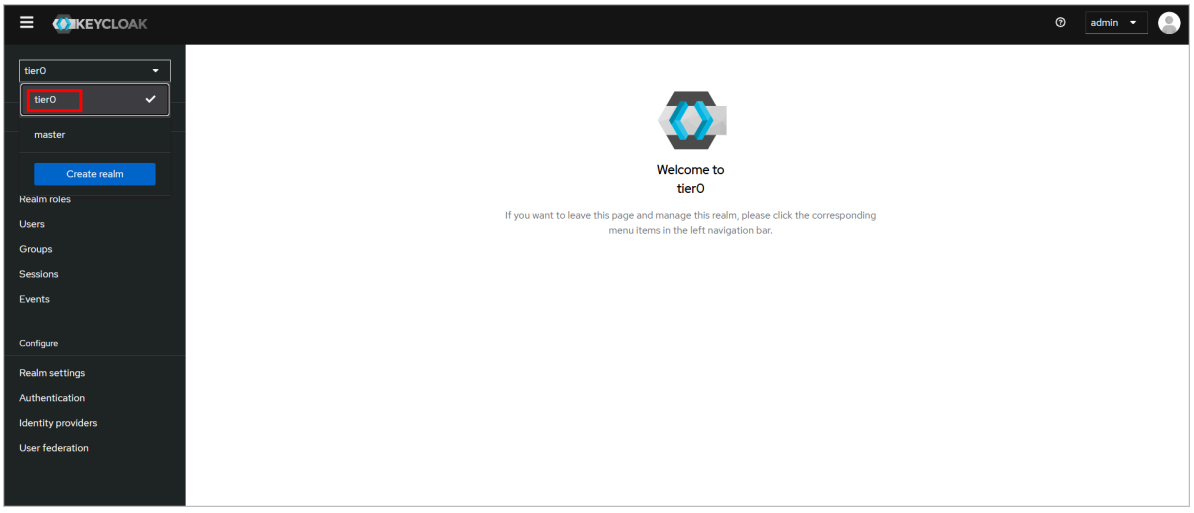


- You can disable, edit, delete user and reset password of it.
- The default user cannot be disabled or deleted.

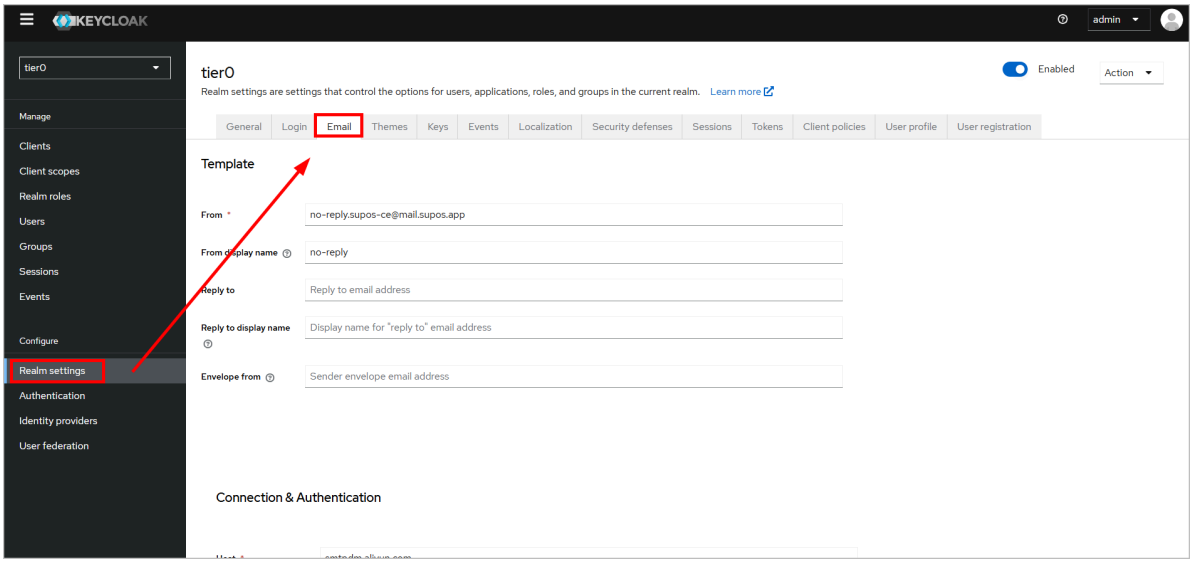
Registering User

For user registration, email is necessary. You need to set up an Email server for sending validation email.

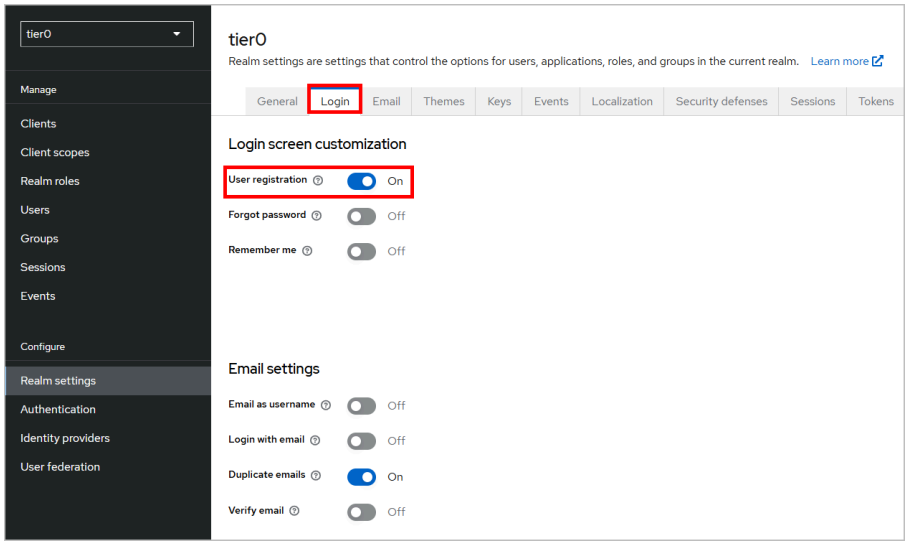
1. Select **System** > **Authentication**, and then log in to Keycloak with **admin/tier0**.
2. Switch the realm to **tier0**, and then select **Realm settings**.



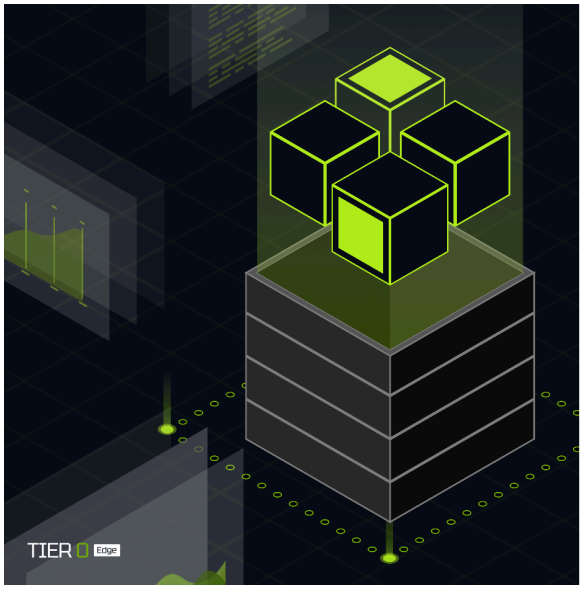
3. Click **Email**, and then enter the email template and server information.

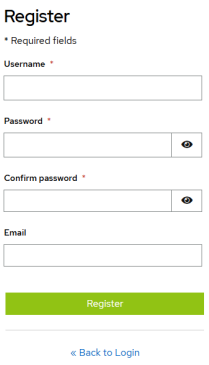


4. Save the settings, and then click **Login** to enable **User registration**.



5. Go to the Tier0 login page, and then click **Register**.
6. Enter the information to apply for a user account.





Open Data

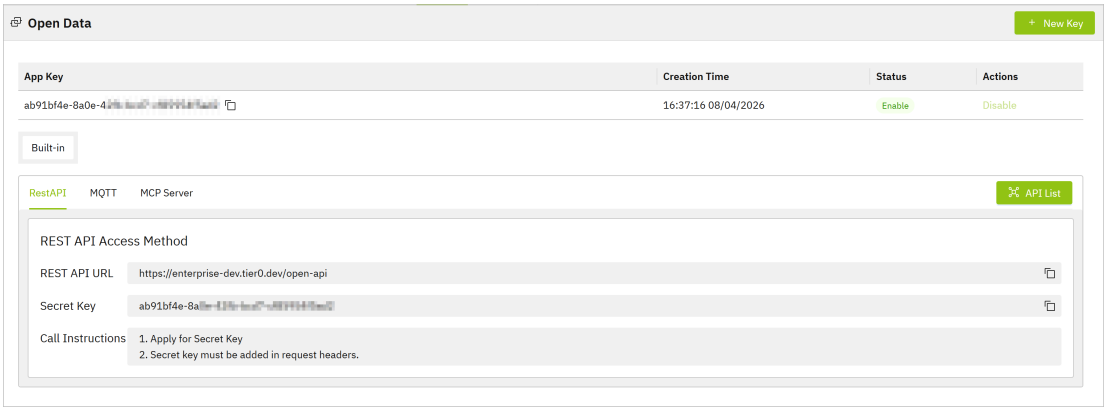
How to Access Data in Tier0

Tier0 provides 3 types of data subscription and invocation methods: API, MQTT, and MCP.

API

To access data through API, you need to use the URL and secret key provided in the platform.

1. Log in to Tier0, and select **System** > **Open Data**.
2. Click **New Key** to create a secret key.



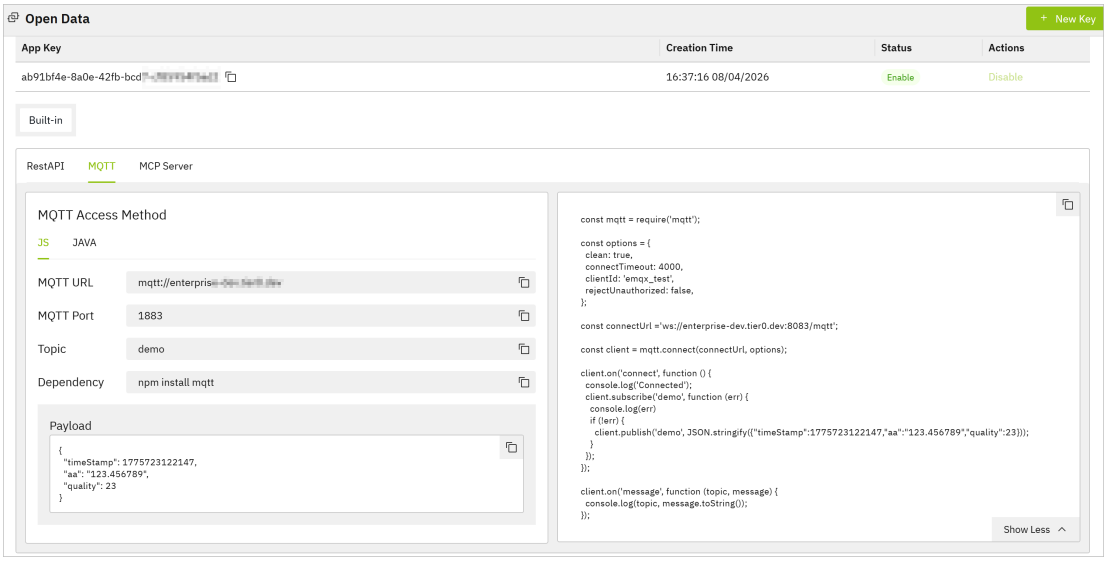
3. Use the URL displayed on the bottom, and the created key to obtain data from Tier0.

> ⓘ **info**
>
> Click **API List** to read the API documentation.

MQTT

You can use third-party clients like MQTTX, MQTT Explorer, or custom code to subscribe to data through MQTT.

1. Log in to Tier0, and select **System** > **Open Data**.
2. Click **MQTT** to view the MQTT connection information, including the broker address, port and topic.

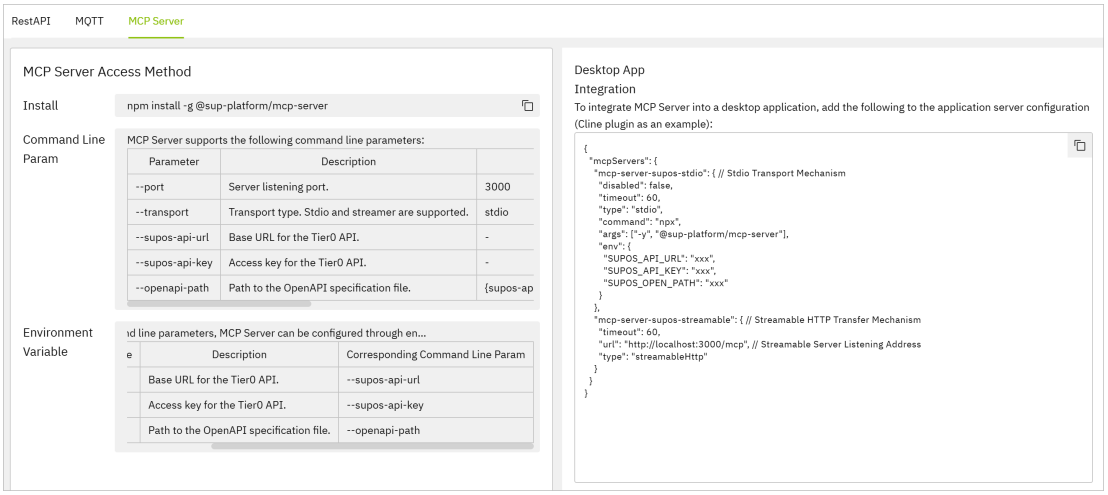


> ⓘ **info**
>
> Tier0 provides script samples in both JavaScript and Java to help you get started.

MCP Server

Tier0 enables you to use an MCP server to get data from the platform through a custom LLM.

1. Log in to Tier0, and select **System** > **Open Data**.
2. Click **MCP Server** to view the MCP connection details and example.



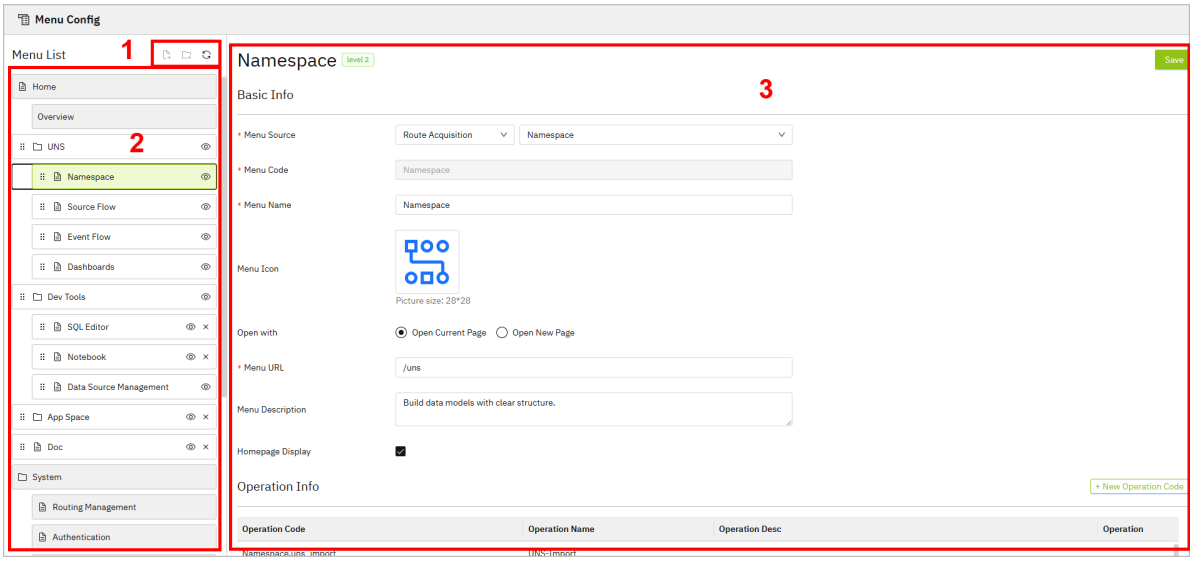
> ⓘ **info**
>
> - Cline plugin usage is used as an example.
> - The required URL and key can be obtained on the **RestAPI** tab.
> - You can click **API List** on the **RestAPI** tab, and copy the address as the **SUPOS_OPEN_PATH** in the scriptto help AI understand.

Menu Config

Tier0 enables basic configuration on function modules in the style of menus, including module position, visibility, and detailed information. Meanwhile, it can be used as an application manager where you can add and configure your web app on Tier0.

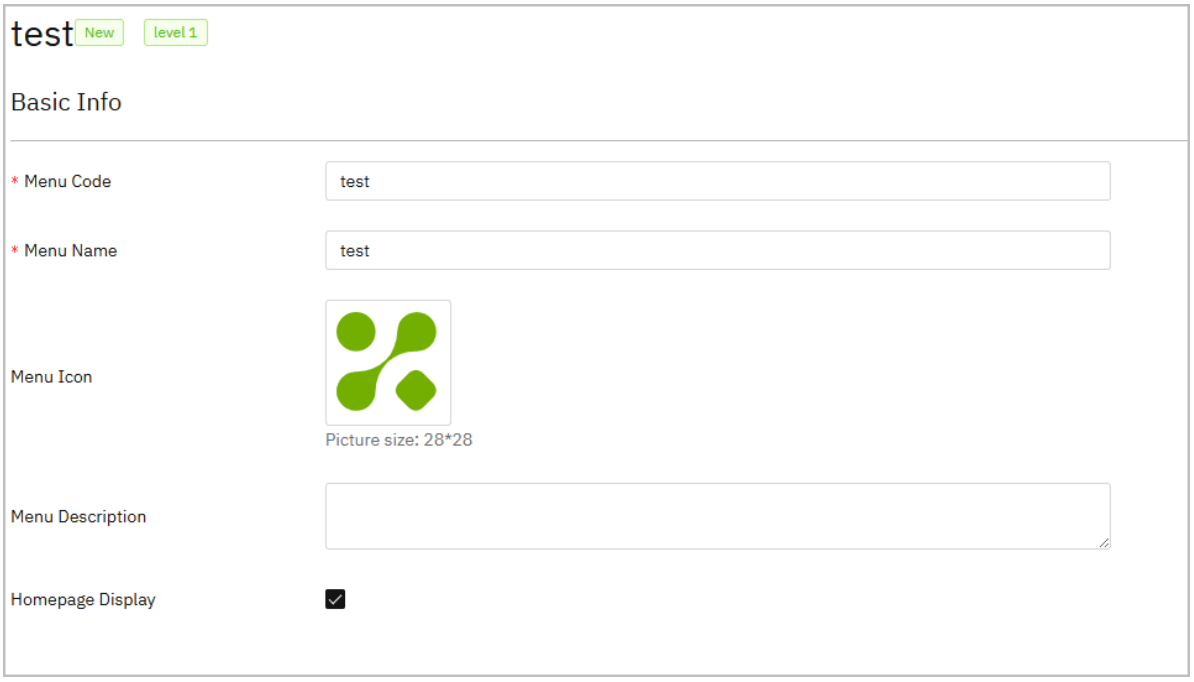
Adding Menu

1. Log in to Tier0, and then select **System** > **Menu Config**.



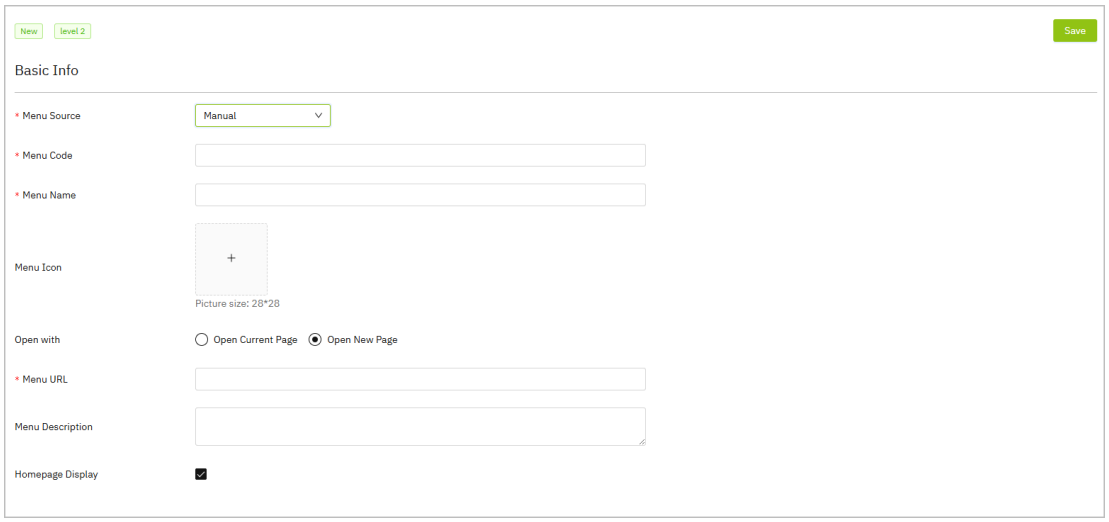
No.	Name	Description
1	Operation	Add menu in floder/file structure and refresh the menu list.
2	List	Lists existing structured menus.
3	Information	Displays information of the selected menu.

2. Click on the top to add a menu folder.
3. Enter the folder information and then click **Save**.



> ⓘ **info**
>
> Menu code cannot be duplicated.

4. Select the created folder, and then click to add a menu file.
5. Enter the menu information and then click **Save**.

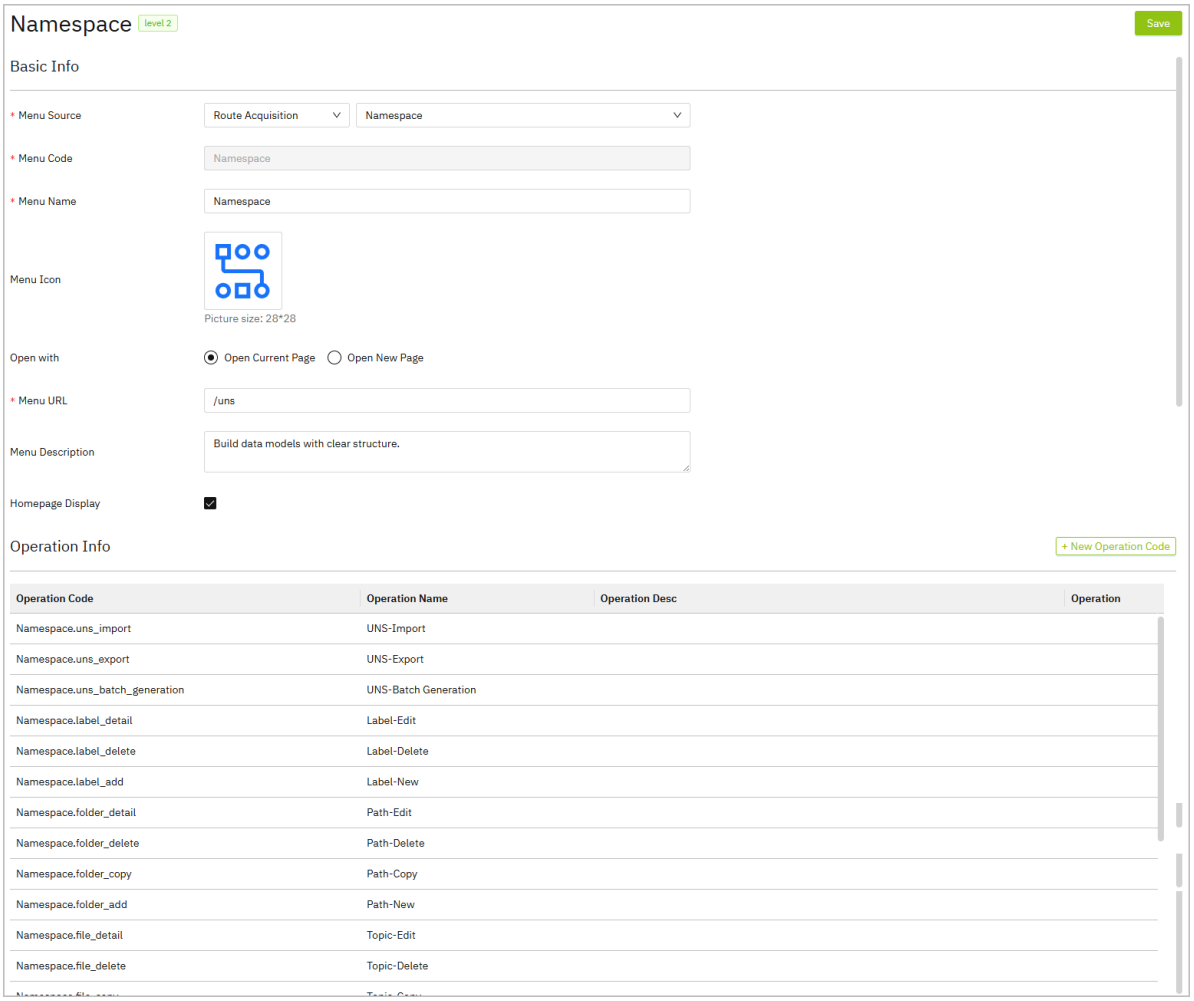


Parameter	Description
Menu Source	Select menu source between manual addition and existing options.
Menu Code	Unique identification of the menu.
Open with	Directly open the menu on the current page or on a new page.
Menu URL	Access address of the added function module (menu).
Homepage Display	Whether or not to display the menu on the homepage.

6. (Optional) Go back to **Home** to check the added menu if it is set to display on homepage.

Editing Existing Menus

1. Log in to Tier0, and then select **System** > **Menu Config**.
2. Select a menu from the list, eidt its information on the right side, and then click **Save**.



> ⚠ **warning**
>
> For original menus provided by Tier0, do not change the menu source, URL and operation code. Otherwise the module may not work properly.

Managing Menu Position and Visibility

- On the left list, drag a menu to change its position in the folder/file structure, and it will show on the homepage automatically.
- Click next to a menu to hide it from homepage.
- Click × to delete a menu.

> ⚠ **warning**
>
> - Deleting a folder will delete all menus under it.
> - Deleting a menu cannot be restored. Please proceed with caution.

Glossary

Key Terms and Definitions

Term	Definition
Tier0	An open-source industrial data integration platform built on the Unified Namespace (UNS) methodology, designed for managing, processing, and optimizing industrial data.
UNS (Unified Namespace)	A data management method that centralizes data from various devices and systems, providing a structured and unified framework to prevent data silos.
Node-RED	An open-source automation tool that enables users to wire together devices, APIs, and online services using a flow-based programming approach.
Grafana	An open-source platform for monitoring and data visualization, which allows users to create dashboards to view real-time data from various sources.
MQTT	A lightweight messaging protocol for small sensors and mobile devices optimized for high-latency or unreliable networks, commonly used for IoT communication.
JSON	JavaScript Object Notation, a lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate.
TimescaleDB	An open-source time-series database built on PostgreSQL, designed for handling large-scale time-series data, especially useful for monitoring systems and IoT applications.
PostgreSQL	An advanced open-source relational database management system that uses SQL for querying and is designed for high performance and reliability.
API (Application Programming Interface)	A set of protocols and tools that allows different software applications to communicate and interact with each other.
Topic	A unique identifier for data in the Tier0 system that helps classify and organize data from various sources for easy access and management.

Term	Definition
OEE (Overall Equipment Effectiveness)	A metric used to assess the efficiency and effectiveness of equipment in a manufacturing environment, considering availability, performance, and quality.
AGV (Automated Guided Vehicle)	An automated vehicle used in manufacturing or warehouse environments to transport materials, reducing human intervention in logistics operations.
REST API	A web service interface that allows applications to communicate over HTTP, using standard methods like GET, POST, PUT, and DELETE to access resources.
Time-Series Data	Data that is collected and recorded at regular intervals over time, such as equipment run time, temperature, and other performance metrics.
JSON Payload	Data transferred in the form of a JSON object that contains relevant information, such as sensor readings or device states, to be processed or stored.
Authentication	The process of verifying the identity of a user or system to ensure that only authorized entities can access resources.
Event-Driven	A programming paradigm in which the flow of the program is determined by events, such as user actions, sensor data, or messages from other systems.
JSONB	A PostgreSQL data type that stores JSON data in a binary format, offering efficient indexing, faster querying, and improved performance compared to plain JSON fields.